

Block

2

Probability and Distributions

UNIT 4

Basics of Probability	99
------------------------------	-----------

UNIT 5

Random Variables and Expectation	135
---	------------

UNIT 6

Discrete Probability Distributions	173
---	------------

UNIT 7

Continuous Probability Distributions and Exact Sampling Distributions	199
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BLOCK-2 INTRODUCTION

This block focuses on the concept of probability and probability distribution. This block first introduces you to some of the basic laws of probability, including Bayes's theorem. It also introduces the concept of discrete and continuous random variables, including bivariate random variables. The block discusses some of the basic discrete and continuous distributions. In addition, it introduces you to some of the basic exact sampling distributions. This block is divided into four units:

Unit 4 gives a detailed introduction to the basic concepts of probability. It defines the concept of sample space and event, mathematical probability, the axioms of probability and the addition law. Further, the unit introduces you to the concept of conditional probability, independent events and Bayes' Theorem.

Unit 5 discusses the concept of random variables and expectation. This unit defines the discrete random variable and probability mass function, and the continuous random variable and probability density function. It also discusses bivariate discrete and bivariate continuous random variables. The unit also introduces the expectation of a random variable. This unit briefly discusses the concept of moments in terms of expectation. It defines variance and covariance, and the addition and multiplication theorems of expectation

Unit 6 introduces you to several discrete probability distributions. It explains the Bernoulli distribution and its properties, the binomial probability function, and the Poisson distribution. discrete uniform distribution and hypergeometric distribution.

Unit 7 is devoted to continuous probability distributions and exact sampling distributions. The unit describes the normal distribution and the continuous uniform distribution. It also introduces you to χ^2 -distribution, t-distribution and f-distribution

The content of each unit is adopted from several units, listed in the reference section of each unit.

UNIT 4 BASICS OF PROBABILITY

Structure

- 4.0 Introduction
- 4.1 Objectives
- 4.2 Sample Space and Events
- 4.3 Computing Mathematical Probability
- 4.4 Axioms of Probability
- 4.5 Addition Law
- 4.6 Conditional Probability
- 4.7 Independent Events
- 4.8 Bayes' Theorem
- 4.9 Summary
- 4.10 Solutions /Answers
- 4.11 References

4.0 INTRODUCTION

In our daily lives, we face many situations when we are unable to forecast the future with complete certainty. That is, in many decisions, uncertainty is faced. Need to cope up with uncertainty leads to the study and use of the probability theory. The first attempt to give quantitative measure of probability was made by Galileo (1564-1642), an Italian mathematician, when he was answering the following question on the request of his patron, the Grand Duke of Tuscany, who wanted to improve his performance at the gambling tables: "With three dice a total of 9 and 10 can each be produced by six different combinations, and yet experience shows that the number 10 is oftener thrown than the number 9?" To the mind of his patron the cases were (1, 2, 6), (1, 3, 5), (1, 4, 4), (2, 2, 5), (2, 3, 4), (3, 3, 3) for 9 and (1, 3, 6), (1, 4, 5), (2, 2, 6), (2, 3, 5), (2, 4, 4), (3, 3, 4) for 10 and hence he was thinking that why they do not occur equally frequently i.e. why their chances are not the same? Galileo makes a careful analysis of all the cases which can occur, and he showed that out of the 216 possible cases 27 are favourable to the appearance of the number 10 since permutations of (1, 3, 6) are (1, 3, 6), (1, 6, 3), (3, 1, 6), (3, 6, 1), (6, 1, 3), (6, 3, 1) i.e. number of permutations of (1, 3, 6) is 6; similarly, the number of permutations of (1, 4, 5), (2, 2, 6), (2, 3, 5), (2, 4, 4), (3, 3, 4) is 6, 3, 6, 3, 3 respectively and hence the total number of cases come out to be $6 + 6 + 3 + 6 + 3 + 3 = 27$ whereas the number of favourable cases for getting a total of 9 on three dice are $6 + 6 + 3 + 3 + 6 + 1 = 25$. Hence, this was the reason for 10 appearing oftener thrown than 9. But the first foundation was laid by the two mathematicians Pascal (1623-62) and Fermat (1601-65) due to a gambler's dispute in 1654, which led to the creation of a mathematical theory of probability by them. Later, important contributions were made by various researchers including Huyghens (1629 - 1695), Jacob Bernoulli (1654-1705), Laplace (1749-1827), Abraham De-Moivre (1667-1754), and Markov (1856-1922). Thomas Bayes (died in 1761, at the age of 59) gave an important technical result known as Bayes' theorem, published after his death in 1763, using which probabilities can be revised on the basis of some new information. Thereafter, the probability, an important branch of Statistics, is being used worldwide. We will start this unit with very elementary ideas. We will go step

by step clearing the basic ideas which are required to understand the probability. In this unit, we will first present the various terms which are used in the definition of probability and then we will give the classical definition of probability and simple problems. This will be followed by discussion on topics like Axioms of probability, conditional probability and Bayes' theorem.

4.1 OBJECTIVES

After studying this unit, you would be able to

- define and give examples of random experiment and trial;
- define and give examples of sample space, sample point and event;
- explain mutually exclusive cases
- explain the mathematical definition of probability;
- define and use axioms of probability;
- define and use the addition law, conditional probability and independent events;
- define and use Bayes' theorem;
- solve problems of probability based in addition law, conditional probability, independent events and Bayes' theorem.

4.2 SAMPLE SPACES AND EVENTS

In this section, first, we will define the terms Random Experiment and Trial, which form the basis of probability theory. This will be followed by a discussion on sample spaces and events.

Random Experiment: An experiment in which all the possible outcomes are known in advance, but we cannot predict as to which of them will occur when we perform the experiment, e.g. Experiment of tossing a coin is random experiment as the possible outcomes, "Heads" and "Tails" are known in advance, but which one will turn up is not known. Similarly, 'Throwing a die' and 'Drawing a card from a well shuffled pack of 52 playing cards' are the examples of random experiments.

Trial: Performing an experiment is called a trial, e.g.

- (i) Tossing a coin is a trial.
- (ii) Throwing a die is a trial.

Sample Space: Set of all possible outcomes of a random experiment is known as sample space and is usually denoted by S , and the total number of elements in the sample space is known as size of the sample space and is denoted by $n(S)$, for example:

- (i) If we toss a coin, then the sample space is $S = \{H, T\}$, where H and T denote Heads and Tails, respectively and $n(S) = 2$.
- (ii) If a die is thrown, then the sample space is $S = \{1, 2, 3, 4, 5, 6\}$ and $n(S) = 6$, as the die has six faces numbered 1, 2, 3, 4, 5, 6.
- (iii) If a coin and a die are thrown simultaneously, then the sample space is $S = \{H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6\}$ and $n(S) = 12$. where $H1$ denotes that the coin shows Heads and die shows 1, etc.

Note: Unless stated, the coin means an unbiased coin (i.e. the coin which favours neither Heads nor Tails)

- (iv) If a coin is tossed twice or two coins are tossed simultaneously, then the sample space is $S = \{HH, HT, TH, TT\}$, where HH means both the coins show heads, HT means the first coin shows heads and the second shows tails, etc. Here, $n(S) = 4$.
- (v) If a coin is tossed thrice or three coins are tossed simultaneously, then the sample space is $S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$ and $n(S) = 8$.
- (vi) If a coin is tossed 4 times or four coins are tossed simultaneously, then the sample space is $S = \{HHHH, HHHT, HHTH, HTHH, THHH, HHTT, HTHT, HTTH, THHT, THTH, TTHH, HTTT, THTT, TTHT, TTTH, TTTT\}$ and $n(S) = 16$.
- (vii) If a die is thrown twice or a pair of dice is thrown simultaneously, then the sample space is $S = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), (4, 1), (4, 2), (4, 3), (4, 4), (5, 5), (4, 6), (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$ here, e.g., (1, 4) means the first die shows 1 and the second die shows 4. Here, $n(S) = 36$.
- (viii) If a family contains two children then the sample space is $S = \{B_1B_2, B_1G_2, G_1B_2, G_1G_2\}$ where B_i denotes that i^{th} birth is of boy, $i = 1, 2$, and G_i denotes that i^{th} birth is of girl, $i = 1, 2$. This sample space can also be written as $S = \{BB, BG, GB, GG\}$
- (ix) If a bag contains 2 red and 3 black balls, and
- One ball is drawn from the bag, then the sample space is $\{R_1, R_2, B_1, B_2, B_3\}$, where R_1, R_2 denote the two red balls and B_1, B_2, B_3 denote the three black balls in the bag.
 - Two balls are drawn one by one without replacement from the bag, then the sample space is

$$\{ R_1 R_2, R_1 B_1, R_1 B_2, R_1 B_3, \\ R_2 R_1, R_2 B_1, R_2 B_2, R_2 B_3, \\ B_1 R_1, B_1 R_2, B_1 B_2, B_1 B_3, \\ B_2 R_1, B_2 R_2, B_2 B_1, B_2 B_3, \\ B_3 R_1, B_3 R_2, B_3 B_1, B_3 B_2 \}$$

Note: It is very simple to write the above sample space – first write all other balls with R_1 and then with R_2 , then with B_1 and so on.

Remark 1: If a random experiment with x possible outcomes is performed n times, then the total number of elements in the sample is n^x i.e. $n(S) = n^x$, e.g. if a coin is tossed twice, then $n(S) = 2^2 = 4$; if a die is thrown thrice, then $n(S) = 3^6 = 216$.

Sample Point: Each outcome of an experiment is visualised as a sample point in the sample space. e.g.

- If a coin is tossed then getting head or tail is a sample point.
- If a die is thrown twice, then getting (1, 1) or (1, 2) or (1, 3) or ... or (6, 6) is a sample point.

Event: Set of one or more possible outcomes of an experiment constitutes what is known as event. Thus, an event can be defined as a subset of the sample space, e.g.

- i) In a die throwing experiment, event of getting a number less than 5 is the set $\{1, 2, 3, 4\}$, which refers to the combination of 4 outcomes and is a sub-set of the sample space $S = \{1, 2, 3, 4, 5, 6\}$.
- ii) If a card is drawn from a well-shuffled pack of playing cards, then the event of getting a card of a spade suit is $\{1s, 2s, 3s, \dots, 9s, 10s, Js, Qs, Ks\}$ where suffix s under each character in the set denotes that the card is of spade and J, Q and K represent jack, queen and king respectively.

Different Types of Events: The following are different types of events in Probability computation. However, we present here three simple but important types.

Simple Event: Simple event consists of only one sample point is a simple event. For example, getting the number 2 on roll of a die.

Composite Event: An event consisting of at least two sample points. For example, getting a number less than 5 is the set of getting numbers $\{1, 2, 3, 4\}$

Mutually Exclusive Events: Those events out of which only one event can occur at a time are called mutually exclusive events. For example, on throwing a dice, the events: Event 1: “Getting a number less than 3 with sample space $\{1, 2\}$ ”, Event 2: “Getting the number 4 or 5 with sample space $\{4, 5\}$ ” and Event 3: Getting a number 6; are mutually exclusive.

Let us take up some examples which define the events as subsets of sample space.

Example 1: If a fair die is thrown once, what is the event of:

- (i) getting an even number
- (ii) getting a prime number
- (iii) getting a number multiple of 3
- (iv) getting an odd prime
- (v) getting an even prime
- (vi) getting a number greater than 4
- (vii) getting a number multiple of 2 and 3

Solution: When a die is thrown, then the sample space is:

$$S = \{1, 2, 3, 4, 5, 6\}$$

- (i) Let E_1 be the event of getting an even number, $\therefore E_1 = \{2, 4, 6\}$
- (ii) Let E_2 be the event of getting a prime number, $\therefore E_2 = \{2, 3, 5\}$
- (iii) Let E_3 be the event of getting a number multiple of 3 $\therefore E_3 = \{3, 6\}$
- (iv) Let E_4 be the event of getting an odd prime, $\therefore E_4 = \{3, 5\}$
- (v) Let E_5 be the event of getting an even prime, $\therefore E_5 = \{2\}$
- (vi) Let E_6 be the event of getting a number greater than 4, $\therefore E_6 = \{5, 6\}$
- (vii) Let E_7 be the event of getting a number multiple of 2 and 3, $\therefore E_7 = \{6\}$

Example 2: If a pair of fair dice is thrown, what is the event of

- (i) getting a doublet
- (ii) getting sum as 11
- (iii) getting sum less than 5
- (iv) getting sum greater than 16
- (v) getting 3 on the first die
- (vi) getting a number multiple of 3 on second die

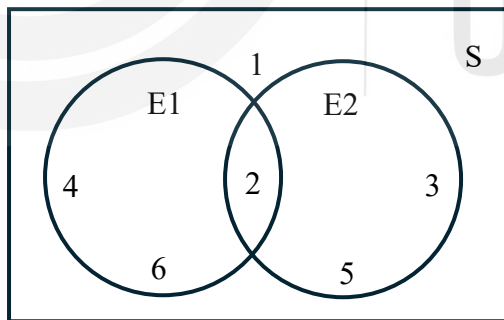
- (vii) getting a number multiple of 2 on first die and a multiple of 3 on second die.

Solution: The sample space, when two dice are thrown, is given in Sample Space point (vii) of this section.

- (i) Let E_1 be the event of getting a doublet. $\therefore E_1 = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$
- (ii) Let E_2 be the event of getting sum 11. $\therefore E_2 = \{(5, 6), (6, 5)\}$
- (iii) Let E_3 be the event of getting sum less than 5 i.e. sum can be 2 or 3 or 4 $\therefore E_3 = \{(1, 1), (1, 2), (2, 1), (1, 3), (3, 1), (2, 2)\}$
- (iv) Let E_4 be the event of getting sum greater than 16. $\therefore E_4 = \{\}$, i.e. E_4 is a null event.
- (v) Let E_5 be the event of getting 3 on the first die i.e. 3 on first die and second die may have any number $\therefore E_5 = \{(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6)\}$
- (vi) Let E_6 be the event of getting a number multiple of 3 on second die i.e. first die may have any number and the second has 3 or 6. $\therefore E_6 = \{(1, 3), (2, 3), (3, 3), (4, 3), (5, 3), (6, 3), (1, 6), (2, 6), (3, 6), (4, 6), (5, 6), (6, 6)\}$
- (vii) Let E_7 be the event of getting a multiple of 2 on the first die and a multiple of 3 on the second die i.e. 2 or 4 or 6 on first die and 3 or 6 on the second. $\therefore E_7 = \{(2, 3), (4, 3), (6, 3), (2, 6), (4, 6), (6, 6)\}$

Venn Diagram and Graphical Representation of Events

Events can be represented graphically using Venn Diagrams. This helps in a better understanding of the sample space and helps in computing the probability. The following diagram illustrates the use of a Venn Diagram. Consider the two events E_1 and E_2 of example 1, which have a sample space $S = \{1, 2, 3, 4, 5, 6\}$. The events are defined as: $E_1 = \{2, 4, 6\}$ and $E_2 = \{2, 3, 5\}$. The following Venn diagrams can be used to represent these events:



Venn Diagram

Please note the following points about the diagram:

- $E_1 \cap E_2 = \{2\}$
- $E_1 \cup E_2 = \{2, 3, 4, 5, 6\}$
- $S = \{1, 2, 3, 4, 5, 6\}$

You can learn more about Venn diagrams from references.

4.3 COMPUTING MATHEMATICAL PROBABILITY

Let there be 'n' exhaustive cases in a random experiment which are mutually exclusive as well as equally likely. Let 'm' out of them be favourable for the happening of an event A (say), then the probability of happening event A (denoted by $P(A)$) is defined as:

$$P(A) = \frac{\text{Number of favourable cases for event } A}{\text{Number of exhaustive cases}} = \frac{m}{n} \quad \dots(1)$$

Probability of non-happening of the event A is denoted by $P(\bar{A})$ and is defined as:

$$P(\bar{A}) = \frac{\text{Number of favourable cases for event } \bar{A}}{\text{Number of exhaustive cases}} = \frac{n - m}{n} = 1 - \frac{m}{n}$$

$$\text{So, } P(\bar{A}) = 1 - P(A)$$

$$\text{Or } P(A) + P(\bar{A}) = 1 \quad \dots(2)$$

Therefore, we conclude that the sum of the probabilities of happening an event and that of its complementary event is 1.

$$\text{Also, since } 0 \leq m \leq n, \text{ therefore } 0 \leq P(A) \leq 1 \quad \dots(3)$$

Remark: Probability of an impossible event is always zero and that of certain event is 1, e.g. probability of getting 7 when we throw a die is zero, as getting 7 here is an impossible event and probability of getting either of the six faces is 1, as it is a certain event.

This definition of probability fails if

- (i) The cases are not equally likely, e.g. probability of a candidate passing a test is not defined, as passing or failing in a test are not equally likely cases
- (ii) The number of exhaustive cases is indefinitely large, e.g. probability of drawing an integer say 2, from the set of integers i.e., $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ by definition, is $1/\infty = 0$. But, in actuality, it is not so, drawing 2 is not impossible, i.e. there are some chances of drawing 2. Hence, this definition fails here also.

Simple Problems On Probability

Now, let us give some examples so that you become familiar as to how and when this definition of probability is used.

Example 3: A bag contains 4 red, 5 black and 2 green balls. One ball is drawn from the bag. Find the probability that (i) It is a red ball, (ii) It is not black (iii) It is green or black

Solution: Let R_1, R_2, R_3, R_4 denote 4 red balls in the bag. Similarly, B_1, B_2, B_3, B_4, B_5 denote 5 black balls and G_1, G_2 denote two green balls in the bag. Then the sample space for drawing a ball is given by $\{R_1, R_2, R_3, R_4, B_1, B_2, B_3, B_4, B_5, G_1, G_2\}$

In these examples, we will use equation 1, which is given below:

$$P(A) = \frac{\text{Number of favourable cases for event } A}{\text{Number of exhaustive cases}}$$

- (i) Let A be the event of getting a red ball, then $A = \{R_1, R_2, R_3, R_4\}$
 $\therefore$ Number of favourable cases = 4, and
 Number of exhaustive cases = 11.
 So, $P(A) = 4/11$.
- (ii) Let B be the event that drawn ball is not black, then $B = \{R_1, R_2, R_3, R_4, G_1, G_2\}$
 $\therefore$ Number of favourable cases = 6
 Number of exhaustive cases = 11.
 So, $P(B) = 6/11$.
- (iii) Let C be the event that drawn ball is green or black, then $C = \{B_1, B_2, B_3, B_4, B_5, G_1, G_2\}$
 $\therefore$ Number of favourable cases = 7
 Number of exhaustive cases = 11.
 So, $P(C) = 7/11$.

Example 4: Three unbiased coins are tossed simultaneously. Find the probability of getting (i) at least two heads (ii) at most two heads (iii) all heads (iv) exactly one head (v) exactly one tail

Solution: The sample space in this case is $S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$. (We have used equation 1 to answer this question)

- (i) Let E1 be the event of getting at least 2 heads, then
 $E1 = \{HHT, HTH, THH, HHH\}$
 $\therefore$ Number of favourable cases = 4
 Number of exhaustive cases = 8
 $P(E1) = 4/8 = 1/2$
- (ii) Let E2 be the event of getting at most 2 heads, then
 $E2 = \{TTT, TTH, THT, HTT, HHT, HTH, THH\}$
 $\therefore$ Number of favourable cases = 7
 Number of exhaustive cases = 8
 $P(E2) = 7/8$
- (iii) Let E3 be the event of getting all heads, then $E3 = \{HHH\}$
 $\therefore$ Number of favourable cases = 1
 Number of exhaustive cases = 8
 $P(E3) = 1/8$
- (iv) Let E4 be the event of getting exactly one head, then
 $E4 = \{HTT, THT, TTH\}$
 $P(E4) = 3/8$
- (v) Let E5 be the event of getting exactly one tail, then
 $E5 = \{HHT, HTH, THH\}$
 $P(E5) = 3/8$

Example 5: In an experiment of throwing two fair dice, find the probability of getting (i) a doublet (ii) sum 7 (iii) sum greater than 8 (iv) 3 on first die and a multiple of 2 on second die (v) prime number on the first die and odd prime on the second die.

Solution: The sample space has already been given in (vii) of Sec. 4.2. Here, the sample space contains 36 elements i.e. **number of exhaustive cases** is 36.

- (i) Let E_1 be the event of getting a doublet (i.e. same number on both dice), then $E_1 = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$.
 $\therefore$ Number of favourable cases = 6
 $P(E_1) = 6/36 = 1/6$
- (ii) Let E_2 be the event of getting sum 7, then
 $E_2 = \{(1, 6), (6, 1), (2, 5), (5, 2), (3, 4), (4, 3)\}$
 $\therefore$ Number of favourable cases = 6
 $P(E_2) = 6/36 = 1/6$
- (iii) Let E_3 be the event of getting sum greater than 8, then
 $E_3 = \{(3, 6), (6, 3), (4, 5), (5, 4), (4, 6), (6, 4), (5, 5), (5, 6), (6, 5), (6, 6)\}$
 $\therefore$ Number of favourable cases = 10
 $P(E_3) = 10/36 = 5/18$
- (iv) Let E_4 be the event of getting 3 on first die and multiple of 2 on second die, then $E_4 = \{(3, 2), (3, 4), (3, 6)\}$
 $\therefore$ Number of favourable cases = 3
 $P(E_4) = 3/36 = 1/12$
- (v) Let E_5 be the event of getting prime number on first die and odd prime on second die, then
 $E_5 = \{(2, 3), (2, 5), (3, 3), (3, 5), (5, 3), (5, 5)\}$
 $\therefore$ Number of favourable cases = 6
 $P(E_5) = 6/36 = 1/6$

Example 6: Out of 52 well shuffled playing cards, one card is drawn at random. Find the probability of getting (i) a red card (ii) a face card (iii) a card of spade (iv) a card other than club (v) a king

Solution: Here, the number of exhaustive cases is 52 and a pack of playing cards contains 13 cards of each suit (spade, club, diamond, heart). (

- i) Let A be the event of getting a red card. We know that there are 26 red cards,
 $\therefore$ Number of favourable cases = 26
 Number of exhaustive cases = 52
 So, $P(A) = 26/52 = 1/2$
- ii) Let B be the event of getting a face card. We know that there are 12 face cards (jack, queen and king in each suit),
 $\therefore$ Number of favourable cases = 12
 Number of exhaustive cases = 52
 So, $P(B) = 12/52 = 3/13$
- iii) Let C be the event of getting a card of spades. We know that there are 13 cards of spades
 $\therefore$ Number of favourable cases = 13
 Number of exhaustive cases = 52
 So, $P(C) = 13/52 = 1/4$
- iv) Let D be the event of getting a card other than club. As there are 39 cards other than that of club,
 $\therefore$ Number of favourable cases = 39
 Number of exhaustive cases = 52
 So, $P(D) = 39/52 = 3/4$

- v) Let E be the event of getting a king. We know that there are 4 kings,
 $\therefore$ Number of favourable cases = 4
 Number of exhaustive cases = 52
 So, $P(E) = 4/52 = 1/13$

Example 7: Find the probability of getting 53 Sundays in a randomly selected non-leap year.

Solution: We know that there are 365 days in a non-leap year.

$365 \text{ days} / 7 = 52 \text{ weeks } 1/7 \text{ days, i.e. one non-leap year} = (52 \text{ complete weeks} + \text{one over day}).$

This over day may be one of the days Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday

So, the number of exhaustive cases = 7

Let A be the event of getting 53 Sundays. There will be 53 Sundays in a non-leap year if and only if the over day is Sunday.

$\therefore$ Number of favourable cases for event A = 1

$\therefore P(A) = 1/7$

Example 8: A single letter selected at random from the word 'STATISTICS'. What is the probability that it is a vowel?

Solution: Here, as the total number of letters in the word 'STATISTICS' is $n = 10$, and the number of vowels in the word is $m = 3$ (vowels are a, i, i),
 $\therefore$ The required probability = $3/10$

Example 9: Three horses A, B and C are in a race. A is twice as likely to win as B and B is twice as likely to win as C. What are the respective probabilities of their winning?

Solution: Let p be the probability that A wins the race. Probability that B wins the race = twice the probability of A's winning, i.e. $= 2(p) = 2p$; and probability of C's winning = twice the probability of B's winning $= 2(2p) = 4p$

Now, as the sum of the probability of happening an event and that of its complementary event(s) is 1. Here, the complementary of A is the happening of B or C] $\therefore p + 2p + 4p = 1$, and hence $p = 1/7$.

Thus, Probability of A winning the race = $1/7$; Probability of B winning the race = $2/7$ and probability of C winning the race = $4/7$.

Now, let us take up some problems on probability that are based on permutations/combinations. You may refer to these topics from the Programme Post-Graduate Diploma in Applied Statistics (PGDAST) in Course MST001: Foundation in Mathematics and Statistics, Unit 4.

Example 10: Out of 52 well shuffled playing cards, two cards are drawn at random. Find the probability of getting. (i) One red and one black (ii) Both cards of the same suit (iii) One jack and other king (iv) One red and the other of clubs.

Solution: Out of 52 playing cards, two cards can be drawn in ${}^{52}C_2$ ways i.e. $52 \times 51 / 2! = 26 \times 51$ ways. So, **Number of exhaustive cases = 26×51**

- (i) Let A be the event of getting one red and one black card, then the number of favourable cases for the event A are ${}^{26}C_1 \times {}^{26}C_1$ [As one red card out of 26 red cards can be drawn in ${}^{26}C_1$ ways and one black card out of 26 black cards can be drawn in ${}^{26}C_1$ ways.]

$\therefore P(A) = {}^{26}C_1 \times {}^{26}C_1 / (26 \times 51) = (26 \times 26) / (26 \times 51) = 26/51$

- (ii) Let B be the event of getting both the cards of the same suit and i.e. two cards of spades or two cards of clubs or 2 cards of diamonds or 2 cards of hearts.
 $\therefore$ Number of favourable cases for event B = ${}^{13}C_2 + {}^{13}C_2 + {}^{13}C_2 + {}^{13}C_2$
 $= 4 \times {}^{13}C_2 = 4 \times 13 \times 12 / 2! = 2 \times 13 \times 12$
 $P(B) = (2 \times 13 \times 12) / (26 \times 51) = 4/17$
- (iii) Let C be the event of getting a jack and a king.
 $\therefore$ Number of favourable cases ${}^4C_1 \times {}^4C_1 = (4 \times 4)$
 $P(C) = (4 \times 4) / (26 \times 51) = 8/663$
- (iv) Let D be the event of getting one red and one card of club. $\therefore$ Number of favourable cases $P(D) = {}^{26}C_1 \times {}^{13}C_1 = 26 \times 13$
 $P(D) = (26 \times 13) / (26 \times 51) = 13/51$

Example 11: If the letters of the word STATISTICS are arranged randomly, then find the probability that all three T's are together.

Solution: Let E be the event that selected word contains 3 T's together. There are 10 letters in the word STATISTICS. If we consider three T's as a single letter TTT, then we have 8 letters i.e. 1 'TTT'; 3 'S'; 1 'A'; 2 'I' and 1 'C'.

Number of possible arrangements with three T's coming together = $8! / (2! \cdot 3!)$
 (Please note you have 3 Ss and 2 Is in the word STATISTICS, therefore, $8!$ Is being divided by $3!$ (for 3 S) and $2!$ (for 2 I).

Number of favourable cases for event E = $8! / (2! \cdot 3!)$ and

Number of exhaustive cases = Total number of permutations of 10 letters in the word STATISTICS = $10! / (2! \cdot 3! \cdot 3!)$ [as out of 10 letters, 3 are T's, 2 are I's and 3 are S's]

The probability that all three T's are together in a random arrangement of letters of the word STATISTICS $P(A) = 8! / (2! \cdot 3!) \div 10! / (2! \cdot 3! \cdot 3!)$

$$P(A) = 8! \cdot (2! \cdot 3! \cdot 3!) / (2! \cdot 3!) \cdot 10! = 3 \times 2 / 10 \times 9 = 1/15$$

Example 12: In a lottery, one has to choose six numbers at random out of the numbers from 1 to 30. He/ she will get the prize only if all the six chosen numbers matched with the six numbers already decided by the lottery committee. Find the probability of winning the prize.

Solution: Out of 30 numbers, 6 can be drawn in ${}^{30}C_6$ ways, = $30! / (24! \cdot 6!)$

$$= (30 \times 29 \times 28 \times 27 \times 26 \times 25) / (6 \times 5 \times 4 \times 3 \times 2) = 593775 \text{ ways}$$

$$\therefore \text{Number of exhaustive cases} = 593775$$

Out of these 593775 ways, there is only one way to win the prize (i.e. choose those six numbers that are already fixed by the committee). Here, the number of favourable cases is 1.

Hence, Favourable cases 1

$$P(\text{winning the prize}) = 1 / \text{Exhaustive cases} = 1/593775$$

Check Your Progress 1

1) Write the sample space if we draw a card from a pack of 52 playing cards.

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- 2) If a die and a coin are tossed simultaneously, write the event of getting
- (i) head and prime number
 - (ii) tail and an even number
 - (iii) head and multiple of 3

- 3) If two coins are tossed, then find the probability of getting.

- (i) At least one head
- (ii) head and tail
- (iii) At most one head

- 4) If three dice are thrown, then find the probability of getting

- (i) triplet
- (ii) sum 5
- (iii) sum at least 17
- (iv) prime number on first die and odd prime number on second and third die.

- 5) Find the probability of getting 53 Mondays in a randomly selected leap year.

4.4 AXIOMS OF PROBABILITY

The axioms are fundamental to probability and provide us with an approach to probability, i.e., an axiomatic approach to probability. It defines the probability function as follows:

Definition: Let S be a sample space for a random experiment and A be an event which is subset of S , then $P(A)$ is called probability function if it satisfies the following axioms:

- (i) $P(A)$ is real and $P(A) \geq 0$
- (ii) $P(S) = 1$
- (iii) If $A_1, A_2, \dots$ is any finite or infinite sequence of disjoint events in S , then $P(A_1 \text{ or } A_2 \text{ or } \dots \text{ or } A_n) = P(A_1) + P(A_2) + \dots + P(A_n)$

Now, let us give some results using probability function. But before taking up these results, we discuss some statements with their meanings in terms of set theory. If A and B are two events, then in terms of set theory, we write:

- i) 'At least one of the events A or B occurs' as $A \cup B$
- ii) 'Both the events A and B occurs' as $A \cap B$
- iii) 'Neither A nor B occurs' as $\bar{A} \cap \bar{B}$

- iv) 'Event A occurs and B does not occur' as $A \cap \bar{B}$
- v) 'Exactly one of the events A or B occurs' as $(\bar{A} \cap B) \cup (A \cap \bar{B})$
- vi) 'Not more than one of the events A or B occurs' as $(\bar{A} \cap B) \cup (A \cap \bar{B}) \cup (\bar{A} \cap \bar{B})$

Similarly, you can write the meanings in terms of set theory for such statement in case of three or more events e.g. in case of three events A, B and C, happening of at least one of the events is written as $A \cup B \cup C$.

Some Results Using Probability Function:

1 Prove that probability of the impossible event is zero

Proof: Let S be the sample space and ϕ be the set of impossible event.

$$\therefore S \cup \phi = S$$

$$\Rightarrow P(S \cup \phi) = P(S)$$

$$\Rightarrow P(S) + P(\phi) = P(S) \quad [\text{By axiom (iii)}]$$

$$\Rightarrow 1 + P(\phi) = 1 \quad [\text{By axiom (ii)}]$$

$$\Rightarrow P(\phi) = 0$$

2 Probability of non-happening of an event A i.e. complementary event $\bar{A}$ of A is given by $P(\bar{A}) = 1 - P(A)$

Proof: If S is the sample space then

$$A \cup \bar{A} = S \quad [\because A \text{ and } \bar{A} \text{ are mutually disjoint events}]$$

$$\Rightarrow P(A \cup \bar{A}) = P(S)$$

$$\Rightarrow P(A) + P(\bar{A}) = P(S) \quad [\text{Using axiom (iii)}]$$

$$= 1 \quad [\text{Using axiom (ii)}]$$

$$\Rightarrow P(\bar{A}) = 1 - P(A)$$

3. Prove that

$$(i) P(A \cap \bar{B}) = P(A) - P(A \cap B)$$

$$(ii) P(\bar{A} \cap B) = P(B) - P(A \cap B)$$

Proof

If S is the sample space and $A, B \subset S$ then

$$(i) A = (A \cap \bar{B}) \cup (A \cap B)$$

$$\Rightarrow P(A) = P((A \cap \bar{B}) \cup (A \cap B))$$

$$= P(A \cap \bar{B}) + P(A \cap B)$$

[Using axiom (iii) as $A \cap \bar{B}$ and $A \cap B$ are mutually disjoint]

$$\Rightarrow P(A \cap \bar{B}) = P(A) - P(A \cap B)$$

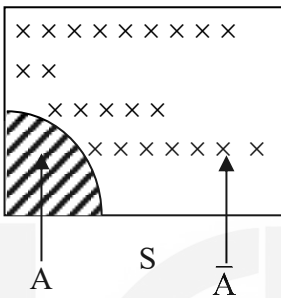
$$(ii) B = (A \cap B) \cup (\bar{A} \cap B)$$

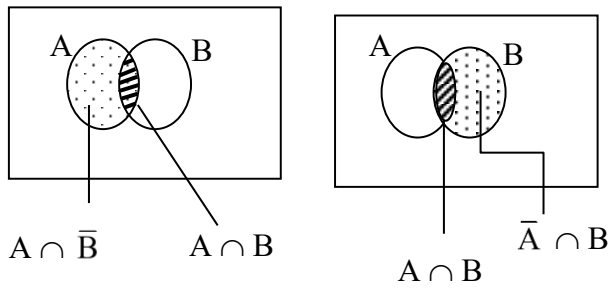
$$P(B) = P((A \cap B) \cup (\bar{A} \cap B))$$

$$= P(A \cap B) + P(\bar{A} \cap B)$$

[Using axiom (iii) as $A \cap B$ and $\bar{A} \cap B$ are mutually disjoint]

$$\Rightarrow P(\bar{A} \cap B) = P(B) - P(A \cap B)$$





Example 13: A, B and C are three mutually exclusive and exhaustive events associated with a random experiment. Find $P(A)$ given that :

$$P(B) = \frac{3}{4}P(A) \text{ and } P(C) = \frac{1}{3}P(B)$$

Solution:

As A, B and C are mutually exclusive and exhaustive events,

$$\therefore A \cup B \cup C = S$$

$$\Rightarrow P(A \cup B \cup C) = P(S)$$

$$\Rightarrow P(A) + P(B) + P(C) = 1 \quad \left[\because \text{using axiom (iii) as A, B, C are mutually disjoint events} \right]$$

$$\Rightarrow P(A) + \frac{3}{4}P(A) + \frac{1}{3}P(B) = 1$$

$$\Rightarrow P(A) + \frac{3}{4}P(A) + \frac{1}{3} \left(\frac{3}{4}P(A) \right) = 1$$

$$\Rightarrow \left(1 + \frac{3}{4} + \frac{1}{4} \right) P(A) = 1$$

$$\Rightarrow 2P(A) = 1$$

$$\Rightarrow P(A) = \frac{1}{2}$$

Example 14: If two dice are thrown, what is the probability that sum is

- greater than 9, and
- neither 10 or 12.

Solution:

$$\text{a) } P[\text{sum} > 9] = P[\text{sum} = 10 \text{ or sum} = 11 \text{ or sum} = 12]$$

$$= P[\text{sum} = 10] + P[\text{sum} = 11] + P[\text{sum} = 12]$$

[using axiom (iii)]

$$= \frac{3}{36} + \frac{2}{36} + \frac{1}{36} = \frac{6}{36} = \frac{1}{6}$$

[$\therefore$ for sum = 10, there are three favourable cases (4, 6), (5, 5) and (6, 4). Similarly for sum = 11 and 12, there are two and one favourable cases respectively.]

Let A denotes the event for sum = 10 and B denotes the event for sum = 12,

$\therefore$ Required probability = $P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B})$ [Using De- Morgan's law

(see Unit 1 of Course MST-001)]

$$= 1 - P(A \cup B)$$

$$= 1 - [P(A) + P(B)] \quad \text{[Using axiom (iii)]}$$

$$= 1 - \left[\frac{3}{36} + \frac{1}{36} \right] = 1 - \frac{4}{36} = 1 - \frac{1}{9} = \frac{8}{9}$$

4.5 ADDITION LAW

Addition Theorem on Probability for Two Events

Statement

Let S be the sample space of a random experiment, and events A and $B \subseteq S$ then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

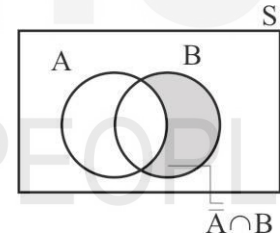
Proof: From the Venn diagram, we have

$$A \cup B = A \cup (\bar{A} \cap B)$$

$$\therefore P(A \cup B) = P(A) + P(\bar{A} \cap B)$$

[By axiom (iii) $\therefore$ A and $\bar{A} \cap B$ are mutually disjoint]

$$= P(A) + P(B) - P(A \cap B) \quad \text{[Refer to section 4.4]}$$



Hence proved

Corollary: If events A and B are mutually exclusive events, then

$$P(A \cup B) = P(A) + P(B).$$

[This is known as the addition theorem for mutually exclusive events]

Proof: For any two events A and B, we know that

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Now, if the events A and B are mutually exclusive then

$$A \cap B = \phi$$

Also, we know that probability of impossible event is zero i.e.

$$P(A \cap B) = P(\phi) = 0.$$

Hence,

$$P(A \cup B) = P(A) + P(B)$$

Similarly, for three non-mutually exclusive events A, B and C, we have

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

and for three mutually exclusive events A, B and C, we have

$$P(A \cup B \cup C) = P(A) + P(B) + P(C).$$

The result can similarly be extended for more than 3 events.

Applications of the Addition Theorem of Probability

Example 15: 25 lottery tickets are marked with first 25 numerals. A ticket is drawn at random. Find the probability that it is a multiple of 5 or 7.

Solution: Let A be the event that the drawn ticket bears a number multiple of 5 and B be the event that it bears a number multiple of 7.

Therefore,

$$A = \{5, 10, 15, 20, 25\}$$

$$B = \{7, 14, 21\}$$

Here, as $A \cap B = \phi$,

$\therefore$ A and B are mutually exclusive, and hence,

$$P(A \cup B) = P(A) + P(B) = \frac{5}{25} + \frac{3}{25} = \frac{8}{25}$$

Example 16: There are 40 pages in a book. A page is opened at random. Find the probability that the number of this opened page is a multiple of 3 or 5.

Solution Let A be the event that the number of the opened page is a multiple of 3, and B be the event that it is a multiple of 5.

$$\therefore A = \{3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, 39\},$$

$$B = \{5, 10, 15, 20, 25, 30, 35, 40\}, \text{ and}$$

$$A \cap B = \{15, 30\}$$

As A and B are non-mutually exclusive,

$$\therefore \text{the required probability} = P(A \cup B)$$

$$= P(A) + P(B) - P(A \cap B)$$

$$= \frac{13}{40} + \frac{8}{40} - \frac{2}{40} = \frac{19}{40}.$$

Check Your Progress 2

1) If A, B and C are any three events, write down the expressions in terms of set theory:

- Only A occurs
- A and B occur, but C does not
- A, B and C all the three occur

- d) at least two occur
- e) exactly two do not occur
- f) none occurs

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- 2) Fourteen balls are serially numbered and placed in a bag. Find the probability that a ball is drawn bears a number multiple of 3 or 5.

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- 3) A card is drawn from a pack of 52 playing cards. Find the probability that it is either a king or a red card.

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- 4) Two dice are thrown together. Find the probability that the sum of the numbers turned up is either 6 or 8.

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4.6 CONDITIONAL PROBABILITY

We have discussed earlier that $P(A)$ represents the probability of happening event A for which the number of exhaustive cases is the number of elements in the sample space S . $P(A)$ dealt earlier was the unconditional probability. Here, we are going to deal with conditional probability.

Let us start with taking the following example:

Suppose a card is drawn at random from a pack of 52 playing cards. Let A be the event of drawing a black colour face card. Then $A = \{J_s, Q_s, K_s, J_c, Q_c, K_c\}$ and hence

$$P(A) = 6/52 = 3/26.$$

Let B be the event of drawing a card of spade i.e.

$$B = \{1_s, 2_s, 3_s, 4_s, 5_s, 6_s, 7_s, 8_s, 9_s, 10_s, J_s, Q_s, K_s\}.$$

If after a card is drawn from the pack of cards, we are given the information that card of spade has been drawn i.e., B has happened, then the probability of event

A will no more be $3/26$ because here in this case, we have the information that the card drawn is of spade (i.e. from amongst 13 cards) and hence there are 13 exhaustive cases and not 52. From amongst these 13 cards of spades, there are 3 black colour face cards and hence probability of having black colour face card given that it is a card of spades i.e. $P(A|B) = 3/13$, which is the conditional probability of A given that B has already happened.

Note: Here, the symbol ‘|’ used in $P(A|B)$ should be read as ‘given’ and not ‘upon’. $P(A|B)$ is the conditional probability of happening A given that B has already happened.

So, the conditional probability $P(A|B)$ is also the probability of happening A but here the information is given that the event B has already happened. $P(A|B)$ refers to the sample space B and not S.

Remark 1: $P(A|B)$ is meaningful only when $P(B) \neq 0$ i.e. when the event B is not an impossible event.

Multiplication Law of Probability

Statement: For two events A and B,

$$P(A \cap B) = P(A) P(B|A), P(A) > 0 \quad \dots (1)$$

$$= P(B) P(A|B), P(B) > 0, \quad \dots (2)$$

where $P(B|A)$ is the conditional probability of B given that A has already happened and $P(A|B)$ is the conditional probability of A given that B has already happened.

Proof: Let n be the number of exhaustive cases corresponding to the sample space S and m_1, m_2, m_3 be the number of favourable cases for events A, B and $A \cap B$ respectively.

$$\therefore P(A) = \frac{m_1}{n}, P(B) = \frac{m_2}{n}, P(A \cap B) = \frac{m_3}{n}$$

Now, as $B|A$ represents the event of happening B given that A has already happened and hence it refers to the sample space A ($\because$ we have with us the information that A has already happened) and thus the number of exhaustive cases for $B|A$ is m_1 (i.e. the number of cases favourable to “A relative to sample space S”). The number of cases favourable to $B|A$ is the number of those elements of B which are in A i.e. the number of favourable cases to $B|A$ is the number of favourable cases to “ $A \cap B$ relative to S”. So, the number of favourable cases to $B|A$ is m_3

$$\therefore P(B|A) = \frac{m_3}{m_1}$$

$$= \frac{m_3/n}{m_1/n} \quad \left[\begin{array}{l} \text{Dividing the numerator} \\ \text{and denominator by } n \end{array} \right]$$

$$= \frac{P(A \cap B)}{P(A)}, P(A) \neq 0$$

$$\Rightarrow P(A \cap B) = P(A) P(B|A), P(A) \neq 0$$

Similarly, you can prove that

$$P(A \cap B) = P(B) P(A|B), P(B) \neq 0.$$

This result can be extended for 3 or more events e.g. for three events A, B and C, we have

$$P(A \cap B \cap C) = P(A) P(B|A) P(C|A \cap B),$$

where $P(C|A \cap B)$ represents the probability of happening C given that A and B both have already happened.

4.7 INDEPENDENT EVENTS

Before defining the independent events, let us again consider the concept of conditional probability, taking the following example:

Suppose we draw a card from a pack of 52 playing cards, then probability of drawing a card of spades is $13/52$. Now, if we do not replace the card back and draw the next card. Then, the probability of drawing the second card 'a card of spades' given that the first card was spades would be $12/51$ and it is the conditional probability. Now, if the first card had been replaced back then this conditional probability would have been $13/52$. So, if sampling is done without replacement, the probability of second draw and that of subsequent draws made following the same way is affected but if it is done with replacement, then the probability of second draw and subsequent draws made following the same way remains unaltered.

In the example above, if the next draw is made with replacement, then the draw is not affected by the preceding draws. Let us now define independent events.

Independent Events

Events are said to be independent if happening or non-happening of any one event is not affected by the happening or non-happening of other events. For example, if a coin is tossed certain number of times, then happening of head in any trial is not affected by any other trial i.e. all the trials are independent.

Two events A and B are independent if and only if $P(B|A) = P(B)$ i.e. there is no relevance of giving any information. Here, if A has already happened, even then it does not alter the probability of B. e.g. Let A be the event of getting head in the 4th toss of a coin and B be the event of getting head in the 5th toss of the coin. Then the probability of getting head in the 5th toss is $1/2$, irrespective of the case whether we know or don't know the outcome of 4th toss, i.e. $P(B|A) = P(B)$.

Multiplicative Law for Independent Events:

If A and B are independent events, then

$$P(A \cap B) = P(A) P(B).$$

This is because if A and B are independent, then $P(B|A) = P(B)$ and hence the equation (1) discussed in the previous section becomes $P(A \cap B) = P(A) P(B)$.

Similarly, if A, B and C are three independent events, then

$$P(A \cap B \cap C) = P(A) P(B) P(C).$$

The result can be extended for more than three events also.

Remark 2: Mutually exclusive events can never be independent.

Proof: Let A and B be two mutually exclusive events with positive probabilities (i.e. $P(A) > 0$, and $P(B) > 0$)

$$\therefore P(A \cap B) = 0 \quad [\because A \cap B = \phi, \text{ as the events are mutually exclusive}]$$

Also, by multiplication law of probability, we have

$$P(A \cap B) = P(A) P(B|A), P(A) \neq 0.$$

$$\therefore 0 = P(A) P(B|A).$$

$$\text{Now as } P(A) \neq 0, \therefore P(B|A) = 0$$

$$\text{But } P(B) \neq 0 \quad [\because P(B) > 0]$$

$$\therefore P(B|A) \neq P(B),$$

Hence, A and B are not independent.

Result: If events A and B are independent, then prove that

- (i) A and $\bar{B}$ are independent
- (ii) $\bar{A}$ and B are independent
- (iii) $\bar{A}$ and $\bar{B}$ are independent

Proof:

(i) We know that

$$\begin{aligned} P(A \cap \bar{B}) &= P(A) - P(A \cap B) && [\text{Already proved in this Unit}] \\ &= P(A) - P(A)P(B) && [\because \text{events A and B are independent.}] \\ &= P(A)(1 - P(B)) \\ &= P(A)P(\bar{B}) \end{aligned}$$

$\Rightarrow$ Events A and $\bar{B}$ are independent

(ii) We know that

$$\begin{aligned} P(\bar{A} \cap B) &= P(B) - P(A \cap B) && [\text{Already proved in this unit}] \\ &= P(B) - P(A)P(B) && [\because \text{events A and B are independent.}] \\ &= P(B)(1 - P(A)) \\ &= P(B)P(\bar{A}) \\ &= P(\bar{A})P(B) \end{aligned}$$

$\Rightarrow$ Events $\bar{A}$ and B are independent

$$(iii) P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B}) \quad [\text{By De-Morgan's law}]$$

$$= 1 - P(A \cup B) \quad [\because P(E) + P(\bar{E}) = 1]$$

$$= 1 - [P(A) + P(B) - P(A \cap B)] \quad [\text{Using addition law on probability}]$$

$$= 1 - [P(A) + P(B) - P(A)P(B)] \quad [\because \text{events A and B are independent.}]$$

$$= 1 - P(A) - P(B) + P(A)P(B)$$

$$= (1 - P(A)) - P(B)(1 - P(A))$$

$$= (1 - P(A))(1 - P(B))$$

$$= P(\bar{A})P(\bar{B})$$

$\Rightarrow$ Events $\bar{A}$ and $\bar{B}$ are independent

Now let us take up some examples on conditional probability, multiplicative law and independent events:

Example 17: A die is rolled. If the outcome is a number greater than 3, what is the probability that it is a prime number?

Solution: The sample space of the experiment is

$$S = \{1, 2, 3, 4, 5, 6\}$$

Let A be event that the outcome is a number greater than 3 and B be the event that it is a prime number.

$$\therefore A = \{4, 5, 6\}, B = \{2, 3, 5\} \text{ and hence } A \cap B = \{5\}.$$

$$\Rightarrow P(A) = 3/6, P(B) = 3/6, P(A \cap B) = 1/6.$$

Now, the required probability = $P(B|A)$

$$= \frac{P(A \cap B)}{P(A)} \quad [\text{Refer to equation 1}]$$

$$= \frac{1/6}{3/6} = \frac{1}{3}$$

Example 18: A couple has 2 children. What is the probability that both children are boys, if it is known that?

- (i) younger child is a boy
- (ii) older child is a boy
- (iii) at least one of them is boy

Solution: Let B_i, G_i denote that i^{th} birth is of boy and girl respectively, $i = 1, 2$.

Then, for a couple having two children, the sample space is

$$S = \{B_1B_2, B_1G_2, G_1B_2, G_1G_2\}$$

Let A be the event that both children are boys then

$$A = \{B_1 B_2\}$$

(i) Let B be the event of getting younger child as boy i.e.

$$B = \{B_1 B_2, G_1 B_2\}. \text{ Hence } A \cap B = \{B_1 B_2\}$$

$$\therefore \text{required probability } P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{1/4}{2/4} = \frac{1}{2}$$

(ii) Let C be the event of getting older child as boy, then $C = \{B_1 B_2, B_1 G_2\}$

$$\text{and hence } A \cap C = \{B_1 B_2\}.$$

$$\therefore \text{required probability } P(A | C) = \frac{P(A \cap C)}{P(C)} = \frac{1/4}{2/4} = \frac{1}{2}.$$

(iii) Let D be the event of getting at least one of the children as boy, then

$$D = \{B_1 B_2, B_1 G_2, G_1 B_2\} \text{ and hence}$$

$$A \cap D = \{B_1 B_2\}.$$

$$\therefore \text{required probability } P(A | D) = \frac{P(A \cap D)}{P(D)} = \frac{1/4}{3/4} = \frac{1}{3}$$

Example 19: An urn contains 4 red and 7 blue balls. Two balls are drawn one by one without replacement. Find the probability of getting 2 red balls.

Solution: Let A be the event that first ball drawn is red and B be the event that the second ball drawn is red.

$$\therefore P(A) = 4/11 \text{ and } P(B|A) = 3/10 \left[\begin{array}{l} \because \text{it is given that one red ball} \\ \text{has already been drawn} \end{array} \right]$$

$$\therefore \text{The required probability} = P(A \cap B)$$

$$= P(A) P(B|A)$$

$$= \left(\frac{4}{11}\right) \left(\frac{3}{10}\right) = \frac{6}{55}$$

Example 20: Three cards are drawn one by one without replacement from a well shuffled pack of 52 playing cards. What is the probability that first card is jack, second is queen and the third is again a jack.

Solution: Define the following events

E_1 be the event of getting a jack in the first draw,

E_2 be the event of getting a queen in second draw, and

E_3 be the event of getting a jack in third draw,

$$\therefore \text{Required probability} = P(E_1 \cap E_2 \cap E_3)$$

$$= P(E_1)P(E_2 | E_1)P(E_3 | E_1 \cap E_2)$$

[$\because$ cards are drawn without replacement and
hence the events are not independent]

$$= \frac{4}{52} \times \frac{4}{51} \times \frac{3}{50} = \frac{1}{13} \times \frac{2}{17} \times \frac{1}{25} = \frac{2}{5525}.$$

Example 21: Solve the following for the events A and B.

- (i) If A and B are independent events with $P(A \cup B) = 0.8$ and $P(B) = 0.4$, then find $P(A)$.
- (ii) If A and B are independent events with $P(A) = 0.2$, $P(B) = 0.5$, then find $P(A \cup B)$.
- (iii) If A and B are independent events and $P(A) = 0.4$ and $P(B) = 0.3$, then find $P(A|B)$ and $P(B|A)$.
- (iv) If A and B are independent events with $P(A) = 0.4$ and $P(B) = 0.2$, then find

$$P(\bar{A} \cap B), P(A \cap \bar{B}), P(\bar{A} \cap \bar{B})$$

Solution:

- (i) We are given

$$P(A \cup B) = 0.8, P(B) = 0.4.$$

We know that

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad [\text{By Addition theorem of probability}]$$

$$= P(A) + P(B) - P(A)P(B) \quad [\because \text{events A and B are independent}]$$

$$\Rightarrow 0.8 = P(A) + 0.4 - 0.4P(A)$$

$$\Rightarrow 0.4 = (1 - 0.4) P(A)$$

$$= 0.6 P(A)$$

$$\Rightarrow P(A) = 0.4/0.6 = 2/3$$

- (ii) We are given that $P(A) = 0.2$, $P(B) = 0.5$.

We know that

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad [\text{Addition theorem of probability}]$$

$$= P(A) + P(B) - P(A)P(B) \quad [\because \text{events A and B are independent}]$$

$$= 0.2 + 0.5 - 0.2 \times 0.5$$

$$= 0.7 - 0.10 = 0.6$$

- (iii) We are given that $P(A) = 0.4$, $P(B) = 0.3$.

Now, as A and B are independent events,

$$\therefore P(A \cap B) = P(A)P(B)$$

$$= 0.4 \times 0.3 = 0.12$$

And hence from conditional probability, we have

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.12}{0.3} = \frac{12}{30} = \frac{2}{5} = 0.4,$$

$$\text{and } P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{0.12}{0.4} = \frac{12}{40} = \frac{3}{10} = 0.3.$$

(iv) We are given that $P(A) = 0.4$, $P(B) = 0.2$.

We know that if two events A and B are independent then

$\bar{A}$ and B; A and $\bar{B}$; $\bar{A}$ and $\bar{B}$ are also independent events.

$\therefore P(\bar{A} \cap B) = P(\bar{A})P(B)$ [Using the concept of independent events]

$$= (1 - P(A))(P(B))$$

$$= (1 - 0.4)(0.2)$$

$$= (0.6)(0.2)$$

$$= 0.12$$

$$P(A \cap \bar{B}) = P(A)P(\bar{B}) \quad [\because A \text{ and } \bar{B} \text{ are independent}]$$

$$= (0.4)(1 - 0.2) = 0.32$$

$$P(\bar{A} \cap \bar{B}) = P(\bar{A})P(\bar{B}) = (1 - 0.4)(1 - 0.2) = (0.6)(0.8) = 0.48.$$

Example 22: Three unbiased coins are tossed simultaneously. In which of the following cases are the events A and B independent?

(i) A be the event of getting exactly one head

B be the event of getting exactly one tail

(ii) A be the event that first coin shows head

B be the event that third coin shows tail

(iii) A be the event that shows exactly two tails

B be the event that third coin shows head

Solution: When three unbiased coins are tossed simultaneously, then the sample space is given by

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

(i) $A = \{HTT, THT, TTH\}$

$$B = \{HHT, HTH, THH\}$$

$$A \cap B = \{\} = \phi$$

$$\therefore P(A) = \frac{3}{8}, P(B) = \frac{3}{8}, P(A \cap B) = \frac{0}{8} = 0$$

$$\text{Hence, } P(A)P(B) = \frac{3}{8} \times \frac{3}{8} = \frac{9}{64}$$

$$\Rightarrow P(A \cap B) \neq P(A)P(B)$$

$\Rightarrow$ Events A and B are not independent.

$$(ii) A = \{HHH, HHT, HTH, HTT\}$$

$$B = \{HHT, HTT, THT, TTT\}$$

$$A \cap B = \{HHT, HTT\}$$

$$\therefore P(A) = \frac{4}{8} = \frac{1}{2}, P(B) = \frac{4}{8} = \frac{1}{2}, P(A \cap B) = \frac{2}{8} = \frac{1}{4}$$

$$P(A)P(B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} = P(A \cap B)$$

$\Rightarrow$ Events A and B are independent.

$$(iii) A = \{HTT, THT, TTH\}$$

$$B = \{HHH, HTH, THH, TTH\}$$

$$A \cap B = \{TTH\}$$

$$P(A) = \frac{3}{8}, P(B) = \frac{4}{8} = \frac{1}{2}, P(A \cap B) = \frac{1}{8}.$$

$$\text{Hence, } P(A)P(B) = \frac{3}{8} \times \frac{1}{2} = \frac{3}{16} \neq P(A \cap B)$$

$\Rightarrow$ Events A and B are not independent.

Example 23: Two cards are drawn from a pack of cards in succession with replacement of first card. Find the probability that both are the cards of 'hearts'.

Solution: Let A be the event that the first card drawn is a hearts card and B be the event that second card is a hearts card.

As the cards are drawn with replacement,

$\therefore$ A and B are independent and hence the required probability

$$= P(A \cap B) = P(A)P(B) = \left(\frac{13}{52}\right)\left(\frac{13}{52}\right) = \frac{1}{4} \times \frac{1}{4} = \frac{1}{16}.$$

Example 24: A class consists of 10 boys and 40 girls. 5 of the students are rich and 15 students are brilliant. Find the probability of selecting a brilliant rich boy.

Solution: Let A be the event that the selected student is brilliant, B be the event that he/she is rich and C be the event that the student is boy.

$$\therefore P(A) = \frac{15}{50}, P(B) = \frac{5}{50}, P(C) = \frac{10}{50} \text{ and hence}$$

the required probability = $P(A \cap B \cap C)$

$$= P(A)P(B)P(C) \quad [\because A, B \text{ and } C \text{ are independent}]$$

$$= \left(\frac{15}{50}\right)\left(\frac{5}{50}\right)\left(\frac{10}{50}\right) = \left(\frac{3}{10}\right)\left(\frac{1}{10}\right)\left(\frac{1}{5}\right) = \frac{3}{500}$$

Check Your Progress 3

- 1) A card is drawn from a well-shuffled pack of cards. If the card drawn is a face card, what is the probability that it is a king?

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- 2) Two cards are drawn one by one without replacement from a well shuffled pack of 52 cards. What is the probability that both the cards are red?

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- 3) A bag contains 10 good and 4 defective items, two items are drawn one by one without replacement. What is the probability that first drawn item is defective and the second one is good?

.....

- 4) The odds in favour of passing driving test by a person X are 3 : 5 and odds in favour of passing the same test by another person Y are 3 : 2. What is the probability that both will pass the test?

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4.8 BAYES' THEOREM

One of the important theorems of probability that is useful in data science is known as Bayes' theorem, given by Thomas Bayes, a British Mathematician.. In this unit we state this theorem without proof.

Statement: Let S be the sample space and $E_1, E_2, \dots, E_n$ be n mutually exclusive and exhaustive events with $P(E_i) \neq 0; i = 1, 2, \dots, n$. Let A be any event which is a sub-set of $E_1 \cup E_2 \cup \dots \cup E_n$ (i.e. at least one of the events $E_1, E_2, \dots, E_n$) with $P(A) > 0$ [Notice that up to this line the statement is the same as that of law of total probability], then

$$P(E_i | A) = \frac{P(E_i)P(A|E_i)}{P(A)}, i = 1, 2, \dots, n$$

where $P(A) = P(E_1) P(A|E_1) + P(E_2) P(A|E_2) + \dots + P(E_n) P(A|E_n)$.

Applications Of Bayes' Theorem

Example 25(a): There are two bags. First bag contains 5 red, 6 white balls and the second bag contains 3 red, 4 white balls. One bag is selected at random and a ball is drawn from it. What is the probability that it is (i) red, (ii) white.

Solution: Let E_1 be the event that first bag is selected and E_2 be the event that second bag is selected.

$$\therefore P(E_1) = P(E_2) = \frac{1}{2}.$$

(i) Let R be the event of getting a red ball from the selected bag.

$$\therefore P(R | E_1) = \frac{5}{11}, \text{ and } P(R | E_2) = \frac{3}{7}.$$

Thus, the required probability is given by

$$P(R) = P(E_1)P(R | E_1) + P(E_2)P(R | E_2)$$

$$\begin{aligned} &= \frac{1}{2} \times \frac{5}{11} + \frac{1}{2} \times \frac{3}{7} \\ &= \frac{5}{22} + \frac{3}{14} = \frac{35 + 33}{154} = \frac{68}{154} = \frac{34}{77} \end{aligned}$$

(ii) Let W be the event of getting a white ball from the selected bag.

$$\therefore P(W | E_1) = \frac{6}{11}, \text{ and } P(W | E_2) = \frac{4}{7}.$$

Thus, the required probability is given by

$$P(W) = P(E_1)P(W | E_1) + P(E_2)P(W | E_2)$$

$$= \frac{1}{2} \times \frac{6}{11} + \frac{1}{2} \times \frac{4}{7} = \frac{3}{11} + \frac{2}{7} = \frac{21 + 22}{77} = \frac{43}{77}.$$

Example 25(b): Let us consider the problem given in Example 25(a) after replacing the question asked therein (i.e. the last sentence) by the following question:

If it is found to be red, what is the probability of?

i) selecting the first bag

ii) selecting the second bag

Solution: First, we have to give the solution exactly as given for Example 25(a). After that, we are to proceed as follows:

i) Probability of selecting the first bag given that the ball drawn is red

$$= P(E_1 | R)$$

$$= \frac{P(E_1)P(R | E_1)}{P(R)} \quad \left[\begin{array}{l} \text{Applying Bayes'} \\ \text{theorem} \end{array} \right]$$

$$= \frac{\frac{1}{2} \times \frac{5}{11}}{\frac{34}{77}} = \frac{5}{22} \times \frac{77}{34} = \frac{35}{68}$$

ii) Probability of selecting the second bag given that the ball drawn is red

$$P(E_2|R) = \frac{P(E_2)P(R|E_2)}{P(R)} = \frac{\frac{1}{2} \times \frac{3}{7}}{\frac{34}{77}} = \frac{3}{14} \times \frac{77}{34} = \frac{33}{68}$$

Example 26(a): A factory produces certain type of output by 3 machines. The respective daily production figures are - machine X: 3000 units, machine Y: 2500 units and machine Z: 4500 units. Past experience shows that 1% of the output produced by machine X is defective. The corresponding fractions of defectives for the other two machines are 1.2 and 2 percent respectively. An item is drawn from the day's production. What is the probability that it is defective?

Solution: Let E_1 , E_2 and E_3 be the events that the drawn item is produced by machine X, machine Y and machine Z, respectively. Let A be the event that the drawn item is defective.

As the total number of units produced by all the machines is

$$3000 + 2500 + 4500 = 10000,$$

$$\therefore P(E_1) = \frac{3000}{10000} = \frac{3}{10}, P(E_2) = \frac{2500}{10000} = \frac{1}{4}, P(E_3) = \frac{4500}{10000} = \frac{9}{20}.$$

$$P(A|E_1) = \frac{1}{100} = 0.01, P(A|E_2) = \frac{1.2}{100} = 0.012, P(A|E_3) = \frac{2}{100} = 0.02.$$

Thus, the required probability = Probability that the drawn item is defective

$$\begin{aligned} &= P(A) \\ &= P(E_1) P(A|E_1) + P(E_2) P(A|E_2) + P(E_3) P(A|E_3) \\ &= \frac{3}{10} \times 0.01 + \frac{1}{4} \times 0.012 + \frac{9}{20} \times 0.02 \\ &= \frac{3}{1000} + \frac{3}{1000} + \frac{9}{1000} \\ &= \frac{15}{1000} = 0.015. \end{aligned}$$

Example 26(b): Consider the problem given in Example 26(a) after replacing the question asked therein (i.e. the last sentence) by the following question:

If the drawn item is found to be defective, what is the probability that it has been produced by machine Y?

Solution: Proceed exactly in the manner the Example 26(a) has been solved and then as under:

Probability that the drawn item has been produced by machine Y, given that it is defective

$$\begin{aligned}
 &= P(E_2 | A) \\
 &= \frac{P(E_2)P(A | E_2)}{P(A)} \quad \left[\begin{array}{l} \text{Applying Bayes'} \\ \text{theorem} \end{array} \right] \\
 &= \frac{\frac{1}{4} \times 0.012}{0.015} = \frac{0.003}{0.015} = \frac{1}{5}.
 \end{aligned}$$

Example 27(a): There are two coins-one unbiased and the other two-headed, otherwise they are identical. One of the coins is taken at random without seeing it and tossed. What is the probability of getting a head?

Solution: Let E_1 and E_2 be the events of selecting the unbiased coin and the two-headed coin, respectively. Let A be the event of getting head on the tossed coin.

$$\therefore P(E_1) = \frac{1}{2}, P(E_2) = \frac{1}{2} \quad [\because \text{selection of each of the coin is equally likely}]$$

$$P(A | E_1) = \frac{1}{2} \quad [\because \text{if it is unbiased coin, then head and tail are equally likely}]$$

$$P(A | E_2) = 1 \quad [\because \text{if it is two-headed coin, then getting the head is certain}]$$

Thus, the required probability = $P(A)$

$$\begin{aligned}
 &= P(E_1) P(A | E_1) + P(E_2) P(A | E_2) \\
 &= \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times 1 \\
 &= \frac{1}{4} + \frac{1}{2} = \frac{3}{4}.
 \end{aligned}$$

Example 27(b): Consider the problem given in Example 27(a), after replacing the question asked therein (i.e. the last sentence) by the following question:

If head turns up, what is the probability that

- i) it is the two-headed coin
- ii) it is the unbiased coin.

Solution: First give the solution of Example 27(a) and then proceed as under:

Now, the probability that the tossed coin is two-headed given that head turned up

$$\begin{aligned}
 &= P(E_2 | A) \\
 &= \frac{P(E_2)P(A | E_2)}{P(A)} \quad \left[\begin{array}{l} \text{Applying Bayes'} \\ \text{theorem} \end{array} \right]
 \end{aligned}$$

$$= \frac{\frac{1}{2} \times 1}{\frac{3}{4}} = \frac{1}{2} \times \frac{4}{3} = \frac{2}{3}.$$

ii) The probability that the tossed coin is unbiased given that head turned up

$$= P(E_1 | A)$$

$$= \frac{P(E_1)P(A | E_1)}{P(A)} \quad \left[\begin{array}{l} \text{Applying Bayes' } \\ \text{theorem} \end{array} \right]$$

$$= \frac{\frac{1}{2} \times \frac{1}{2}}{\frac{3}{4}} = \frac{1}{4} \times \frac{4}{3} = \frac{1}{3}$$

Check Your Progress 4

- 1) A bag contains 4 red and 5 white balls. Another bag contains 2 red and 3 white balls. A ball is drawn from the first bag and is transferred to the second bag. A ball is then drawn from the second bag and is found to be red, what is the probability that red ball was transferred from first to second bag?

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- 2) An insurance company insured 1000 scooter drivers, 3000 car drivers and 6000 truck drivers. The probabilities that scooter, car and truck drivers meet an accident are 0.02, 0.04, 0.25, respectively. One of the insured persons meets with an accident. What is the probability that he is a

(i) car driver

(ii) truck driver

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- 3) By examining the chest X-ray, the probability that T.B is detected when a person is actually suffering from T.B. is 0.99. The probability that the doctor diagnoses incorrectly that a person has T.B. on the basis of X-ray is 0.002. In a certain city, one in 1000 persons suffers from T.B. A person is selected at random and is diagnosed to have T.B., what is the chance that he actually has T.B.?

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4.9 SUMMARY

In this unit, we have discussed:

1. How to create a sample space for a problem.
2. Define an event and create its instances.
3. How to define probability.
4. What are the axioms of probability, and how can these axioms be used
5. How to use the addition law, conditional probability and independent events.
6. What is Bayes' theorem and how can it be used to compute probability.

4.10 SOLUTIONS / ANSWERS

Check Your Progress 1

1) Let suffices C, D, S, H denote that corresponding card is a club, diamond, spade, heart, respectively then sample space of drawing a card can be written as

$$\{1_C, 2_C, 3_C, \dots, 9_C, 10_C, J_C, Q_C, K_C, \\ 1_D, 2_D, 3_D, \dots, 9_D, 10_D, J_D, Q_D, K_D, \\ 1_S, 2_S, 3_S, \dots, 9_S, 10_S, J_S, Q_S, K_S, \\ 1_H, 2_H, 3_H, \dots, 9_H, 10_H, J_H, Q_H, K_H, \}$$

2) If a die and a coin is thrown then sample space is

$$\{H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6\}$$

- (i) Let A be the event of getting head and prime number, then
 $A = \{H2, H3, H5\}$
- (ii) Let B be the event of getting tail and even number, then
 $B = \{T2, T4, T6\}$
- (iii) Let C be the event of getting head and multiple of 3, then
 $C = \{H3, H6\}$

3) When two coins are tossed simultaneously then sample space is

$$S = \{HH, HT, TH, TT\}$$

- (i) Let A be the event of getting at least one head, then
 $A = \{HH, HT, TH\}$ and $P(A) = \frac{3}{4}$
- (ii) Let B = event of getting both head and tail, then
 $B = \{HT, TH\}$ and $P(B) = \frac{2}{4} = \frac{1}{2}$
- (iii) Let C be the event of getting at most one head, then
 $C = \{TT, TH, HT\}$ and $P(C) = \frac{3}{4}$

4) When 3 dice are thrown, then the sample space is

$$S = \{ (1, 1, 1), (1, 1, 2), (1, 1, 3), \dots, (1, 1, 6), \\ (1, 2, 1), (1, 2, 2), (1, 2, 3), \dots, (1, 2, 6), \\ (1, 3, 1), (1, 3, 2), (1, 3, 3), \dots, (1, 3, 6), \\ \dots \\ (6, 6, 1), (6, 6, 2), (6, 6, 3), \dots, (6, 6, 6) \}$$

$$\text{Number of elements in the sample space} = 6 \times 6 \times 6 = 216$$

$\left[\because \text{We have to fill up 3 positions } (.,.,.) \text{ and each position can be filled with 6 options, } \therefore \text{this can be done in } 6 \times 6 \times 6 = 216 \text{ ways} \right]$

- (i) Let A be the event of getting triplet i.e. same number on each die.

$$\therefore A = \{ (1, 1, 1), (2, 2, 2), (3, 3, 3), (4, 4, 4), (5, 5, 5), (6, 6, 6) \}$$

$$\text{and hence } P(A) = \frac{6}{216} = \frac{1}{36}$$

- (ii) Let B be the event of getting sum 5

$$\therefore B = \{ (1, 1, 3), (1, 3, 1), (3, 1, 1), (1, 2, 2), (2, 1, 2), (2, 2, 1) \}$$

$$\text{and } P(B) = \frac{6}{216} = \frac{1}{36}$$

- (iii) Let C be the event of getting sum at least 17 i.e. sum 17 or 18

$$\therefore C = \{ (5, 6, 6), (6, 5, 6), (6, 6, 5), (6, 6, 6) \}$$

$$\text{and hence } P(C) = \frac{4}{216} = \frac{1}{54}$$

- (iv) Let D be the event of getting prime number on first die and odd prime number on second and third die. i.e. first die can show 2 or 3 or 5 and second, third dice can show 3 or 5

$$\therefore D = \{ (2, 3, 3), (2, 3, 5), (2, 5, 3), (2, 5, 5), (3, 3, 3), (3, 3, 5), \\ (3, 5, 3), (3, 5, 5), (5, 3, 3), (5, 3, 5), (5, 5, 3), (5, 5, 5) \}$$

$$\text{and hence } P(D) = \frac{12}{216} = \frac{1}{18}$$

5) We know that there are 366 days in a leap year. i.e. $\frac{365}{7} = 52 \frac{2}{7}$ weeks

i.e. one leap year = (52 complete weeks + two over days)

i.e. 52 complete weeks and two over days. These two over days may be

- (i) Sunday and Monday
- (ii) Monday and Tuesday
- (iii) Tuesday and Wednesday
- (iv) Wednesday and Thursday
- (v) Thursday and Friday
- (vi) Friday and Saturday
- (vii) Saturday and Sunday

$\therefore$ Number of exhaustive Cases = 7

Let A be the event of getting 53 Mondays

There will be 53 Mondays in a leap year if and only if these two over days are

“Sunday and Monday” or “Monday and Tuesday”

$\therefore$ Number of favourable cases for event A are 2

$$\text{and } P(A) = \frac{\text{Number of favourable cases}}{\text{Number of exhaustive cases}} = \frac{2}{7}$$

Check Your Progress 2

1) a) $A \cap \bar{B} \cap \bar{C}$

b) $A \cap B \cap \bar{C}$

c) $A \cap B \cap C$

d) $(A \cap B \cap \bar{C}) \cup (A \cap \bar{B} \cap C) \cup (\bar{A} \cap B \cap C) \cup (A \cap B \cap C)$

e) $(\bar{A} \cap \bar{B} \cap C) \cup (\bar{A} \cap B \cap \bar{C}) \cup (A \cap \bar{B} \cap \bar{C})$

f) $\bar{A} \cap \bar{B} \cap \bar{C}$

- 2) Let A be the event that the drawn ball bears a number multiple of 3 and B be the event that it bears a number multiple of 5, then

$$A = \{3, 6, 9, 12\} \text{ and } B = \{5, 10\}$$

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= 4/14 + 2/14 - 0 = 3/7 \end{aligned}$$

- 3) Let A be the event of getting a card of king and B be the event of getting a red card.

$\therefore$ the required probability

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad [\text{By addition theorem}]$$

$$= \frac{4}{52} + \frac{26}{52} - \frac{2}{52} \left[\because \text{there are two cards which are both king as well as red} \right]$$

$$= \frac{4}{52} + \frac{26}{52} - \frac{2}{52} = \frac{4+26-2}{52} = \frac{28}{52} = \frac{7}{13}$$

- 4) Here, the number of exhaustive cases = $6 \times 6 = 36$.

Let A be the event that the sum is 6 and B be the event that the sum is 8.

$$\therefore A = \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}, \text{ and}$$

$$B = \{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\}.$$

Here, as $A \cap B = \phi$,

$\therefore$ A and B are mutually exclusive and hence

the required probability = $P(A \cup B) = P(A) + P(B)$

$$= \frac{5}{36} + \frac{5}{36} = \frac{10}{36} = \frac{5}{18}$$

Check Your Progress 3

- 1) Let A be the event that the card drawn is face card and B be the event that it is a king.

$$\therefore A = \{J_s, Q_s, K_s, J_h, Q_h, K_h, J_d, Q_d, K_d, J_c, Q_c, K_c\}$$

$$B = \{K_s, K_h, K_c, K_d\}$$

$$\Rightarrow A \cap B = \{K_s, K_h, K_c, K_d\}$$

$$\Rightarrow P(A) = \frac{12}{52}, P(B) = \frac{4}{52}, \text{ and } P(A \cap B) = \frac{4}{52}.$$

$$\therefore \text{the required probability} = P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{4/52}{12/52} = \frac{4}{12} = \frac{1}{3}.$$

- 2) Let A be the event that first card drawn is red and B be the event that second card is red.

$$\therefore P(A) = \frac{26}{52} \text{ and } P(B|A) = \frac{25}{51}.$$

Thus, the desired probability = $P(A \cap B)$

$$\begin{aligned} &= P(A) P(B|A) \left[\begin{array}{l} \text{By multiplication} \\ \text{theorem} \end{array} \right] \\ &= \left(\frac{26}{52} \right) \left(\frac{25}{51} \right) = \frac{25}{102}. \end{aligned}$$

- 3) Let E_1 be the event of getting a defective item in first draw and E_2 be the event of getting a good item in the second draw.

$\therefore$ the required probability = $P(E_1 \cap E_2)$

$$\begin{aligned} &= P(E_1)P(E_2 | E_1) \left[\begin{array}{l} \text{By multiplication} \\ \text{theorem} \end{array} \right] \\ &= \frac{4}{14} \times \frac{10}{13} = \frac{20}{91}. \end{aligned}$$

- 4) Let A be the event that person X passes the test and B be the event that person Y passes the test.

$$\therefore P(A) = \frac{3}{3+5} = \frac{3}{8}$$

$$\text{and } P(B) = \frac{3}{3+2} = \frac{3}{5}$$

Now, as both the person take the test independently,

$\therefore$ the required probability = $P(A \cap B)$

$$= P(A)P(B) = \left(\frac{3}{8} \right) \left(\frac{3}{5} \right) = \frac{9}{40}$$

Check Your Progress 4

- 1) Let E_1 be the event that a red ball is drawn from the first bag and E_2 be the event that the drawn ball from the first bag is white. Let A be the event of drawing a red ball from the second bag after transferring the ball drawn from first bag into it.

$$\therefore P(E_1) = \frac{4}{9}, P(E_2) = \frac{5}{9}$$

$$P(A|E_1) = \frac{3}{6}, P(A|E_2) = \frac{2}{6}$$

$\therefore$ By law of total probability,

$$\begin{aligned} P(A) &= P(E_1)P(A|E_1) + P(E_2)P(A|E_2) \\ &= \frac{4}{9} \times \frac{3}{6} + \frac{5}{9} \times \frac{2}{6} = \frac{11}{27}. \end{aligned}$$

Thus, the required probability = $P(E_1|A)$

$$= \frac{P(E_1)P(A|E_1)}{P(A)} \quad \left[\begin{array}{l} \text{Applying Bayes'} \\ \text{theorem} \end{array} \right]$$

$$= \frac{\frac{4}{9} \times \frac{3}{6}}{\frac{11}{27}} = \frac{6}{11}$$

2) Total number of drivers = 1000 + 3000 + 6000 = 10000.

Let E_1, E_2, E_3 be the events that the selected insured person is a scooter driver, car driver, truck driver respectively. Let A be the event that a driver meets with an accident.

$$\therefore P(E_1) = \frac{1000}{10000} = \frac{1}{10}, P(E_2) = \frac{2000}{10000} = \frac{1}{5}, P(E_3) = \frac{6000}{10000} = \frac{6}{10} = \frac{3}{5}.$$

$$P(A|E_1) = 0.02 = \frac{2}{100}, P(A|E_2) = 0.04 = \frac{4}{100}, P(A|E_3) = 0.25 = \frac{25}{100}$$

$\therefore$ By total probability theorem, we have

$$\begin{aligned} P(A) &= P(E_1)P(A|E_1) + P(E_2)P(A|E_2) + P(E_3)P(A|E_3) \\ &= \frac{1}{10} \times \frac{2}{100} + \frac{1}{5} \times \frac{4}{100} + \frac{3}{5} \times \frac{25}{100} \\ &= \frac{1}{500} + \frac{4}{500} + \frac{75}{500} = \frac{80}{500} = \frac{4}{25} \end{aligned}$$

i) The required probability = $P(E_2|A)$

$$= \frac{P(E_2)P(A|E_2)}{P(A)} \quad \left[\begin{array}{l} \text{Applying Bayes'} \\ \text{theorem} \end{array} \right]$$

$$= \frac{\frac{1}{5} \times \frac{4}{100}}{\frac{4}{25}} = \frac{4}{500} \times \frac{25}{4} = \frac{1}{20}.$$

ii) The required probability = $P(E_3 | A)$

$$\begin{aligned}
 &= \frac{P(E_3)P(A | E_3)}{P(A)} \left[\begin{array}{l} \text{Applying Bayes' } \\ \text{theorem} \end{array} \right] \\
 &= \frac{\frac{3}{5} \times \frac{25}{100}}{\frac{4}{25}} \\
 &= \frac{75}{500} \times \frac{25}{4} = \frac{75}{80} = \frac{15}{16}.
 \end{aligned}$$

- 3) Let E_1 be the event that a person selected from the population of the city is suffering from the T.B. and E_2 be the event that he/she is not suffering from the T.B. Let A be the event that the selected person is diagnosed to have T.B. Therefore, according to given,

$$P(E_1) = \frac{1}{1000} = 0.001, \quad P(E_2) = 1 - 0.001 = 0.999 = \frac{999}{1000},$$

$$P(A | E_1) = 0.99 = \frac{99}{100}, \quad P(A | E_2) = 0.002 = \frac{2}{1000}.$$

$\therefore$ By total probabilities theorem, we have

$$\begin{aligned}
 P(A) &= P(E_1)P(A|E_1) + P(E_2)P(A|E_2) \\
 &= \frac{1}{1000} \times \frac{99}{100} + \frac{999}{1000} \times \frac{2}{1000} \\
 &= \frac{99}{100000} + \frac{1998}{1000000} = \frac{990 + 1998}{1000000} = \frac{2988}{1000000}
 \end{aligned}$$

Thus, the required probability = $P(E_1|A)$

$$\begin{aligned}
 &= \frac{P(E_1)P(A | E_1)}{P(A)} \\
 &= \frac{\frac{1}{1000} \times \frac{99}{100}}{\frac{2988}{1000000}} \\
 &= \frac{990}{2988} = \frac{55}{166}.
 \end{aligned}$$

4.11 REFERENCES

IGNOU Course Material

1. Post-Graduate Diploma In Applied Statistics (PGDAST), MST 003: Probability Theory, Block 1: Basic Concepts in Probability, UNIT 1: Measures of Dispersion (for Sections 4.2, 4.3 and related portions in Sections 4.0, 4.1, 4.9, 4.10 and Check Your Progress questions)
2. PGDAST, MST 003: Probability Theory, Block 1: Basic Concepts in Probability, UNIT 2: Different Approaches to Probability Theory (for Sections 4.4 and related portions in Sections 4.0, 4.1, 4.9, 4.10 and Check Your Progress questions)
3. PGDAST, MST 003: Probability Theory, Block 1: Basic Concepts in Probability, UNIT 3: Laws of Probability (for Sections 4.5 to Section 4.7 and related portions in Sections 4.0, 4.1, 4.9, 4.10 and Check Your Progress questions)
4. PGDAST, MST 003: Probability Theory, Block 1: Basic Concepts in Probability, UNIT 4: Bayes' Theorem (for Sections 4.8 and related portions in Sections 4.0, 4.1, 4.9, 4.10 and Check Your Progress questions)

Reference Book

5. "Introduction to Probability and Statistics for Engineers and Scientists", 3rd Edition, Sheldon M. Ross, Elsevier Academic Press, 2004

UNIT 5 RANDOM VARIABLES AND EXPECTATION

Structure

- 5.0. Introduction
- 5.1. Objectives
- 5.2. Random Variable
- 5.3. Discrete Random Variable and Probability Mass Function
- 5.4. Continuous Random Variable and Probability Density Function
- 5.5. Distribution Function
- 5.6. Bivariate Random Variables
 - 5.6.1 Bivariate Discrete Random Variables
 - 5.6.2 Bivariate Continuous Random Variables
- 5.7. Expectation of a Random Variable
- 5.8. Moments and Other Measures in Terms of Expectations
 - 5.8.1 Moments
 - 5.8.2 Variance and Covariance
 - 5.8.3 Addition and Multiplication Theorems of Expectation
- 5.9. Summary
- 5.10. Solutions/Answers
- 5.11. References

5.0 INTRODUCTION

In the previous units, we have studied the computation of probabilities of events in detail. In those units, we were interested in knowing the occurrence of outcomes. In the present unit, we will be interested in the numbers associated with such outcomes of the random experiments. Such an interest leads to studying the concept of random variable. In this unit, we will introduce the concept of random variable, discrete and continuous random variables their probability functions

There may also be situations where we have to study two-dimensional random variables in connection with a random experiment. For example, we may be interested in recording the number of boys and girls born in a hospital on a particular day. Here, 'the number of boys' and 'the number of girls' are random variables taking the values 0, 1, 2, ..., and both these random variables are also discrete. In this unit, we define two-dimensional discrete and continuous random variables. The joint, marginal and conditional probability mass functions of two-dimensional discrete random variables. The Unit also defines the distribution function and the marginal distribution function for bivariate continuous random variable

You have studied the methods of finding mean, variance and other measures in context of frequency distributions in Block 1 of this course. In the present unit, we discuss the expectations of random variables and their properties. Addition and multiplication laws of expectation have also been defined.

5.1 OBJECTIVES

A study of this unit would enable you to:

- define a random variable, discrete and continuous random variables;

- specify the probability mass function, i.e. probability distribution of discrete random variables;
- specify the probability density function, i.e. probability function of continuous random variables;
- define the distribution function and two two-dimensional discrete and continuous random variables;
- find the expected values of random variables;
- obtain various measures for probability distributions; and

5.2 RANDOM VARIABLE

The last unit discusses the process of computing probabilities for events (subsets of a sample space). In many experiments, we may be interested in a numerical characteristic associated with outcomes of a random experiment. Like the outcome, the value of such a numerical characteristic cannot be predicted in advance.

For example, suppose a die is tossed twice and we are interested in number of times an odd number appears. Let X be the number of appearances of odd number. If a die is thrown twice, an odd number may appear '0' times (i.e. we may have even number both the times) or once (i.e. we may have odd number in one throw and even number in the other throw) or twice (i.e. we may have odd number both the times). Here, X can take the values 0, 1, 2 and is a variable quantity behaving randomly and hence we may call it a 'random variable'. Also notice that its values are real and are defined on the sample space

$\{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6),$
 $(2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6),$
 $(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6),$
 $(4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6),$
 $(5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6),$
 $(6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$

i.e.

$$X = \begin{cases} 0, & \text{if the outcome is } (2, 2), (2, 4), (2, 6), (4, 2), (4, 4), (4, 6), (6, 2), \\ & (6, 4), (6, 6) \\ 1, & \text{if the outcome is } (1, 2), (1, 4), (1, 6), (2, 1), (2, 3), (2, 5), (3, 2), \\ & (3, 4), (3, 6), (4, 1), (4, 3), (4, 5), (5, 2), (5, 4), \\ & (5, 6), (6, 1), (6, 3), (6, 5) \\ 2, & \text{if the outcome is } (1, 1), (1, 3), (1, 5), (3, 1), (3, 3), (3, 5), (5, 1), \\ & (5, 3), (5, 5) \end{cases}$$

$$\text{So, } P[X = 0] = \frac{9}{36} = \frac{1}{4}, P[X = 1] = \frac{18}{36} = \frac{1}{2}, P[X = 2] = \frac{9}{36} = \frac{1}{4},$$

$$\text{and } P[X = 0] + P[X = 1] + P[X = 2] = \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1.$$

Observe that a probability can be assigned to the event that X assumes a particular value. It can also be observed that the sum of the probabilities corresponding to different values of X is one.

So, a random variable can be defined as below:

Definition: A random variable is a real-valued function whose domain is a set of possible outcomes of a random experiment and range is a sub-set of the set of real numbers and has the following properties:

- i) Each particular value of the random variable can be assigned some probability
- ii) Uniting all the probabilities associated with all the different values of the random variable gives the value 1(unity).

Remark 1: We shall denote random variables by capital letters like X, Y, Z , etc. and write **r.v.** for random variable.

5.3 DISCRETE RANDOM VARIABLE AND PROBABILITY MASS FUNCTION

Discrete Random Variable

A random variable is said to be discrete if it has either a finite or a countable number of values. Countable number of values means the values which can be arranged in a sequence, i.e. the values which have one-to-one correspondence with the set of natural numbers, i.e., on the basis of three or four successive known terms, we can catch a rule and hence can write the subsequent terms. For example, suppose X is a random variable taking the values say 2, 5, 8, 11, ... then we can write the fifth, sixth, ... values, because the values have one-to-one correspondence with the set of natural numbers and have the general term as $3n - 1$, i.e. on taking $n = 1, 2, 3, 4, 5, \dots$ we have 2, 5, 8, 11, 14, So, X in this example is a discrete random variable. The number of students present each day in a class during an academic session is an example of discrete random variable as the number cannot take a fractional value.

Probability Mass Function

Let X be a r.v. which takes the values $x_1, x_2, \dots$ and let $P[X = x_i] = p(x_i)$. This function $p(x_i)$, $i = 1, 2, \dots$ defined for the values $x_1, x_2, \dots$ assumed by X is called probability mass function of X satisfying $p(x_i) \geq 0$ and $\sum_i p(x_i) = 1$. The set

$\{(x_1, p(x_1)), (x_2, p(x_2)), \dots\}$ specifies the probability distribution of a discrete r.v. X . Probability distribution of r.v. X can also be exhibited in the following manner:

X	x_1	x_2	x_3	$\dots$
$p(x)$	$p(x_1)$	$p(x_2)$	$p(x_3)$	$\dots$

Now, let us take up some examples concerning probability mass function.

Example 1: State, giving reasons, which of the following are not probability distributions:

(i)

X	0	1
p(x)	$\frac{1}{2}$	$\frac{3}{4}$

(ii)

X	0	1	2
p(x)	$\frac{3}{4}$	$-\frac{1}{2}$	$\frac{3}{4}$

(iii)

X	0	1	2
p(x)	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

(iv)

X	0	1	2	3
p(x)	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{1}{4}$	$\frac{1}{8}$

Solution:

(i) Here $p(x_i) \geq 0$, $i = 1, 2$; but

$$\sum_{i=1}^2 p(x_i) = p(x_1) + p(x_2) = p(0) + p(1) = \frac{1}{2} + \frac{3}{4} = \frac{5}{4} > 1.$$

So, the given distribution is not a probability distribution as $\sum_{i=1}^2 p(x_i)$ is greater than 1.

(ii) It is not probability distribution as $p(x_2) = p(1) = -\frac{1}{2}$ i.e. negative

(iii) Here, $p(x_i) \geq 0$, $i = 1, 2, 3$

$$\text{and } \sum_{i=1}^3 p(x_i) = p(x_1) + p(x_2) + p(x_3) = p(0) + p(1) + p(2) = \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1.$$

$\therefore$ The given distribution is probability distribution.

(iv) Here, $p(x_i) \geq 0$, $i = 1, 2, 3, 4$; but

$$\begin{aligned} \sum_{i=1}^4 p(x_i) &= p(x_1) + p(x_2) + p(x_3) + p(x_4) \\ &= p(0) + p(1) + p(2) + p(3) = \frac{1}{8} + \frac{3}{8} + \frac{1}{4} + \frac{1}{8} = \frac{7}{8} < 1. \end{aligned}$$

$\therefore$ The given distribution is not probability distribution.

Example 2: For the following probability distribution of a discrete r.v. X , find

- (i) the constant c ,
- (ii) $P[X \leq 3]$ and
- (iii) $P[1 < X < 4]$.

X	0	1	2	3	4	5
$p(x)$	0	c	c	$2c$	$3c$	c

Solution:

- i) As the given distribution is probability distribution,

$$\therefore \sum_i p(x_i) = 1$$

$$\Rightarrow 0 + c + c + 2c + 3c + c = 1 \Rightarrow 8c = 1 \Rightarrow c = \frac{1}{8}$$

- ii) $P[X \leq 3] = P[X = 3] + P[X = 2] + P[X = 1] + P[X = 0]$

$$= 2c + c + c + 0 = 4c = 4 \times \frac{1}{8} = \frac{1}{2}.$$

- iii) $P[1 < X < 4] = P[X = 2] + P[X = 3] = c + 2c = 3c = 3 \times \frac{1}{8} = \frac{3}{8}.$

Example 3: Find the probability distribution of the number of heads when three fair coins are tossed simultaneously.

Solution: Let X be the number of heads in the toss of three fair coins.

As the random variable, “the number of heads” in a toss of three coins may be 0 or 1 or 2 or 3 associated with the sample space

$\{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\},$

$\therefore X$ can take the values 0, 1, 2, 3, with

$$P[X = 0] = P[TTT] = \frac{1}{8}$$

$$P[X = 1] = P[HTT, THT, TTH] = \frac{3}{8}$$

$$P[X = 2] = P[HHT, HTH, THH] = \frac{3}{8}$$

$$P[X = 3] = P[HHH] = \frac{1}{8}.$$

$\therefore$ Probability distribution of X , i.e. the number of heads when three coins are tossed simultaneously is

X	0	1	2	3
$p(x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

which is the required probability distribution.

Let us define and explain a continuous random variable and its probability function in the next section.

5.4 CONTINUOUS RANDOM VARIABLE AND PROBABILITY DENSITY FUNCTION

In Sec. 5.3 of this unit, we have defined the discrete random variable as a random variable having countable number of values, i.e. whose values can be arranged in a sequence. But, if a random variable is such that its values cannot be arranged in a sequence, it is called continuous random variable. Temperature of a city at various points of time during a day is an example of continuous random variable as the temperature takes uncountable values, i.e. it can take fractional values also. So, a random variable is said to be continuous if it can take all possible real (i.e. integer as well as fractional) values between two certain limits. For example, let us denote the variable, “Difference between the rainfall (in cm) of a city and that of another city on every rainy day in a rainy season”, by X , then X here is a continuous random variable as it can take any real value between two certain limits. It can be noticed that for a continuous random variable, the chance of occurrence of a particular value of the variable is very small, so instead of specifying the probability of taking a particular value by the variable, we specify the probability of its lying within an interval. For example, chance that an athlete will finish a 100-meter race in say exactly 10 seconds is very-very small, i.e. almost zero as it is very rare to finish the race in a fixed time. Here, the probability is specified for an interval, i.e. we may be interested in finding as to what is the probability of finishing the race by the athlete in an interval of say 10 to 12 seconds.

So, continuous random variable is represented by different representation known as **probability density function** unlike the discrete random variable which is represented by probability mass function.

Probability Density Function

Let $f(x)$ be a continuous function of x . Suppose the shaded region ABCD shown in the following figure represents the area bounded by $y = f(x)$, x -axis and the ordinates at the points x and $x + \delta x$, where δx is the length of the interval $(x, x + \delta x)$.

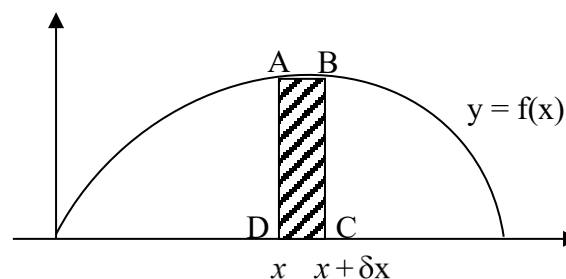


Fig. 5.1: Probability Density Function

Now, if δx is very-very small, then the curve AB will act as a line and hence the shaded region will be a rectangle whose area will be $AD \times DC$ i.e. $f(x) \delta x$

[$\because AD =$ the value of y at x i.e. $f(x)$, $DC =$ length δx of the interval $(x, x + \delta x)$]

Also, this area = probability that X lies in the interval $(x, x + \delta x)$

$$= P[x \leq X \leq x + \delta x]$$

Hence,

$$P[x \leq X \leq x + \delta x] = f(x) \delta x$$

$$\Rightarrow \frac{P[x \leq X \leq x + \delta x]}{\delta x} = f(x), \text{ where } \delta x \text{ is very-very small}$$

$$\Rightarrow \lim_{\delta x \rightarrow 0} \frac{P[x \leq X \leq x + \delta x]}{\delta x} = f(x).$$

$f(x)$, so defined, is called probability density function.

Probability density function has the same properties as that of probability mass function. So, $f(x) \geq 0$ and sum of the probabilities of all possible values that the random variable can take, has to be 1. But, here, as X is a continuous random variable, the summation is made possible through 'integration' and hence

$$\int_R f(x) dx = 1,$$

where integral has been taken over the entire range R of values of X .

Remark 2

- i) Summation and integration have the same meanings but in mathematics there is still difference between the two and that is that the former is used in case of discrete values, i.e. countable values and the latter is used in continuous case.
- ii) An essential property of a continuous random variable is that there is zero probability that it takes any specified numerical value, but the probability that it takes a value in specified intervals is non-zero and is calculable as a definite integral of the probability density function of the random variable and hence the probability that a continuous r.v. X will lie between two values a and b is given by

$$P[a < X < b] = \int_a^b f(x) dx.$$

Example 4: A continuous random variable X has the probability density function: $f(x) = Ax^3$, $0 \leq x \leq 1$. Determine

- i) A
- ii) $P[0.2 < X < 0.5]$

Solution:

(i) As $f(x)$ is probability density function,

$$\therefore \int_R f(x) dx = 1$$

$$\Rightarrow \int_0^1 f(x) dx = 1 \Rightarrow \int_0^1 Ax^3 dx = 1$$

$$\Rightarrow A \left[\frac{x^4}{4} \right]_0^1 = 1 \Rightarrow A \left(\frac{1}{4} - 0 \right) = 1 \Rightarrow A = 4$$

$$\begin{aligned} \text{(ii) } P[0.2 < X < 0.5] &= \int_{0.2}^{0.5} f(x) dx = \int_{0.2}^{0.5} Ax^3 dx = 4 \left[\frac{x^4}{4} \right]_{0.2}^{0.5} = [(0.5)^4 - (0.2)^4] \\ &= 0.0625 - 0.0016 = 0.0609 \end{aligned}$$

5.5 DISTRIBUTION FUNCTION

A function F defined for all values of a random variable X by $F(x) = P[X \leq x]$ is called the distribution function. It is also known as the cumulative distribution function (c.d.f.) of X since it is the cumulative probability of X up to and including the value x . As X can take any real value, therefore the domain of the distribution function is set of real numbers and as $F(x)$ is a probability value, therefore the range of the distribution function is $[0, 1]$.

Remark 3: Here, X denotes the random variable and x represents a particular value of random variable. $F(x)$ may also be written as $F_X(x)$, which means that it is a distribution function of random variable X .

Discrete Distribution Function

Distribution function of a discrete random variable is said to be discrete distribution function or cumulative distribution function (c.d.f.). Let X be a discrete random variable taking the values $x_1, x_2, x_3, \dots$ with respective probabilities $p_1, p_2, p_3, \dots$

$$\begin{aligned} \text{Then } F(x_i) &= P[X \leq x_i] = P[X = x_1] + P[X = x_2] + \dots + P[X = x_i] \\ &= p_1 + p_2 + \dots + p_i. \end{aligned}$$

The distribution function of X , in this case, is given as in the following table:

X	$F(x)$
x_1	p_1
x_2	$p_1 + p_2$
x_3	$p_1 + p_2 + p_3$
$\dots$	$\dots$
x_i	$p_1 + p_2 + p_3 + \dots + p_i$

The value of $F(x)$ corresponding to the last value of the random variable X is always 1, as it is the sum of all the probabilities. $F(x)$ remains 1 beyond this last value of X also, as it being a probability, it can never exceed one.

For example, let X be a random variable having the following probability distribution:

X	0	1	2
$p(x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

Notice that $p(x)$ will be zero for other values of X . Then, Distribution function of X is given by

X	$F(x) = P[X \leq x]$
0	$\frac{1}{4}$
1	$\frac{1}{4} + \frac{1}{2} = \frac{3}{4}$
2	$\frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1$

Here, for the last value, i.e. for $X = 2$, we have $F(x) = 1$.

Also, if we take a value beyond 2 say 4, then we get

$$\begin{aligned} F(4) &= P[X \leq 4] \\ &= P[X = 4] + P[X = 3] + P[X \leq 2] \\ &= 0 + 0 + 1 = 1. \end{aligned}$$

Example 5: A random variable X has the following probability function:

X	0	1	2	3	4	5	6	7
$p(x)$	0	$\frac{1}{10}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{3}{10}$	$\frac{1}{100}$	$\frac{1}{50}$	$\frac{17}{100}$

Determine the distribution function of X .

Solution: Here,

$$F(0) = P[X \leq 0] = P[X = 0] = 0,$$

$$F(1) = P[X \leq 1] = P[X = 0] + P[X = 1] = 0 + \frac{1}{10} = \frac{1}{10},$$

$$F(2) = P[X \leq 2] = P[X = 0] + P[X = 1] + P[X = 2] = 0 + \frac{1}{10} + \frac{1}{5} = \frac{3}{10},$$

and so on.

Continuous Distribution Function

Distribution function of a continuous random variable is called the continuous distribution function or cumulative distribution function (c.d.f.).

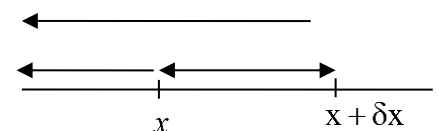
Let X be a continuous random variable having the probability density function $f(x)$, as defined in the last section of this unit, then the distribution function $F(x)$ is given by

$$F(x) = P[X \leq x] = \int_{-\infty}^x f(x) dx.$$

Also, in the last section, we have defined the p.d.f. $f(x)$ as

$$f(x) = \lim_{\delta x \rightarrow 0} \frac{P[x \leq X \leq x + \delta x]}{\delta x},$$

$$\therefore f(x) = \lim_{\delta x \rightarrow 0} \frac{P[X \leq x + \delta x] - P[X \leq x]}{\delta x}$$



$$\Rightarrow f(x) = \lim_{\delta x \rightarrow 0} \frac{F(x + \delta x) - F(x)}{\delta x},$$

$$\Rightarrow f(x) = \text{Derivative of } F(x) \text{ with respect to } x \quad \left[\begin{array}{l} \text{By definition of} \\ \text{the derivative} \end{array} \right]$$

$$\Rightarrow f(x) = F'(x)$$

$$\Rightarrow f(x) = \frac{d}{dx}(F(x))$$

$$\Rightarrow dF(x) = f(x) dx$$

Here, $dF(x)$ is known as the probability differential.

$$\text{So, } F(x) = \int_{-\infty}^x f_x(x) dx \text{ and } F'(x) = f(x).$$

Example 6: The diameter 'X' of a cable is assumed to be a continuous random variable with p.d.f. $f(x) = \begin{cases} 6x(1-x), & 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$.

Obtain the c.d.f. of X.

Solution: For $0 \leq x \leq 1$, the c.d.f. of X is given by

$$\begin{aligned} F(x) &= P[X \leq x] = \int_0^x f(x) dx \\ &= \int_0^x 6x(1-x) dx \\ &= 6 \int_0^x (x - x^2) dx \\ &= 6 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^x = 3x^2 - 2x^3 \end{aligned}$$

$\therefore$ The c.d.f. of X is given by

$$F(x) = \begin{cases} 0, & x < 0 \\ 3x^2 - 2x^3, & 0 \leq x \leq 1. \\ 1, & x > 1 \end{cases}$$

Remark 4: In the above example, $F(x)$ is taken as 0 for $x < 0$ since $p(x) = 0$ for $x < 0$; and $F(x)$ is taken as 1 for $x > 1$ since $F(1) = 1$ and therefore, for $x > 1$ also $F(x)$ will remain 1.

Check Your Progress 1

- 1) 2 bad articles are mixed with 5 good ones. Find the probability distribution of the number of bad articles, if 2 articles are drawn at random.

.....
.....

- 2) Given the probability distribution:

X	0	1	2	3
p(x)	$\frac{1}{10}$	$\frac{3}{10}$	$\frac{1}{2}$	$\frac{1}{10}$

Let $Y = X^2 + 2X$. Find the probability distribution of Y.

.....
.....

- 3) An urn contains 3 white and 4 red balls. 3 balls are drawn one by one with replacement. Find the probability distribution of the number of red balls drawn.

.....
.....

- 4) The life (in hours) X of a certain type of light bulb may be supposed to be a continuous random variable with p.d.f.:

$$f(x) = \begin{cases} \frac{A}{x^3}, & 1500 < x < 2500 \\ 0, & \text{elsewhere} \end{cases}$$

Determine the constant A and compute the probability that $1600 \leq X \leq 2000$.

.....
.....

- 5) A random variable X has the following probability distribution:

X	0	1	2	3	4	5	6	7	8
p(x)	k	3k	5k	7k	9k	11k	13k	15k	17k

- (i) Determine the value of k.
(ii) Find the distribution function of X.

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.....

- 6) Let X be continuous random variable with p.d.f. given by.

$$f(x) = \begin{cases} \frac{x}{2}, & 0 \leq x < 1 \\ \frac{1}{2}, & 1 \leq x < 2 \\ \frac{1}{2}(3-x), & 2 \leq x < 3 \\ 0, & \text{elsewhere.} \end{cases}$$

Determine F(x), the c.d.f. of X.

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.....

5.6 BIVARIATE RANDOM VARIABLES

In the previous sections, the concept of a single-dimensional random variable has been studied. Proceeding in analogy with the one-dimensional case, is the concept of two-dimensional random variables

5.6.1 Bivariate Discrete Random Variables

Definition: Let X and Y be two discrete random variables defined on the sample space S of a random experiment then the function (X, Y) defined on the same sample space is called a two-dimensional discrete random variable. In other words, (X, Y) is a two-dimensional random variable if the possible values of (X, Y) are finite or countably infinite. Here, each value of X and Y is represented as a point (x, y) in the xy -plane.

As an illustration, let us consider the following example:

Let three balls b_1, b_2, b_3 be placed randomly in three cells. The possible outcomes of placing the three balls in three cells are shown in Table 5.1.

Table 5.1: Possible Outcomes of Placing the Three Balls in Three Cells

Arrangement Number	Placement of the Balls in		
	Cell 1	Cell 2	Cell 3
1	b_1	b_2	b_3
2	b_1	b_3	b_2
3	b_2	b_1	b_3
4	b_2	b_3	b_1
5	b_3	b_1	b_2
6	b_3	b_2	b_1
7	b_1, b_2	b_3	-
8	b_1, b_2	-	b_3
9	-	b_1, b_2	b_3
10	b_1, b_3	b_2	-
11	b_1, b_3	-	b_2
12	-	b_1, b_3	b_2
13	b_2, b_3	b_1	-
14	b_2, b_3	-	b_1
15	-	b_2, b_3	b_1
16	b_1	b_2, b_3	-
17	b_1	-	b_2, b_3
18	-	b_1	b_2, b_3
19	b_2	b_3, b_1	-
20	b_2	-	b_3, b_1
21	-	b_2	b_3, b_1
22	b_3	b_1, b_2	-
23	b_3	-	b_1, b_2
24	-	b_3	b_1, b_2

25	b_1, b_2, b_3	-	-
26	-	b_1, b_2, b_3	-
27	-	-	b_1, b_2, b_3

Now, let X denote the number of balls in Cell 1 and Y be the number of cells occupied. Notice that X and Y are discrete random variables where X take on the values 0, 1, 2, 3 ($\because$ number of balls in Cell 1 may be 0 or 1 or 2 or 3) and Y take on the values 1, 2, 3 ($\because$ number of occupied cells may be 1 or 2 or 3). The possible values of two-dimensional random variable (X, Y) , therefore, are all ordered pairs of the values x and y of X and Y , respectively, i.e. are (0, 1), (0, 2), (0, 3), (1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3).

Now, to each possible value (x_i, y_j) of (X, Y) , we can associate a number $p(x_i, y_j)$ representing $P(X = x_i, Y = y_j)$ as follows:

$p(0, 1) = P[X = 0, Y = 1] = P[\text{no ball in Cell 1 and 1 cell occupied}]$ $= P[\text{Arrangement numbers 26, 27}] = \frac{2}{27}$
$p(0, 2) = P[X = 0, Y = 2] = P[\text{no ball in Cell 1 and 2 cells occupied}]$ $= P[\text{Arrangement numbers 9, 12, 15, 18, 21, 24}]$ $= \frac{6}{27}$
$p(0, 3) = P[X = 0, Y = 3] = P[\text{no ball in Cell 1 and 3 cells occupied}]$ $= P[\text{Impossible event}] = 0$
$p(1, 1) = P[X = 1, Y = 1] = P[\text{one ball in Cell 1 and 1 cell occupied}]$ $= P[\text{Impossible event}] = 0$
$p(1, 2) = P[X = 1, Y = 2] = P[\text{one ball in Cell 1 and 2 cells occupied}]$ $= P[\text{Arrangement numbers 16, 17, 19, 20, 22, 23}]$ $= \frac{6}{27}$
$p(1, 3) = P[X = 1, Y = 3] = P[\text{one ball in Cell 1 and 3 cells occupied}]$ $= P[\text{Arrangement numbers 1 to 6}] = \frac{6}{27}$
$p(2, 1) = P[X = 2, Y = 1] = P[\text{two balls in Cell 1 and 1 cell occupied}]$ $= P[\text{Impossible event}] = 0$
$p(2, 2) = P[X = 2, Y = 2] = P[\text{two balls in Cell 1 and 2 cells occupied}]$ $= P[\text{Arrangement numbers 7, 8, 10, 11, 13, 14}]$ $= \frac{6}{27}$
$p(2, 3) = P[X = 2, Y = 3] = P[\text{two balls in Cell 1 and 3 cells occupied}]$ $= P[\text{Impossible event}] = 0$
$p(3, 1) = P[X = 3, Y = 1] = P[\text{three balls in Cell 1 and 1 cell occupied}]$ $= P[\text{Arrangement number 25}] = \frac{1}{27}$

$p(3, 2) = P[X = 3, Y = 2] = P[\text{three balls in Cell 1 and 2 cells occupied}]$ $= P[\text{Impossible event}] = 0$
$p(3, 3) = P[X = 3, Y = 3] = P[\text{three balls in Cell 1 and 3 cells occupied}]$ $= P[\text{Impossible event}] = 0$

The values of (X, Y) together with the number associated as above constitute what is known as joint probability distribution of (X, Y) which can be written in the tabular form also as shown below:

Table 5.2: The Joint Probability Distribution for Table 5.1

Y \ X	1	2	3	Total
0	2 / 27	6 / 27	0	8 / 27
1	0	6 / 27	6 / 27	12 / 27
2	0	6 / 27	0	6 / 27
3	1 / 27	0	0	1 / 27
Total	3 / 27	18 / 27	6 / 27	1

We are now in a position to define joint, marginal and conditional probability mass functions.

Joint Probability Mass Function

Let (X, Y) be a two-dimensional discrete random variable. With each possible outcome (x_i, y_j) , we associate a number $p(x_i, y_j)$ representing $P[X = x_i, Y = y_j]$ or $P[X = x_i \cap Y = y_j]$ and satisfying following conditions:

$$(i) \quad p(x_i, y_j) \geq 0$$

$$(ii) \quad \sum_i \sum_j p(x_i, y_j) = 1$$

The function p defined for all (x_i, y_j) is in analogy with one-dimensional case and called the **joint probability mass function of X and Y**. It is usually represented in the form of the table as shown in the Table 5.2.

Marginal Probability Function

Let (X, Y) be a discrete two-dimensional random variable which take up finite or countably infinite values (x_i, y_j) . For each such two-dimensional random variable (X, Y), we may be interested in the probability distribution of X or the probability distribution of Y, individually.

Let us again consider the example of the random placement of three balls in there cells wherein X and Y are the discrete random variables representing “the number of balls in Cell 1” and “the number of occupied cells”, respectively. Let us consider Table 5.2 showing the joint distribution of (X, Y). From this table, let us take up the row totals and column totals. The row totals in the table represent the probability distribution of X and the column totals represent the probability distribution of Y, individually. That is,

$$P[X = 0] = \frac{2}{27} + \frac{6}{27} + 0 = \frac{8}{27}$$

$$P[X = 1] = 0 + \frac{6}{27} + \frac{6}{27} = \frac{12}{27}$$

$$P[X = 2] = 0 + \frac{6}{27} + 0 = \frac{6}{27}$$

$$P[X = 3] = \frac{1}{27} + 0 + 0 = \frac{1}{27}$$

and

$$P[Y = 1] = \frac{2}{27} + 0 + 0 + \frac{1}{27} = \frac{3}{27}$$

$$P[Y = 2] = \frac{6}{27} + \frac{6}{27} + \frac{6}{27} + 0 = \frac{18}{27}$$

$$P[Y = 3] = 0 + \frac{6}{27} + 0 + 0 = \frac{6}{27}$$

These distributions of X and Y , individually, are called the **marginal probability distributions** of X and Y , respectively.

So, if (X, Y) is a discrete two-dimensional random variable which take up the values (x_i, y_j) , then the probability distribution of X is determined as follows:

$$\begin{aligned} p(x_i) &= P[X = x_i] \\ &= P[(X = x_i \cap Y = y_1) \text{ or } (X = x_i \cap Y = y_2) \text{ or } \dots] \\ &= P[X = x_i \cap Y = y_1] + P[X = x_i \cap Y = y_2] + P[X = x_i \cap Y = y_3] + \dots \\ &= \sum_j P[X = x_i \cap Y = y_j] \\ &= \sum_j p(x_i, y_j) \left[\begin{array}{l} \because p(x_i, y_j), \text{ the joint probability mass} \\ \text{function, is } P[X = x_i \cap Y = y_j] \end{array} \right] \end{aligned}$$

which is known as the marginal probability mass function of X . Similarly, the probability distribution of Y is

$$\begin{aligned} p(y_j) &= P[Y = y_j] \\ &= P[X = x_1 \cap Y = y_j] + P[X = x_2 \cap Y = y_j] + \dots \\ &= \sum_i P[X = x_i \cap Y = y_j] \\ &= \sum_i p(x_i, y_j) \end{aligned}$$

and is known as the marginal probability mass function of Y .

5.6.2 Bivariate Continuous Random Variables

Definition: If X and Y are continuous random variables defined on the sample space S of a random experiment, then (X, Y) defined on the same sample space S is called bivariate continuous random variable if (X, Y) assigns a point in xy -plane defined on the sample space S . Notice that it (unlike discrete random

variable) assumes values in some non-countable set. Some examples of bivariate continuous random variables are:

1. A gun is aimed at a certain point (say origin of the coordinate system). Because of the random factors, suppose the actual hit point is any point (X, Y) in a circle of radius unity about the origin.

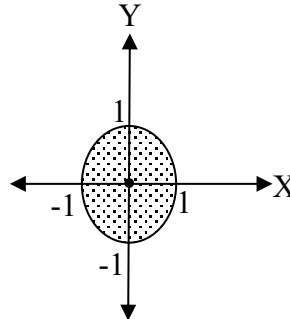


Fig. 5.2: Actual Hit Point When a Gun is Aimed at a Certain Point

Then (X, Y) assumes all the values in the circle $\{(x, y) : x^2 + y^2 \leq 1\}$ i.e. (X, Y) assumes all values corresponding to each and every point in the circular region as shown in Fig. 5.2. Here, (X, Y) is bivariate continuous random variable.

2. Assuming all values in the rectangle, represented as (X, Y) $\{(x, y) : a \leq x \leq b, c \leq y \leq d\}$ is a bivariate continuous random variable. Here, (X, Y) assumes all values corresponding to each and every point in the rectangular region as shown in Fig. 5.3.

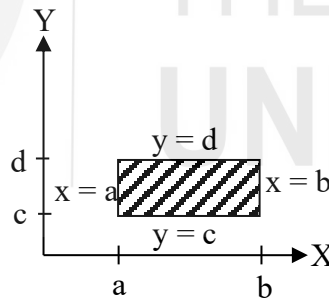


Fig. 5.3: (X, Y) Assuming All Values in the Rectangle $\{(x, y) : a \leq x \leq b, c \leq y \leq d\}$

3. In a statistical survey, let X denotes the daily number of hours a child watches television and Y denotes the number of hours he/she spends on studies. Here, (X, Y) is a two-dimensional continuous random variable.

Joint and marginal distribution and density functions for continuous variables are defined using integration formulas. A detailed discussion on these topics is beyond the scope of this Unit. You may refer to further reading for more details on these topics.

Check Your Progress 2

- 1) The joint probability distribution of a pair of random variables is given by the following table:

X \ Y	1	2	3
1	1/12	0	1/18
2	1/6	1/9	1/4
3	0	1/5	2/15

(i) Evaluate marginal distribution of X.

(ii) Obtain $P[X + Y < 5]$.

.....
.....

2) For the following joint probability distribution of (X, Y),

X \ Y	1	2	3
1	1/20	1/10	1/10
2	1/20	1/10	1/10
3	1/10	1/10	1/20
4	1/10	1/10	1/20

(i) find the probability that $Y = 2$ given that $X = 4$,

(ii) find the probability that $Y = 2$

.....
.....

3) Let the joint density function of a two-dimensional random variable (X, Y) be:

$$f(x, y) = \begin{cases} x + y & \text{for } 0 \leq x < 1 \text{ and } 0 \leq y < 1 \\ 0, & \text{otherwise} \end{cases}$$

Find the conditional density function of Y given X.

.....
.....

5.7 EXPECTATION OF A RANDOM VARIABLE

Let us now define a similar measure for the probability distribution of a random variable X, which assumes the values say $x_1, x_2, \dots, x_n$ with their associated probabilities $p_1, p_2, \dots, p_n$. This measure is known as the expected value of X and is given as

$$x_1(p_1) + x_2(p_2) + \dots + x_n(p_n) = \sum_{i=1}^n x_i p_i$$

The expected value of X is written as $E(X)$.

The above aspect can be viewed in the following way also:

Mean of a frequency distribution of X is $\frac{\sum_{i=1}^n x_i f_i}{\sum_{i=1}^n f_i}$, similarly mean of a

probability distribution of r.v. X is $\frac{\sum_{i=1}^n x_i p_i}{\sum_{i=1}^n p_i}$.

Now, as we know that $\sum_{i=1}^n p_i = 1$ for a probability distribution, therefore the

mean of the probability distribution becomes $\sum_{i=1}^n x_i p_i$.

$\therefore$ Expected value of a random variable X is $E(X) = \sum_{i=1}^n x_i p_i$.

The above formula for finding the expected value of a random variable X is used only if X is a discrete random variable which takes the values $x_1, x_2, \dots, x_n$ with probability mass function $p(x_i) = P[X = x_i]$, $i = 1, 2, \dots, n$.

But, if X is a continuous random variable having the probability density function $f(x)$, then in place of summation we will use integration and in this case, the expected value of X is defined as

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx,$$

The expectation, as defined above, agrees with the logical/theoretical argument also as is illustrated in the following example.

Suppose, a fair coin is tossed twice, then answer to the question, “How many heads do we expect theoretically/logically in two tosses?” is obviously 1 as the coin is unbiased and hence we will undoubtedly expect one head in two tosses. Expectation actually means “what we get on an average”? Now, let us obtain the expected value of the above question using the formula.

Let X be the number of heads in two tosses of the coin and we are to obtain $E(X)$, i.e. expected number of heads. As X is the number of heads in two tosses of the coin, therefore X can take the values 0, 1, 2 and its probability distribution is given as

$$\begin{array}{ccc} X: & 0 & 1 & 2 \\ p(x): & \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{array}$$

$$\therefore E(X) = \sum_{i=1}^3 x_i p_i$$

$$= x_1 p_1 + x_2 p_2 + x_3 p_3$$

$$= (0)\left(\frac{1}{4}\right) + (1)\left(\frac{1}{2}\right) + (2)\left(\frac{1}{4}\right) = 0 + \frac{1}{2} + \frac{1}{2} = 1$$

So, we get the same answer, i.e. 1 using the formula also.

So, expectation of a random variable is nothing but the average (mean) taken over all the possible values of the random variable or it is the value which we get on an average when a random experiment is performed repeatedly.

Remark 5: Sometimes summations and integrals as considered in the above definitions may not be convergent and hence expectations in such cases do not exist. But we will deal only those summations (series) and integrals which are convergent as the topic regarding checking the convergence of series or integrals is out of the scope of this course. You need not to bother as to whether the series or integral is convergent or not, i.e. as to whether the expectation exists or not as we are dealing with only those expectations which exist.

Example 7: If it rains, a raincoat dealer can earn Rs 500 per day. If it is a dry day, he can lose Rs 100 per day. What is his expectation, if the probability of rain is 0.4?

Solution: Let X be the amount earned on a day by the dealer. Therefore, X can take the values Rs 500, – Rs 100 ($\because$ loss of Rs 100 is equivalent to negative of the earning of Rs 100).

$\therefore$ Probability distribution of X is given as

	Rainy Day	Dry day
X (in Rs.):	500	–100
$p(x)$:	0.4	0.6

Hence, the expectation of the amount earned by him is

$$\begin{aligned} E(X) &= \sum_{i=1}^2 x_i p_i = x_1 p_1 + x_2 p_2 \\ &= (500)(0.4) + (-100)(0.6) = 200 - 60 = 140 \end{aligned}$$

Thus, his expectation is Rs 140, i.e. on an average he earns Rs 140 per day.

Example 8: A player tosses two unbiased coins. He wins Rs 5 if 2 heads appear, Rs 2 if one head appears and Rs 1 if no head appears. Find the expected value of the amount won by him.

Solution: In tossing two unbiased coins, the sample space, is

$$S = \{HH, HT, TH, TT\}.$$

$$\therefore P[2 \text{ heads}] = \frac{1}{4}, \quad P(\text{one head}) = \frac{2}{4}, \quad P(\text{no head}) = \frac{1}{4}.$$

Let X be the amount in rupees won by him

$\therefore X$ can take the values 5, 2 and 1 with

$$P[X = 5] = P(2 \text{ heads}) = \frac{1}{4},$$

$$P[X = 2] = P[1 \text{ Head}] = \frac{2}{4}, \text{ and}$$

$$P[X = 1] = P[\text{no Head}] = \frac{1}{4}.$$

∴ Probability distribution of X is

X:	5	2	1
p(x)	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{1}{4}$

Expected value of X is given as

$$\begin{aligned} E(X) &= \sum_{i=1}^3 x_i p_i = x_1 p_1 + x_2 p_2 + x_3 p_3 \\ &= 5\left(\frac{1}{4}\right) + 2\left(\frac{2}{4}\right) + 1\left(\frac{1}{4}\right) = \frac{5}{4} + \frac{4}{4} + \frac{1}{4} = \frac{10}{4} = 2.5. \end{aligned}$$

Thus, the expected value of amount won by him is Rs 2.5.

Example 9: Find the expectation of the number on an unbiased die when thrown.

Solution: Let X be a random variable representing the number on a die when thrown.

∴ X can take the values 1, 2, 3, 4, 5, 6 with

$$P[X = 1] = P[X = 2] = P[X = 3] = P[X = 4] = P[X = 5] = P[X = 6] = \frac{1}{6}.$$

Thus, the probability distribution of X is given by

X:	1	2	3	4	5	6
p(x):	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

Hence, the expectation of number on the die when thrown is

$$E(X) = \sum_{i=1}^6 x_i p_i = 1 \times \frac{1}{6} + 2 \times \frac{1}{6} + 3 \times \frac{1}{6} + 4 \times \frac{1}{6} + 5 \times \frac{1}{6} + 6 \times \frac{1}{6} = \frac{21}{6} = \frac{7}{2}$$

Example 10: Two cards are drawn successively with replacement from a well shuffled pack of 52 cards. Find the expected value for the number of aces.

Solution: Let A_1, A_2 be the events of getting ace in first and second draws, respectively. Let X be the number of aces drawn. Thus, X can take the values 0, 1, 2 with

$$\begin{aligned} P[X = 0] &= P[\text{no ace}] = P[\bar{A}_1 \cap \bar{A}_2] \\ &= P[\bar{A}_1] P[\bar{A}_2] \quad \left[\begin{array}{l} \because \text{cards are drawn with replacement} \\ \text{and hence the events are independent} \end{array} \right] \\ &= \frac{48}{52} \times \frac{48}{52} = \frac{12}{13} \times \frac{12}{13} = \frac{144}{169}, \end{aligned}$$

$$\begin{aligned} P[X = 1] &= [\text{one Ace and one other card}] \\ &= P[(A_1 \cap \bar{A}_2) \cup (\bar{A}_1 \cap A_2)] \end{aligned}$$

$$= P[A_1 \cap \bar{A}_2] + P[\bar{A}_1 \cap A_2] \quad \left[\begin{array}{l} \text{By Addition theorem of probability} \\ \text{for mutually exclusive events} \end{array} \right]$$

$$= P[A_1]P[\bar{A}_2] + P[\bar{A}_1]P[A_2] \quad \left[\begin{array}{l} \text{By multiplication theorem of} \\ \text{probability for independent events} \end{array} \right]$$

$$= \frac{4}{52} \times \frac{48}{52} + \frac{48}{52} \times \frac{4}{52} = \frac{1}{13} \times \frac{12}{13} + \frac{12}{13} \times \frac{1}{13} = \frac{24}{169}, \text{ and}$$

$$P[X = 2] = P[\text{both aces}] = P[A_1 \cap A_2]$$

$$= P[A_1]P[A_2] = \frac{4}{52} \times \frac{4}{52} = \frac{1}{169}.$$

Hence, the probability distribution of random variable X is

$$\begin{array}{ccc} X: & 0 & 1 & 2 \\ p(x): & \frac{144}{169} & \frac{24}{169} & \frac{1}{169} \end{array}$$

∴ The expected value of X is given by

$$E(X) = \sum_{i=1}^3 x_i p_i = 0 \times \frac{144}{169} + 1 \times \frac{24}{169} + 2 \times \frac{1}{169} = \frac{26}{169} = \frac{2}{13}$$

Example 11: For a continuous distribution whose probability density function is given by:

$$f(x) = \frac{3x}{4}(2-x), \quad 0 \leq x \leq 2, \text{ find the expected value of X.}$$

Solution: Expected value of a continuous random variable X is given by

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} xf(x)dx = \int_0^2 x \frac{3x}{4}(2-x)dx = \frac{3}{4} \int_0^2 x^2(2-x)dx \\ &= \frac{3}{4} \int_0^2 (2x^2 - x^3)dx = \frac{3}{4} \left[2 \frac{x^3}{3} - \frac{x^4}{4} \right]_0^2 = \frac{3}{4} \left[2 \frac{(2)^3}{3} - \frac{(2)^4}{4} - 0 \right] \\ &= \frac{3}{4} \left[\frac{16}{3} - \frac{16}{4} \right] = \frac{3}{4} \times \frac{16}{12} = 1 \end{aligned}$$

Let us now discuss some properties of expectation in the next section.

Properties of Expectation of One-Dimensional Random Variable

Properties of mathematical expectation of a random variable X are given below without proof. You can refer to references for the proof:

1. $E(k) = k$, where k is a constant
2. $E(kX) = kE(X)$, k being a constant.
3. $E(aX + b) = aE(X) + b$, where a and b are constants

Example 12: Given the following probability distribution:

X	-2	-1	0	1	2
p(x)	0.15	0.30	0	0.30	0.25

- Find
- i) $E(X)$
 - ii) $E(2X + 3)$
 - iii) $E(X^2)$
 - iv) $E(4X - 5)$

Solution

$$\begin{aligned} \text{i) } E(X) &= \sum_{i=1}^5 x_i p_i = x_1 p_1 + x_2 p_2 + x_3 p_3 + x_4 p_4 + x_5 p_5 \\ &= (-2)(0.15) + (-1)(0.30) + (0)(0) + (1)(0.30) + (2)(0.25) \\ &= -0.3 - 0.3 + 0 + 0.3 + 0.5 = 0.2 \end{aligned}$$

$$\begin{aligned} \text{ii) } E(2X + 3) &= 2E(X) + 3 && [\text{Using property 3 of Expectation Sec 5.7}] \\ &= 2(0.2) + 3 && [\text{Using solution (i) of the question}] \\ &= 0.4 + 3 = 3.4 \end{aligned}$$

$$\begin{aligned} \text{iii) } E(X^2) &= \sum_{i=1}^5 x_i^2 p_i && [\text{By def.}] \\ &= x_1^2 p_1 + x_2^2 p_2 + x_3^2 p_3 + x_4^2 p_4 + x_5^2 p_5 \\ &= (-2)^2 (0.15) + (-1)^2 (0.30) + (0)^2 (0) + (1)^2 (0.30) + (2)^2 (0.25) \\ &= (4)(0.15) + (1)(0.30) + (0) + (1)(0.30) + (4)(0.25) \\ &= 0.6 + 0.3 + 0 + 0.3 + 1 = 2.2 \end{aligned}$$

$$\begin{aligned} \text{iv) } E(4X - 5) &= E[4X + (-5)] \\ &= 4E(X) + (-5) && [\text{Using property 3 of Expectation Sec 5.7}] \\ &= 4(0.2) - 5 \\ &= 0.8 - 5 = -4.2 \end{aligned}$$

Check Your Progress 3

- 1) You toss a fair coin. If the outcome is head, you win Rs 100; if the outcome is tail, you win nothing. What is the expected amount won by you?

.....
.....

- 2) A fair coin is tossed until a tail appears. What is the expectation of number of tosses?

.....
.....

- 3) The distribution of a continuous random variable X is defined by

$$f(x) = \begin{cases} x^3 & , 0 < x \leq 1 \\ (2-x)^3 & , 1 < x \leq 2 \\ 0 & , \text{ elsewhere} \end{cases}$$

Obtain the expected value of X.

.....
.....

- 4) If X is a random variable with mean 'μ' and standard deviation 'σ', then what is the expectation of $Z = \frac{X - \mu}{\sigma}$?

[Note: Here Z so defined is called standard random variate.]

.....
.....

5.8 MOMENTS AND OTHER MEASURES IN TERMS OF EXPECTATIONS

In this section, we present basic definitions of the moments and other measures for a random variable in terms of expectations in the following section

5.8.1 Moments

The moments for probability distributions for the r^{th} order moment about any point 'A' of a random variable X having probability mass function $P[X = x_i] = p(x_i) = p_i$ is defined as

$$\begin{aligned}\mu_r' &= \frac{\sum_{i=1}^n p_i (x_i - A)^r}{\sum_{i=1}^n p_i} \\ &= \sum_{i=1}^n p_i (x_i - A)^r \quad \left[\because \sum_{i=1}^n p_i = 1 \right]\end{aligned}$$

The above formula is valid if X is a discrete random variable. But, if X is a continuous random variable having probability density function $f(x)$, then

r^{th} order moment about A is defined as $\mu_r' = \int_{-\infty}^{\infty} (x - A)^r f(x) dx$.

So, r^{th} order moment about any point 'A' of a random variable X is defined as

$$\begin{aligned}\mu_r' &= \begin{cases} \sum_i p_i (x_i - A)^r, & \text{if X is a discrete r.v.} \\ \int_{-\infty}^{\infty} (x - A)^r f(x) dx, & \text{if X is a continuous r.v.} \end{cases} \\ &= E(X - A)^r\end{aligned}$$

Similarly, r^{th} order moment about mean (μ) i.e. r^{th} order central moment is defined as

$$\mu_r = \begin{cases} \sum_i p_i (x_i - \mu)^r, & \text{if } X \text{ is a discrete r.v.} \\ \int_{-\infty}^{\infty} (x - \mu)^r f(x) dx, & \text{if } X \text{ is a continuous r.v.} \end{cases}$$

$$= E(X - \mu)^r = E[X - E(X)]^r$$

5.8.2 Variance and Covariance

Variance of a random variable X is second order central moment and is defined as

$$\mu_2 = V(X) = E[X - \mu]^2 = E[X - E(X)]^2$$

Also, we know that

$$V(X) = \mu_2' - (\mu_1')^2$$

where μ_1' , μ_2' be the moments about origin.

$$\therefore \text{We have } V(X) = E(X^2) - [E(X)]^2$$

$$\left[\because \mu_1' = E[X - 0]^1 = E(X), \text{ and } \mu_2' = E[X - 0]^2 = E(X^2) \right]$$

Covariance

For a bivariate frequency distribution, you have already studied in Unit 6 of MST-002 that covariance between two variables X and Y is defined as

$$\text{Cov}(X, Y) = \frac{\sum_i f_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i f_i}$$

$\therefore$ For a bivariate probability distribution, $\text{Cov}(X, Y)$ is defined as

$$\text{Cov}(X, Y) = \begin{cases} \sum_i p_{ij} (x_i - \bar{x})(y_j - \bar{y}), & \text{if } (X, Y) \text{ is two-dimensional discrete r.v.} \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \bar{x})(y - \bar{y}) f(x, y) dy dx, & \text{if } (X, Y) \text{ is two dimensional continuous r.v.} \end{cases}$$

$$\text{where } p_{ij} = P[X = x_i, Y = y_j]$$

$$= E(X - \bar{X})(Y - \bar{Y}) \quad [\text{By definition of expectation}]$$

$$= E[(X - E(X))(Y - E(Y))] \quad \left[\begin{array}{l} \because E(X) = \text{Mean of } X \text{ i.e. } \bar{X}, \\ E(Y) = \text{Mean of } Y \text{ i.e. } \bar{Y} \end{array} \right]$$

On simplifying,

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y).$$

Now, if X and Y are independent random variables then, by multiplication theorem,

$$E(XY) = E(X)E(Y) \text{ and hence in this case } \text{Cov}(X, Y) = 0.$$

Remark 6:

i) If X and Y are independent random variables, then

$$V(X + Y) = V(X) + V(Y).$$

ii) If X and Y are independent random variables, then

$$V(X - Y) = V(X) + V(Y).$$

iii) If X and Y are independent random variables, then

$$V(aX + bY) = a^2 V(X) + b^2 V(Y).$$

Mean Deviation about Mean

Mean deviation about mean in context of probability distribution is

$$\frac{\sum_{i=1}^n p_i |x_i - \text{mean}|}{\sum_{i=1}^n p_i} = \sum_{i=1}^n p_i |x_i - \text{mean}|$$

$\therefore$ by definition of expectation, we have

$$\text{M.D. about mean} = E|X - \text{Mean}|$$

$$= E|X - E(X)|$$

$$= \begin{cases} \sum p_i |x - \text{Mean}| & \text{for discrete r.v} \\ \int_{-\infty}^{\infty} |x - \text{Mean}| f(x) dx & \text{for continuous r.v} \end{cases}$$

Example 13: Considering the probability distribution given in Example 12, obtain

i) $V(X)$

ii) $V(2X + 3)$.

Solution:

$$(i) \quad V(X) = E(X^2) - [E(X)]^2$$

$$= 2.2 - (0.2)^2 \quad \left[\begin{array}{l} \text{The values have already been obtained} \\ \text{in the solution of Example 6} \end{array} \right]$$

$$= 2.2 - 0.04 = 2.16$$

$$(ii) \quad V(2X + 3) = (2)^2 V(X) \quad [\text{Using the result of Theorem 8.1}]$$

$$= 4V(X) = 4(2.16) = 8.64$$

Example 14: If X and Y are independent random variables with variances 2 and 3, respectively, find the variance of $3X + 4Y$.

$$\text{Solution: } V(3X + 4Y) = (3)^2 V(X) + (4)^2 V(Y)$$

$$= 9(2) + 16(3) = 66$$

5.8.3 Addition and Multiplication Theorems of Expectation

Two important properties, i.e., the addition and multiplication laws of expectation, are stated next (without withproof).

Addition Theorem of Expectation

Theorem 8.1: If X and Y are random variables, then $E(X + Y) = E(X) + E(Y)$

Remark 7: The result can be similarly extended for more than two random variables.

Multiplication Theorem of Expectation

Theorem 8.2: If X and Y are independent random variables, then

$$E(XY) = E(X) E(Y)$$

Remark 8: The result can be similarly extended for more than two random variables.

Example 15: Two unbiased dice are thrown. Find the expected value of the sum of number of points on them.

Solution: Let X be the number obtained on the first die and Y be the number obtained on the second die, then

$$E(X) = \frac{7}{2} \text{ and } E(Y) = \frac{7}{2}$$

$\therefore$ The required expected value $= E(X + Y)$

$$= E(X) + E(Y) \left[\begin{array}{l} \text{Using addition theorem} \\ \text{of expectation} \end{array} \right]$$

$$= \frac{7}{2} + \frac{7}{2} = 7$$

Remark 9: This example can also be done considering one random variable only as follows:

Let X be the random variable denoting “the sum of numbers of points on the dice”, then the probability distribution in this case is

$X:$	2	3	4	5	6	7	8	9	10	11	12
$p(x):$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

$$\text{and hence } E(X) = 2 \times \frac{1}{36} + 3 \times \frac{2}{36} + \dots + 12 \times \frac{1}{36} = 7$$

Example 16: Two cards are drawn one by one with replacement from 8 cards numbered from 1 to 8. Find the expectation of the product of the numbers on the drawn cards.

Solution: Let X be the number on the first card and Y be the number on the second card. Then probability distribution of X is

X	1	2	3	4	5	6	7	8
$p(x)$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$

and the probability distribution of Y is

Y	1	2	3	4	5	6	7	8
p(y)	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$

$$\begin{aligned}\therefore E(X) &= E(Y) = 1 \times \frac{1}{8} + 2 \times \frac{1}{8} + \dots + 8 \times \frac{1}{8} \\ &= \frac{1}{8}(1 + 2 + 3 + 4 + 5 + 6 + 7 + 8) = \frac{1}{8}(36) = \frac{9}{2}\end{aligned}$$

Thus, the required expected value is

$$E(XY) = E(X)E(Y) \quad [\text{Using multiplication theorem of expectation}]$$

$$= \frac{9}{2} \times \frac{9}{2} = \frac{81}{4}.$$

Check Your Progress 4

- 1) If X is a random variable with mean μ and standard deviation σ , then find the variance of standard random variable $Z = \frac{X - \mu}{\sigma}$.

.....
.....

- 2) Suppose that X is a random variable for which $E(X) = 10$ and $V(X) = 25$. Find the positive values of a and b such that $Y = aX - b$ has expectation.

.....
.....

5.9 SUMMARY

Following main points have been covered in this unit of the course:

- 1) A **random variable** is a function whose domain is a set of possible outcomes and range is a sub-set of the set of reals and has the following properties:
 - i) Each particular value of the random variable can be assigned some probability.
 - ii) Sum of all the probabilities associated with all the different values of the random variable is unity.
- 2) A random variable is said to be **discrete random variable** if it has either a finite number of values or a countable number of values, i.e. the values can be arranged in a sequence.
- 3) If a random variable is such that its values cannot be arranged in a sequence, it is called **continuous random variable**. So, a random variable is said to be continuous if it can take all the possible real (i.e. integer as well as fractional) values between two certain limits.

- 4) It discussed the concept of **probability mass function** and defined the **probability distribution** of discrete random variable and **probability density function** of a continuous random variable
- 5) The unit also defined the **distribution function** also known as the **cumulative distribution function (c.d.f.)** of a random variable. It explains **discrete distribution function** and **continuous distribution function**.
- 6) The unit briefly presents concepts of **two-dimensional discrete random variable** and **bivariate continuous random variable**.
- 7) The unit defined the Expected value of a random variable and its properties.
- 8) It defines Moments, variance and covariance in terms of expectation.
- 9) The unit has presented many related example and questions. If you want to see what our solutions to the check your progress questions, we have given them in the following section.

5.10 SOLUTIONS/ANSWERS

Check Your Progress 1

- 1) Let X be the number of bad articles drawn.

$\therefore$ X can take the values 0, 1, 2 with

$$P[X = 0] = P[\text{No bad article}]$$

= P[Drawing 2 articles from 5 good articles and zero article
from 2 bad articles]

$$= \frac{{}^5C_2 \times {}^2C_0}{{}^7C_2} = \frac{5 \times 4 \times 1}{7 \times 6} = \frac{10}{21},$$

$$P[X = 1] = P[\text{One bad article and 1 good article}]$$

$$= \frac{{}^2C_1 \times {}^5C_1}{{}^7C_2} = \frac{2 \times 5 \times 2}{7 \times 6} = \frac{10}{21},$$

And

$$P[X = 2] = P[\text{Two bad articles and no good article}]$$

$$= \frac{{}^2C_2 \times {}^5C_0}{{}^7C_2} = \frac{1 \times 1 \times 2}{7 \times 6} = \frac{1}{21}$$

$\therefore$ Probability distribution of number of bad articles is:

X	0	1	2
p(x)	$\frac{10}{21}$	$\frac{10}{21}$	$\frac{1}{21}$

2) As $Y = X^2 + 2X$,

$\therefore$ For $X = 0$, $Y = 0 + 0 = 0$;

For $X = 1$, $Y = 1^2 + 2(1) = 3$;

For $X = 2$, $Y = 2^2 + 2(2) = 8$; and

For $X = 3$, $Y = 3^2 + 2(3) = 15$.

Thus, the values of Y are 0, 3, 8, 15 corresponding to the values 0, 1, 2, 3 of X and hence

$$P[Y = 0] = P[X = 0] = \frac{1}{10}, P[Y = 3] = P[X = 1] = \frac{3}{10},$$

$$P[Y = 8] = P[X = 2] = \frac{1}{2} \text{ and } P[Y = 15] = P[X = 3] = \frac{1}{10}.$$

$\therefore$ The probability distribution of Y is

Y	0	3	8	15
p(y)	$\frac{1}{10}$	$\frac{3}{10}$	$\frac{1}{2}$	$\frac{1}{10}$

3) Let X be the number of red balls drawn.

$\therefore$ X can take the values 0, 1, 2, 3.

Let W_i be the event that i^{th} draw gives white ball and R_i be the event that i^{th} draw gives red ball.

$$\begin{aligned} \therefore P[X = 0] &= P[\text{No Red ball}] = P[W_1 \cap W_2 \cap W_3] \\ &= P(W_1) \cdot P(W_2) \cdot P(W_3) \end{aligned}$$

[$\because$ balls are drawn with replacement and hence the draws are independent]

$$= \frac{3}{7} \times \frac{3}{7} \times \frac{3}{7} = \frac{27}{343}$$

$$P[X = 1] = P[\text{One red and two white}]$$

$$= P[(R_1 \cap W_2 \cap W_3) \text{ or } (W_1 \cap R_2 \cap W_3) \text{ or } (W_1 \cap W_2 \cap R_3)]$$

$$= P[R_1 \cap W_2 \cap W_3] + P[W_1 \cap R_2 \cap W_3] + P[W_1 \cap W_2 \cap R_3]$$

$$= P[R_1]P[W_2]P[W_3] + P[W_1]P[R_2]P[W_3] + P[W_1]P[W_2]P[R_3]$$

$$= \frac{4}{7} \times \frac{3}{7} \times \frac{3}{7} + \frac{3}{7} \times \frac{4}{7} \times \frac{3}{7} + \frac{3}{7} \times \frac{3}{7} \times \frac{4}{7} = 3 \times \frac{4}{7} \times \frac{3}{7} \times \frac{3}{7} = \frac{108}{343},$$

$$P[X = 2] = P[\text{Two red and one white}]$$

$$= P[(R_1 \cap R_2 \cap W_3) \text{ or } (R_1 \cap W_2 \cap R_3) \text{ or } (W_1 \cap R_2 \cap R_3)]$$

$$= P[R_1]P[R_2]P[W_3] + P[R_1]P[W_2]P[R_3] + P[W_1]P[R_2]P[R_3]$$

$$= \frac{4}{7} \times \frac{4}{7} \times \frac{3}{7} + \frac{4}{7} \times \frac{3}{7} \times \frac{4}{7} + \frac{3}{7} \times \frac{4}{7} \times \frac{4}{7} = 3 \times \frac{4}{7} \times \frac{4}{7} \times \frac{3}{7} = \frac{144}{343}.$$

$$P[X = 3] = P[\text{Three red balls}]$$

$$= P[R_1 \cap R_2 \cap R_3] = P(R_1) P(R_2) P(R_3) = \frac{4}{7} \times \frac{4}{7} \times \frac{4}{7} = \frac{64}{343}.$$

$\therefore$ Probability distribution of the number of red balls is

X	0	1	2	3
p(x)	$\frac{27}{343}$	$\frac{108}{343}$	$\frac{144}{343}$	$\frac{64}{343}$

4) As $f(x)$ is p.d.f.,

$$\begin{aligned} \therefore \int_{1500}^{2500} \frac{A}{x^3} dx &= 1 \Rightarrow A \int_{1500}^{2500} x^{-3} dx = 1 \Rightarrow A \left[\frac{x^{-2}}{-2} \right]_{1500}^{2500} = 1 \\ &\Rightarrow -\frac{A}{2} \left[\frac{1}{(2500)^2} - \frac{1}{(1500)^2} \right] = 1 \Rightarrow -\frac{A}{20000} \left[\frac{1}{625} - \frac{1}{225} \right] = 1 \\ &\Rightarrow -\frac{A}{20000} \left[\frac{9-25}{5625} \right] = 1 \Rightarrow 16A = 5625 \times 20000 \\ &\Rightarrow A = \frac{5625 \times 20000}{16} = 5625 \times 1250 = 7031250. \end{aligned}$$

$$\begin{aligned} \text{Now, } P[1600 \leq X \leq 2000] &= \int_{1600}^{2000} f(x) dx = A \int_{1600}^{2000} \frac{1}{x^3} dx \\ &= -\frac{A}{2} \left[\frac{1}{x^2} \right]_{1600}^{2000} = -\frac{A}{2} \left[\frac{1}{(2000)^2} - \frac{1}{(1600)^2} \right] \\ &= -\frac{A}{20000} \left[\frac{1}{400} - \frac{1}{256} \right] = -\frac{A}{20000} \left[\frac{16-25}{6400} \right] \\ &= \frac{9 \times 7031250}{20000 \times 6400} = \frac{2025}{4096} \end{aligned}$$

5) i) As the given distribution is probability distribution,

$\therefore$ sum of all the probabilities = 1

$$\Rightarrow k + 3k + 5k + 7k + 9k + 11k + 13k + 15k + 17k = 1$$

$$\Rightarrow 81k = 1 \Rightarrow k = \frac{1}{81}$$

The distribution function of X is given in the following table:

X	F(x) = P[X ≤ x]
0	$k = \frac{1}{81}$
1	$k + 3k = 4k = \frac{4}{81}$
2	$4k + 5k = 9k = \frac{9}{81}$
3	$9k + 7k = 16k = \frac{16}{81}$
4	$16k + 9k = 25k = \frac{25}{81}$
5	$25k + 11k = 36k = \frac{36}{81}$
6	$36k + 13k = 49k = \frac{49}{81}$
7	$49k + 15k = 64k = \frac{64}{81}$
8	$64k + 17k = 81k = 1$

6) For $x < 0$,

$$F(x) = P[X \leq x] = \int_{-\infty}^x f(x) dx = \int_{-\infty}^x 0 dx = 0 \quad [\because f(x) = 0 \text{ for } x < 0].$$

For $0 \leq x < 1$,

$$\begin{aligned} F(x) = P[X \leq x] &= \int_{-\infty}^x f(x) dx = \int_{-\infty}^0 f(x) dx + \int_0^x f(x) dx \quad [\because 0 \leq x < 1] \\ &= 0 + \int_0^x \frac{x}{2} dx \quad \left[\because f(x) = \frac{x}{2} \text{ for } 0 \leq x < 1 \right] \\ &= \frac{1}{2} \left[\frac{x^2}{2} \right]_0^x = \frac{x^2}{4}. \end{aligned}$$

For $1 \leq x < 2$,

$$\begin{aligned} F(x) = P[X \leq x] &= \int_{-\infty}^x f(x) dx \\ &= \int_{-\infty}^0 f(x) dx + \int_0^1 f(x) dx + \int_1^x f(x) dx \\ &= 0 + \int_0^1 \frac{x}{2} dx + \int_1^x \frac{1}{2} dx = \frac{1}{4} [x^2]_0^1 + \frac{1}{2} [x]_1^x \end{aligned}$$

$$= \frac{1}{4} + \frac{x}{2} - \frac{1}{2} = \frac{1}{4}(2x-1)$$

For $2 \leq x < 3$,

$$\begin{aligned} F(x) &= \int_{-\infty}^x f(x) dx \\ &= \int_{-\infty}^0 f(x) dx + \int_0^1 f(x) dx + \int_1^2 f(x) dx + \int_2^x f(x) dx \\ &= 0 + \int_0^1 \frac{x}{2} dx + \int_1^2 \frac{1}{2} dx + \int_2^x \frac{1}{2}(3-x) dx \\ &= \left[\frac{x^2}{4} \right]_0^1 + \frac{1}{2} [x]_1^2 + \frac{1}{2} \left[3x - \frac{x^2}{2} \right]_2^x \\ &= \frac{1}{4} + \frac{1}{2} + \frac{1}{2} \left[\left(3x - \frac{x^2}{2} \right) - (6-2) \right] \\ &= \frac{1}{2} \left(3x - \frac{x^2}{2} \right) - \frac{5}{4} = -\frac{x^2}{4} + \frac{3x}{2} - \frac{5}{4} \end{aligned}$$

For $3 \leq x < \infty$,

$$\begin{aligned} F(x) &= \int_{-\infty}^x f(x) dx \\ &= \int_{-\infty}^0 f(x) dx + \int_0^1 f(x) dx + \int_1^2 f(x) dx + \int_2^3 f(x) dx + \int_3^x f(x) dx \\ &= 0 + \int_0^1 \frac{x}{2} dx + \int_1^2 \frac{1}{2} dx + \int_2^3 \frac{1}{2}(3-x) dx + \int_3^x 0 dx \\ &= \left[\frac{x^2}{4} \right]_0^1 + \left[\frac{x}{2} \right]_1^2 + \frac{1}{2} \left[3x - \frac{x^2}{2} \right]_2^3 + 0 \\ &= \frac{1}{4} + \left(1 - \frac{1}{2} \right) + \frac{1}{2} \left[\left(9 - \frac{9}{2} \right) - (6-2) \right] \\ &= \frac{1}{4} + \frac{1}{2} + \frac{1}{2} \left(\frac{9}{2} - 4 \right) \\ &= \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1 \end{aligned}$$

Hence, the distribution function is given by:

$$F(x) = \begin{cases} 0, & -\infty < x < 0 \\ \frac{x^2}{4}, & 0 \leq x < 1 \\ \frac{2x-1}{4}, & 1 \leq x < 2 \\ -\frac{x^2}{4} + \frac{3x}{2} - \frac{5}{4}, & 2 \leq x < 3 \\ 1, & 3 \leq x < \infty \end{cases}$$

Check Your Progress 2

- 1) Let us compute the marginal totals. Thus, the complete table with marginal totals is given as

X \ Y	1	2	3	p(x)
1	$\frac{1}{12}$	0	$\frac{1}{18}$	$\frac{1}{12} + 0 + \frac{1}{18} = \frac{5}{36}$
2	$\frac{1}{6}$	$\frac{1}{9}$	$\frac{1}{4}$	$\frac{1}{6} + \frac{1}{9} + \frac{1}{4} = \frac{19}{36}$
3	0	$\frac{1}{5}$	$\frac{2}{15}$	$0 + \frac{1}{5} + \frac{2}{15} = \frac{1}{3}$
p(y)	$\frac{1}{4}$	$\frac{14}{45}$	$\frac{79}{180}$	1

Therefore,

- i) Marginal distribution of X is

X	p(x)
1	5/36
2	19/36
3	1/3

- ii) $P[X + Y < 5]$

$$\begin{aligned}
 &= P[X = 1, Y = 1] + P[X = 1, Y = 2] + P[X = 1, Y = 3] \\
 &\quad + P[X = 2, Y = 1] + P[X = 2, Y = 2] + P[X = 3, Y = 1] \\
 &= \frac{1}{12} + 0 + \frac{1}{18} + \frac{1}{6} + \frac{1}{9} + 0 = \frac{15}{36}.
 \end{aligned}$$

2) First compute the marginal totals, then you will be able to find

$$\text{i) } P[X = 4] = \frac{1}{4}, \text{ and hence}$$

$$P[Y = 2 | X = 4] = \frac{P[Y = 2, X = 4]}{P[X = 4]} = \frac{2}{5}$$

$$\text{ii) } P[Y = 2] = \frac{2}{5}$$

3) The conditional density function of Y given X is $f(y|x) = \frac{f(x,y)}{f(x)}$,

where $f(x,y)$ is the joint density function, which is given; and $f(x)$ is the marginal density function which, by definition, is given by

$$\begin{aligned} f(x) &= \int_{-\infty}^{\infty} f(x,y) dy \\ &= \int_0^1 f(x,y) dy \quad [\because 0 \leq y < 1] \\ &= \int_0^1 (x+y) dy \\ &= \left[xy + \frac{y^2}{2} \right]_0^1 \\ &= \left[x(1) + \frac{(1)^2}{2} - 0 \right] = x + \frac{1}{2}, \quad 0 \leq x < 1. \end{aligned}$$

$\therefore$ the conditional density function of Y given X is

$$f(y|x) = \frac{f(x,y)}{f(x)} = \frac{x+y}{x+\frac{1}{2}}, \text{ for } 0 \leq x < 1 \text{ and } 0 \leq y < 1.$$

Check Your Progress 3

1) Let X be the amount (in rupees) won by you.

$\therefore$ X can take the values 100, 0 with $P[X = 100] = P[\text{Head}] = \frac{1}{2}$, and

$$P[X = 0] = P[\text{Tail}] = \frac{1}{2}.$$

$\therefore$ probability distribution of X is

X:	100	0
p(x)	$\frac{1}{2}$	$\frac{1}{2}$

and hence the expected amount won by you is

$$E(X) = 100 \times \frac{1}{2} + 0 \times \frac{1}{2} = 50.$$

2) Let X be the number of tosses till tail turns up.

$\therefore$ X can take values 1, 2, 3, 4... with

$$P[X = 1] = P[\text{Tail in the first toss}] = \frac{1}{2}$$

$$P[X = 2] = P[\text{Head in the first and tail in the second toss}] = \frac{1}{2} \times \frac{1}{2} = \left(\frac{1}{2}\right)^2,$$

$$P[X = 3] = P[\text{HHT}] = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \left(\frac{1}{2}\right)^3, \text{ and so on.}$$

$\therefore$ Probability distribution of X is

X:	1	2	3	4	5...
p(x)	$\frac{1}{2}$	$\left(\frac{1}{2}\right)^2$	$\left(\frac{1}{2}\right)^3$	$\left(\frac{1}{2}\right)^4$	$\left(\frac{1}{2}\right)^5 \dots$

and hence

$$E(X) = 1 \times \frac{1}{2} + 2 \times \left(\frac{1}{2}\right)^2 + 3 \times \left(\frac{1}{2}\right)^3 + 4 \times \left(\frac{1}{2}\right)^4 + \dots \quad \dots (1)$$

Multiplying both sides by $\frac{1}{2}$, we get

$$\begin{aligned} \frac{1}{2}E(X) &= \left(\frac{1}{2}\right)^2 + 2 \times \left(\frac{1}{2}\right)^3 + 3 \times \left(\frac{1}{2}\right)^4 + 4 \times \left(\frac{1}{2}\right)^5 + \dots \\ \Rightarrow \frac{1}{2}E(X) &= \left(\frac{1}{2}\right)^2 + 2 \times \left(\frac{1}{2}\right)^3 + 3 \times \left(\frac{1}{2}\right)^4 + \dots \quad \dots (2) \end{aligned}$$

[Shifting the position one step towards right so that we get the terms having same power at the same positions as that in (1)]

Now, subtracting (2) from (1), we have

$$\begin{aligned} E(X) - \frac{1}{2}E(X) &= \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^4 + \dots \\ \Rightarrow \frac{1}{2}E(X) &= \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^4 + \dots \\ \Rightarrow E(X) &= 1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \dots \end{aligned}$$

(Which is an infinite G.P. with first term $a = 1$ and
common ratio $r = \frac{1}{2}$)

$$= \frac{1}{1 - \frac{1}{2}} \quad [\because S_{\infty} = \frac{a}{1-r} \text{ (see Unit 3 of course MST - 001)}]$$

$$= \frac{1}{\frac{1}{2}} = 2.$$

$$\begin{aligned} 3) \quad E(X) &= \int_{-\infty}^{\infty} x f(x) dx \\ &= \int_{-\infty}^0 x f(x) dx + \int_0^1 x f(x) dx + \int_1^2 x f(x) dx + \int_2^{\infty} x f(x) dx \\ &= \int_{-\infty}^0 x(0) dx + \int_0^1 x(x^3) dx + \int_1^2 x(2-x)^3 dx + \int_2^{\infty} x(0) dx \\ &= 0 + \int_0^1 x^4 dx + \int_1^2 x[8 - x^3 - 6x(2-x)] dx + 0 \\ &= \int_0^1 x^4 dx + \int_1^2 (8x - x^4 - 12x^2 + 6x^3) dx \\ &= \left[\frac{x^5}{5} \right]_0^1 + \left[8\frac{x^2}{2} - \frac{x^5}{5} - 12\frac{x^3}{3} + 6\frac{x^4}{4} \right]_1^2 \\ &= \frac{1}{5} + \left[\left\{ \frac{8(2)^2}{2} - \frac{(2)^5}{5} - \frac{12(2)^3}{3} + \frac{6(2)^4}{4} \right\} - \left\{ \frac{8(1)^2}{2} - \frac{(1)^5}{5} - \frac{12(1)^3}{3} + \frac{6(1)^4}{4} \right\} \right] \\ &= \frac{1}{5} + \left[\left\{ 16 - \frac{32}{5} - 32 + 24 \right\} - \left\{ 4 - \frac{1}{5} - 4 + \frac{3}{2} \right\} \right] \\ &= \frac{1}{5} + \left[\frac{8}{5} - \frac{13}{10} \right] = \frac{1}{5} + \frac{3}{10} = \frac{1}{2}. \end{aligned}$$

4) As X is a random variable with mean μ ,

$$\therefore E(X) = \mu \quad \dots (1)$$

$$\text{Now, } E(Z) = E\left(\frac{X - \mu}{\sigma}\right)$$

$$= E\left[\frac{1}{\sigma}(X - \mu)\right]$$

$$= \frac{1}{\sigma} E[X - \mu] \quad [\text{Using Property 2 of Expectation Sec 5.7}]$$

$$\begin{aligned}
 &= \frac{1}{\sigma} [E(X) - \mu] \quad [\text{Using Property 3 of Expectation Sec 5.7}] \\
 &= \frac{1}{\sigma} [\mu - \mu] \quad [\text{Using Property 1 of Expectation Sec 5.7}] \\
 &= 0
 \end{aligned}$$

Note: Mean of standard random variable is zero.

Check Your Progress 4

1) Variance of standard random variable $Z = \frac{X - \mu}{\sigma}$ is given as

$$\begin{aligned}
 V(Z) &= V\left(\frac{X - \mu}{\sigma}\right) = V\left(\frac{X}{\sigma} - \frac{\mu}{\sigma}\right) \\
 &= V\left[\frac{1}{\sigma}X + \left(-\frac{\mu}{\sigma}\right)\right] \\
 &= \left(\frac{1}{\sigma}\right)^2 V(X) \\
 &= \frac{1}{\sigma^2} V(X) \\
 &= \frac{1}{\sigma^2} (\sigma^2) = 1 \quad \left[\because \text{it is given that the standard deviation of } X \text{ is } \sigma \text{ and hence its variance is } \sigma^2 \right]
 \end{aligned}$$

Note: The mean of standard random variate is '0' and its variance is 1.

2) Given that $E(Y) = 0 \Rightarrow E(aX - b) = 0 \Rightarrow aE(X) - b = 0$

$$\Rightarrow a(10) - b = 0$$

$$\Rightarrow 10a - b = 0 \quad \dots (1)$$

Also as $V(Y) = 1$,

hence $V(aX - b) = 1$

$$\Rightarrow a^2 V(X) = 1 \Rightarrow a^2 (25) = 1 \Rightarrow a^2 = \frac{1}{25}$$

$$\Rightarrow a = \frac{1}{5} \quad [\because a \text{ is positive}]$$

$\therefore$ From (1), we have

$$10\left(\frac{1}{5}\right) - b = 0 \Rightarrow 2 - b = 0 \Rightarrow b = 2$$

Hence, $a = \frac{1}{5}$, $b = 2$.

5.11 REFERENCES

IGNOU Course Material

1. Post-Graduate Diploma In Applied Statistics (PGDAST), MST 003: Probability Theory, Block 2: Random Variables and Expectation, UNIT 5: Random Variables (for Sections 5.2 to 5.5 and related portions in Sections 5.0 , 5.1, 5.9, 5.10 and Check Your Progress questions)
2. PGDAST, MST 003: Probability Theory, Block 2: Random Variables and Expectation, UNIT 6: Bivariate Discrete Random Variables (for Section 5.6 and related portions in Sections 5.0 , 5.1, 5.9, 5.10 and Check Your Progress questions)
3. PGDAST, MST 003: Probability Theory, Block 2: Random Variables and Expectation, UNIT 7: Bivariate Continuous Random Variables (for Section 5.6 and related portions in Sections 5.0 , 5.1, 5.9, 5.10 and Check Your Progress questions)
4. PGDAST, MST 003: Probability Theory, Block 2: Random Variables and Expectation, UNIT 8: Mathematical Expectation (for Section 5.7, 5.8 and related portions in Sections 5.0 , 5.1, 5.9, 5.10 and Check Your Progress questions)

Reference Book

5. “Introduction to Probability and Statistics for Engineers and Scientists”, 3rd Edition, Sheldon M. Ross, Elsevier Academic Press, 2004

UNIT 6 DISCRETE PROBABILITY DISTRIBUTIONS

Structure

- 6.0 Introduction
- 6.1 Objectives
- 6.2 Bernoulli Distribution and its Properties
- 6.3 Binomial Probability Function
- 6.4 Poisson Distribution
- 6.5 Discrete Uniform Distribution
- 6.6 Hypergeometric Distribution
- 6.7 Summary
- 6.8 Solutions/Answers
- 6.9 References
- Appendix 1

6.0 INTRODUCTION

In Unit 5 of the Course, you have studied random variables, their probability functions and distribution functions. You have come to know how the expectations of random variables are obtained. Unit 5 also discussed the definitions and properties of general discrete and probability distributions.

The present unit is devoted to the study of some special discrete distributions, including Bernoulli, Binomial, Poisson, Uniform and Hypergeometric distributions. First, this unit defines the Bernoulli distribution and its properties.

Binomial distribution and its applications are covered next, which is applied in the cases where the probability of success and failure do not differ much from each other and the number of trials in a random experiment is finite. However, there may be practical situations where the probability of success is very small, that is, there may be situations where the event occurs rarely and the number of trials may not be known. For instance, the number of accidents occurring at a particular spot on a road everyday is a rare event. For such rare events, we cannot apply the binomial distribution. To these situations, we apply Poisson distribution. The concept of Poisson distribution was developed by a French mathematician, Simeon Denis Poisson (1781-1840) in the year 1837. Poisson distribution is also discussed in this unit.

Discrete uniform distribution is applicable to those experiments where the different values of random variable are equally likely. If the population is finite and the sampling is done without replacement i.e. if the events are random but not independent, then we use Hypergeometric distribution. This unit discusses discrete uniform distribution and hypergeometric distribution.

6.1 OBJECTIVES

Study of the present unit will enable you to:

- define the Bernoulli distribution and to establish its properties;
- define the binomial distribution and establish its properties;

- identify the situations where these distributions are applied;
- know the situations where Poisson distribution is applied;
- define and explain Poisson distribution;
- define the discrete uniform and hypergeometric distributions;
- compute mean and variance for different discrete distributions.

6.2 BERNOULLI DISTRIBUTION AND ITS PROPERTIES

There are experiments where the outcomes can be divided into two categories with reference to presence or absence of a particular attribute or characteristic. A convenient method of representing the two is to designate either of them as success and the other as failure. For example, head coming up in the toss of a fair coin may be treated as a success and tail as failure, or vice-versa. Accordingly, probabilities can be assigned to the success and failure.

Suppose a piece of a product is tested which may be defective (failure) or non-defective (a success). Let p be the probability that it found non-defective and $q = 1 - p$ be the probability that it is defective. Let X be a random variable such that it takes value 1 when success occurs and 0 if failure occurs.

Therefore,

$$P[X = 1] = p, \text{ and}$$

$$P[X = 0] = q = 1 - p.$$

The above experiment is a Bernoulli trial, the r.v. X defined in the above experiment is a Bernoulli variate and the probability distribution of X as specified above is called the Bernoulli distribution in honour of J. Bernoulli (1654-1705).

Definition

A discrete random variable X is said to follow Bernoulli distribution with parameter p if its probability mass function is given by

$$P[X = x] = \begin{cases} p^x (1-p)^{1-x} & ; x = 0, 1 \\ 0 & ; \text{elsewhere} \end{cases}$$

$$\text{i.e. } P[X = 1] = p^1 (1-p)^{1-1} = p \quad [\text{putting } x = 1]$$

$$\text{and } P[X = 0] = p^0 (1-p)^{1-0} = 1-p \quad [\text{putting } x = 0]$$

The Bernoulli probability distribution, in tabular form, is given as

X	0	1
p(x)	1-p	p

Remark 1: The Bernoulli distribution is useful whenever a random experiment has only two possible outcomes, which may be labelled as success and failure.

Moments of Bernoulli Distribution

The r^{th} moment about origin of a Bernoulli variate X is given as

$$\mu'_r = E(X^r)$$

$$\begin{aligned}
 &= \sum_{x=0}^1 x^r p(x) && [\text{See Unit 5 of this course}] \\
 &= (0)^r p(0) + (1)^r p(1) \\
 &= (0)(1-p) + (1)p \\
 &= p
 \end{aligned}$$

$$\Rightarrow \mu'_1 = p, \mu'_2 = p, \mu'_3 = p, \mu'_4 = p.$$

Hence,

$$\text{Mean} = \mu'_1 = p,$$

$$\text{Variance } (\mu_2) = \mu'_2 - (\mu'_1)^2 = p - p^2 = p(1-p),$$

Example 1: Let X be a random variable having Bernoulli distribution with parameter $p = 0.4$. Find its mean and variance.

Solution:

$$\text{Mean} = p = 0.4,$$

$$\text{Variance} = p(1-p) = (0.4)(1-0.4) = (0.4)(0.6) = 0.24$$

Single trial is taken into consideration in Bernoulli distribution. But, if trials are performed repeatedly a finite number of times and we are interested in the distribution of the sum of independent Bernoulli trials with the same probability of success in each trial, then we need to study binomial distribution which has been discussed in the next section.

6.3 BINOMIAL PROBABILITY FUNCTION

Here, in this section, we will discuss binomial distribution which was discovered by J. Bernoulli (1654-1705) and was first published eight years after his death i.e. in 1713 and is also known as “Bernoulli distribution for n trials”. Binomial distribution is applicable for a random experiment comprising a finite number (n) of independent Bernoulli trials having the constant probability of success for each trial.

Before defining binomial distribution, let us consider the following example: Suppose a man fires 3 times independently to hit a target. Let p be the probability of hitting the target (success) for each trial and $q (= 1-p)$ be the probability of his failure.

Let S denote the success and F the failure. Let X be the number of successes in 3 trials,

$$\begin{aligned}
 P[X = 0] &= \text{Probability that target is not hit at all in any trial} \\
 &= P[\text{Failure in each of the three trials}] \\
 &= P(F \cap F \cap F) \\
 &= P(F).P(F).P(F) \quad [\because \text{trials are independent}] \\
 &= q.q.q \\
 &= q^3
 \end{aligned}$$

This can be written as

$$P[X = 0] = {}^3C_0 p^0 q^{3-0}$$

$$[\because {}^3C_0 = 1, p^0 = 1, q^{3-0} = q^3. \text{ Recall } {}^nC_x = \frac{n!}{x!(n-x)!} \text{ (see Unit 4 of MST-001)}]$$

$P[X = 1]$ = Probability of hitting the target once

= [(Success in the first trial and failure in the second and third trial) or
(success in the second trial and failure in the first and third trials)
or (success in the third trial and failure in the first two trials)]

$$= P[(S \cap F \cap F) \text{ or } (F \cap S \cap F) \text{ or } (F \cap F \cap S)]$$

$$= P(S \cap F \cap F) + P(F \cap S \cap F) + P(F \cap F \cap S)$$

$$= P(S).P(F).P(F) + P(F).P(S).P(F) + P(F).P(F).P(S)$$

[$\because$ trials are independent]

$$= p.q.q + q.p.q + q.q.p$$

$$= pq^2 + pq^2 + pq^2$$

$$= 3pq^2$$

This can also be written as

$$P[X = 1] = {}^3C_1 p^1 q^{3-1} \quad [\because {}^3C_1 = 3, p^1 = p, q^{3-1} = q^2]$$

$P[X = 2]$ = Probability of hitting the target twice

= P[(Success in each of the first two trials and failure in the third trial) or (Success in first and third trial and failure in the second trial) or (Success in the last two trials and failure in the first trial)]

$$= P[(S \cap S \cap F) \cup (S \cap F \cap S) \cup (F \cap S \cap S)]$$

$$= P[S \cap S \cap F] + P[S \cap F \cap S] + P[F \cap S \cap S]$$

$$= P(S).P(S).P(F) + P(S).P(F).P(S) + P(F).P(S).P(S)$$

$$= p.p.q + p.q.p + q.p.p$$

$$= 3p^2q$$

This can also be written as

$$P[X = 2] = {}^3C_2 p^2 q^{3-2} \quad [\because {}^3C_2 = 3, q^{3-2} = q]$$

$P[X = 3]$ = Probability of hitting the target thrice

= [Success in each of the three trials]

$$= P[S \cap S \cap S]$$

$$= P(S).P(S).P(S)$$

$$= p.p.p$$

$$= p^3$$

This can also be written as

$$P[X = 3] = {}^3C_3 p^3 q^{3-3} \quad [\because {}^3C_3 = 1, q^{3-3} = 1]$$

From the above four enrectangled results, we can write

$$P[X = r] = {}^3C_r p^r q^{3-r}; r = 0, 1, 2, 3.$$

which is the probability of r successes in 3 trials. 3C_r , here, is the number of ways in which r successes can happen in 3 trials.

The result can be generalized for n trials in the similar fashion and is given as

$$P[X = r] = {}^nC_r p^r q^{n-r}; r = 0, 1, 2, \dots, n.$$

This distribution is called the binomial probability distribution. The reason behind giving the name binomial probability distribution for this probability distribution is that the probabilities for $x = 0, 1, 2, \dots, n$ are the respective probabilities ${}^nC_0 p^0 q^{n-0}, {}^nC_1 p^1 q^{n-1}, \dots, {}^nC_n p^n q^{n-n}$ which are the successive terms of the binomial expansion $(q + p)^n$.

$$[\because (q + p)^n = {}^nC_0 q^n p^0 + {}^nC_1 q^{n-1} p^1 + \dots + {}^nC_n q^0 p^n]$$

Binomial Expansion:

‘Bi’ means ‘Two’. ‘Binomial expansion’ means ‘Expansion of expression having two terms, e.g.

$$(X + Y)^2 = X^2 + 2XY + Y^2 = {}^2C_0 X^2 Y^0 + {}^2C_1 X^{2-1} Y^1 + {}^2C_2 X^{2-2} Y^2,$$

$$(X + Y)^3 = X^3 + 3X^2 Y + 3XY^2 + Y^3$$

$$= {}^3C_0 X^3 Y^0 + {}^3C_1 X^{3-1} Y^1 + {}^3C_2 X^{3-2} Y^2 + {}^3C_3 X^{3-3} Y^3$$

So, in general,

$$(X + Y)^n = {}^nC_0 X^n Y^0 + {}^nC_1 X^{n-1} Y^1 + {}^nC_2 X^{n-2} Y^2 + \dots + {}^nC_n X^{n-n} Y^n$$

The above discussion leads to the following definition.

Definition:

A discrete random variable X is said to follow binomial distribution with parameters n and p if it assumes only a finite number of non-negative integer values and its probability mass function is given by

$$P[X = x] = \begin{cases} {}^nC_x p^x q^{n-x}; & x = 0, 1, 2, \dots, n \\ 0; & \text{elsewhere} \end{cases}$$

where, n is the number of independent trials,

x is the number of successes in n trials,

p is the probability of success in each trial, and

$q = 1 - p$ is the probability of failure in each trial.

Remark 2:

- i) The binomial distribution is the probability distribution of sum of n independent Bernoulli variates.
- ii) If X is binomially distributed r.v. with parameters n and p , then we may write it as $X \sim B(n, p)$.
- iii) If X and Y are two binomially distributed independent random variables with parameters (n_1, p) and (n_2, p) respectively then their sum also follows a binomial distribution with parameters $n_1 + n_2$ and p . But, if the probability of success is not the same for the two random variables, then this property does not hold.

Example 2: An unbiased coin is tossed six times. Find the probability of obtaining

- (i) exactly 3 heads
- (ii) less than 3 heads

- (iii) more than 3 heads
- (iv) at most 3 heads
- (v) at least 3 heads
- (vi) more than 6 heads

Solution: Let p be the probability of getting head (success) in a toss of the coin and n be the number of trials.

$$\therefore n = 6, p = \frac{1}{2} \text{ and hence } q = 1 - p = 1 - \frac{1}{2} = \frac{1}{2}.$$

Let X be the number of successes in n trials,

$\therefore$ by binomial distribution, we have

$$P[X = x] = {}^nC_x p^x q^{n-x}; x = 0, 1, 2, \dots, n$$

$$= {}^6C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{6-x}; x = 0, 1, 2, \dots, 6$$

$$= {}^6C_x \left(\frac{1}{2}\right)^6; x = 0, 1, 2, \dots, 6.$$

$$= \frac{1}{64} \cdot {}^6C_x; x = 0, 1, 2, \dots, 6.$$

Therefore,

$$(i) P[\text{exactly 3 heads}] = P[X = 3]$$

$$= \frac{1}{64} ({}^6C_3) = \frac{1}{64} \left[\frac{6 \times 5 \times 4}{3 \times 2} \right] = \frac{5}{16}$$

$$[\because \text{Recall } {}^nC_x = \frac{n!}{x!(n-x)!} \text{ (see Unit 4 of MST-001)}]$$

$$(ii) P[\text{less than 3 heads}] = P[X < 3]$$

$$= P[X = 2 \text{ or } X = 1 \text{ or } X = 0]$$

$$= P[X = 2] + P[X = 1] + P[X = 0]$$

$$= \frac{1}{64} \cdot {}^6C_2 + \frac{1}{64} \cdot {}^6C_1 + \frac{1}{64} \cdot {}^6C_0$$

$$= \frac{1}{64} [{}^6C_2 + {}^6C_1 + {}^6C_0] = \frac{1}{64} \left[\frac{6 \times 5}{2} + 6 + 1 \right]$$

$$= \frac{22}{64} = \frac{11}{32}.$$

$$(iii) P[\text{more than 3 heads}] = P[X > 3]$$

$$= P[X = 4 \text{ or } X = 5 \text{ or } X = 6] \left[\because \text{in 6 trials one can have at most 6 heads} \right]$$

$$= P[X = 4] + P[X = 5] + P[X = 6]$$

$$= \frac{1}{64} \cdot {}^6C_4 + \frac{1}{64} \cdot {}^6C_5 + \frac{1}{64} \cdot {}^6C_6$$

$$= \frac{1}{64} [{}^6C_4 + {}^6C_5 + {}^6C_6]$$

$$= \frac{1}{64} \left[\frac{6 \times 5}{2} + 6 + 1 \right] = \frac{22}{64} = \frac{11}{32}.$$

(iv) $P[\text{at most 3 heads}] = P[3 \text{ or less than 3 heads}]$

$$\begin{aligned} &= P[X = 3] + P[X = 2] + P[X = 1] + P[X = 0] \\ &= \frac{1}{64} \cdot {}^6C_3 + \frac{1}{64} \cdot {}^6C_2 + \frac{1}{64} \cdot {}^6C_1 + \frac{1}{64} \cdot {}^6C_0 \\ &= \frac{1}{64} [{}^6C_3 + {}^6C_2 + {}^6C_1 + {}^6C_0] \\ &= \frac{1}{64} [20 + 15 + 6 + 1] = \frac{42}{64} = \frac{21}{32}. \end{aligned}$$

(v) $P[\text{at least 3 heads}] = P[3 \text{ or more heads}]$

$$\begin{aligned} &= P[X = 3] + P[X = 4] + P[X = 5] + P[X = 6] \\ &\quad \text{or} \\ &= 1 - (P[X = 0] + P[X = 1] + P[X = 2]) \\ &\quad \left[\because \text{sum of probabilities of all possible} \right. \\ &\quad \left. \text{values of a random variable is 1} \right] \\ &= 1 - \left(\frac{11}{32} \right) \quad \left[\text{Already obtained in} \right. \\ &\quad \left. \text{part (ii) of this example} \right] \\ &= \frac{21}{32}. \end{aligned}$$

(vi) $P[\text{more than 6 heads}] = P[7 \text{ or more heads}]$

$$\begin{aligned} &= P[\text{an impossible event}] \quad \left[\because \text{in six tosses, it} \right. \\ &\quad \left. \text{is impossible to get} \right. \\ &\quad \left. \text{more than six heads} \right] \\ &= 0 \end{aligned}$$

Example 3: The chances of catching cold by workers working in an ice factory during winter are 25%. What is the probability that out of 5 workers 4 or more will catch cold?

Solution: Let catching cold be the success and p be the probability of success for each worker.

$\therefore$ Here, $n = 5$, $p = 0.25$, $q = 0.75$ and by binomial distribution

$$\begin{aligned} P[X = x] &= {}^nC_x p^x q^{n-x}; x = 0, 1, 2, \dots, n \\ &= {}^5C_x (0.25)^x (0.75)^{5-x}; 0, 1, 2, \dots, 5 \end{aligned}$$

Therefore, the required probability $= P[X \geq 4]$

$$\begin{aligned} &= p[X = 4 \text{ or } X = 5] \\ &= P[X = 4] + P[X = 5] \\ &= {}^5C_4 (0.25)^4 (0.75)^1 + {}^5C_5 (0.25)^5 (0.75)^0 \\ &= (5)(0.002930) + (1)(0.000977) \\ &= 0.014650 + 0.000977 \\ &= 0.015627 \end{aligned}$$

Example 4: Let X and Y be two independent random variables such that $X \sim B(4, 0.7)$ and $Y \sim B(3, 0.7)$. Find $P[X + Y \leq 1]$.

Solution: We know that if X and Y are independent random variables each following binomial distribution with parameters (n_1, p) and (n_2, p) , then $X + Y \sim B(n_1 + n_2, p)$.

Therefore, here $X + Y$ follows binomial distribution with parameters $4 + 3$ and 0.7 , i.e. 7 and 0.7 . So, here, $n = 7$ and $p = 0.7$.

$$\begin{aligned}\text{Thus, the required probability} &= P[X + Y \leq 1] \\ &= P[X + Y = 1] + P[X + Y = 0] \\ &= {}^7C_1 (0.7)^1 (0.3)^6 + {}^7C_0 (0.7)^0 (0.3)^7 \\ &= 7(0.7)(0.000729) + 1(1)(0.0002187) \\ &= 0.0035721 + 0.0002187 \\ &= 0.0037908\end{aligned}$$

Moments of Binomial Distribution

The r^{th} order moment about origin of a binomial variate X is given as

$$\mu'_r = E(X^r) = \sum_{x=0}^n x^r \cdot P[X = x]$$

Using this moment function the following useful results are drawn for the binomial distribution:

$$\begin{aligned}\therefore \text{Mean} &= \text{First order moment about origin} \\ &= \mu'_1 \\ &= np.\end{aligned}$$

Mean = np

$$\begin{aligned}\therefore \text{Variance } (\mu_2) &= \mu'_2 - (\mu'_1)^2 \\ &= n^2 p^2 - n p^2 + np - (np)^2 \\ &= n^2 p^2 - np^2 + np - n^2 p^2 \\ &= np - np^2 \\ &= np(1 - p) \\ &= npq\end{aligned}$$

$$\therefore \text{Variance} = npq$$

Remark 3:

- (i) As $0 < q < 1$
 $\Rightarrow q < 1$ [Multiplying both sides by $np > 0$]
 $\Rightarrow npq < np$
 $\Rightarrow \text{Variance} < \text{Mean}$
Hence, for binomial distribution
Mean $>$ Variance
- (ii) As variance of $X \sim B(n, p)$ is npq ,
 $\therefore$ its standard deviation is $\sqrt{npq}$.

Example 5: For a binomial distribution with $p = \frac{1}{4}$ and $n = 10$, find mean and variance.

Solution: As $p = \frac{1}{4}$, $\therefore q = 1 - \frac{1}{4} = \frac{3}{4}$.

$$\text{Mean} = np = 10 \times \frac{1}{4} = \frac{5}{2},$$

$$\text{Variance} = npq = 10 \times \frac{1}{4} \times \frac{3}{4} = \frac{15}{8}.$$

Example 6: The mean and standard deviation of binomial distribution are 4 and $\frac{2}{\sqrt{3}}$ respectively. Find $P[X \geq 1]$.

Solution: Let $X \sim B(n, p)$, then

$$\text{Mean} = np = 4$$

and variance $= npq = \left(\frac{2}{\sqrt{3}}\right)^2$ [$\because$ S.D. $= \frac{2}{\sqrt{3}}$ and variance is square of S.D.]

Dividing the second equation by the first equation, we have

$$\frac{npq}{np} = \frac{\frac{4}{3}}{4}$$

$$\Rightarrow q = \frac{1}{3}$$

$$\therefore p = 1 - q = 1 - \frac{1}{3} = \frac{2}{3}$$

Putting $p = \frac{2}{3}$ in the equation of mean, we have

$$n\left(\frac{2}{3}\right) = 4 \Rightarrow n = 6$$

$\therefore$ by binomial distribution,

$$\begin{aligned} P[X = x] &= {}^nC_x p^x q^{n-x} \\ &= {}^6C_x \left(\frac{2}{3}\right)^x \left(\frac{1}{3}\right)^{6-x}; \quad x = 0, 1, 2, \dots, 6. \end{aligned}$$

Thus, the required probability

$$\begin{aligned} P[X \geq 1] &= P[X = 1] + P[X = 2] + P[X = 3] + \dots + P[X = 6] \\ &= 1 - P[X = 0] \\ &= 1 - {}^6C_0 \left(\frac{2}{3}\right)^0 \left(\frac{1}{3}\right)^{6-0} = 1 - (1)(1) \frac{1}{729} = \frac{728}{729}. \end{aligned}$$

Check Your Progress 1

- 1) The probability of a man hitting a target is $\frac{1}{4}$. He fires 5 times. What is the probability of his hitting the target at least twice?

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- 2) A policeman fires 6 bullets on a dacoit. The probability that the dacoit will be killed by a bullet is 0.6. What is the probability that the dacoit is still alive?

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- 3) Comment on the following:

The mean of a binomial distribution is 3 and variance is 4.

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- 4) Find the binomial distribution when sum of mean and variance of 5 trials is 4.8.

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- 5) The mean of a binomial distribution is 30 and standard deviation is 5. Find the values of n , p and q .

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6.4 POISSON DISTRIBUTION

In case of binomial distributions, as discussed in the last section, we deal with events whose occurrences and non-occurrences are almost equally important. However, there may be events which do not occur as outcomes of a definite number of trials of an experiment but occur rarely at random points of time and for such events our interest lies only in the number of occurrences and not in its non-occurrences. Examples of such events are:

- Our interest may lie in how many printing mistakes are there on each page of a book but we are not interested in counting the number of words without any printing mistake.
- In production where control of quality is the major concern, it often requires counting the number of defects (and not the non-defects) per item.
- One may intend to know the number of accidents during a particular time interval.

Under such situations, binomial distribution cannot be applied as the value of n is not definite and the probability of occurrence is very small. Other such situations can be thought of yourself. Poisson distribution discovered by S.D. Poisson (1781-1840) in 1837 can be applied to study these situations.

Poisson distribution is a limiting case of binomial distribution under the following conditions:

- n , the number of trials is indefinitely large, i.e. $n \rightarrow \infty$.
- p , the constant probability of success for each trial is very small, i.e. $p \rightarrow 0$.
- np is a finite quantity say ' λ '.

Definition: A random variable X is said to follow Poisson distribution if it assumes indefinite number of non-negative integer values and its probability mass function is given by:

$$p(x) = P(X = x) = \begin{cases} \frac{e^{-\lambda} \cdot \lambda^x}{x!}; & x = 0, 1, 2, 3, \dots \text{ and } \lambda > 0. \\ 0; & \text{elsewhere} \end{cases}$$

where e = base of natural logarithm, whose value is approximately equal to 2.7183 corrected to four decimal places. Value of $e^{-\lambda}$ can be written from the table given in the Appendix 1 at the end of this unit, or, can be seen from the reference.

Remark 4

- i) If X follows Poisson distribution with parameter λ then we shall use the notation $X \sim P(\lambda)$.
- ii) If X and Y are two independent Poisson variates with parameters λ_1 and λ_2 respectively, then $X + Y$ is also a Poisson variate with parameter $\lambda_1 + \lambda_2$. This is known as **additive property of Poisson distribution**.

Moments Of Poisson Distribution

r^{th} order moment about origin of Poisson variate is

$$\mu'_r = E(X^r) = \sum_{x=0}^{\infty} x^r p(x) = \sum_{x=0}^{\infty} x^r \frac{e^{-\lambda} \lambda^x}{x!}$$

$$\therefore \text{Mean} = \lambda$$

$$\begin{aligned} \therefore \text{Variance of } X \text{ is given as } V(X) &= \mu_2 = \mu'_2 - (\mu'_1)^2 \\ &= \lambda^2 + \lambda - (\lambda)^2 \\ &= \lambda \end{aligned}$$

Remark 5

- i) Mean and variance of Poisson distribution are always equal. In fact, this is the only discrete distribution for which Mean = Variance = the third central moment.
- ii) Moments of the Poisson distribution can be deduced from those of the binomial distribution also as explained below:

For a binomial distribution,

$$\text{Mean} = np$$

$$\text{Variance} = npq$$

Now as the Poisson distribution is a limiting form of binomial distribution under the conditions:

$$(i) n \rightarrow \infty, (ii) p \rightarrow 0 \text{ i.e. } q \rightarrow 1, \text{ and } (iii) np = \lambda \text{ (a finite quantity);}$$

$\therefore$ Mean, Variance and other moments of the Poisson distribution are given as:

$$\text{Mean} = \text{Limiting value of } np = \lambda$$

$$\begin{aligned}\text{Variance} &= \text{Limiting value of } npq \\ &= \text{Limiting value of } (np) (q) \\ &= (\lambda) (1) = \lambda\end{aligned}$$

Now, let's give some examples of the Poisson distribution.

Example 7: It is known that the number of heavy trucks arriving at a railway station follows the Poisson distribution. If the average number of truck arrivals during a specified period of an hour is 2, find the probabilities that during a given hour

- no heavy truck arrive,
- at least two trucks will arrive.

Solution: Here, the average number of truck arrivals is 2

i.e. mean = 2

$$\Rightarrow \lambda = 2$$

Let X be the number of trucks arrive during a given hour,

$\therefore$ by Poisson distribution, we have

$$P[X = x] = \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-2} (2)^x}{x!}; x = 0, 1, 2, \dots$$

Thus, the desired probabilities are:

$$(a) P[\text{arrival of no heavy truck}] = P[X = 0]$$

$$= \frac{e^{-2} 2^0}{0!}$$

$$= e^{-2}$$

$$= 0.1353$$

[See the table given
in the Appendix at
the end of this unit]

$$(b) P[\text{arrival of at least two trucks}] = P[X \geq 2]$$

$$= P[X = 2] + P[X = 3] + \dots$$

$$= 1 - [P[X = 1] + P[X = 0]]$$

[$\because$ sum of all the
probabilities is 1]

$$= 1 - \left[\frac{e^{-2} 2^0}{0!} + \frac{e^{-2} 2^1}{1!} \right]$$

$$= 1 - e^{-2} \left[\frac{2^0}{0!} + \frac{2^1}{1!} \right] = 1 - e^{-2} (1 + 2)$$

$$= 1 - (0.1353)(3) = 1 - 0.4059 = 0.5941$$

Note: In most of the cases for Poisson distribution, if we are to compute the probabilities of the type $P[X > a]$ or $P[X \geq a]$, we write them as $P[X > a] = 1 - P[X \leq a]$ and $P[X \geq a] = 1 - P[X < a]$, because n may not be definite and hence we cannot go up to the last value and hence the probability is written in terms of its complementary probability.

Example 8: If the probability that an individual suffers a bad reaction from an injection of a given serum is 0.001, determine the probability that out of 500 individuals

- i) exactly 3,
- ii) more than 2

individuals suffer from bad reaction

Solution: Let X be the Poisson variate, "Number of individuals suffering from bad reaction". Then,

$$n = 1500, p = 0.001,$$

$$\therefore \lambda = np = (1500)(0.001) = 1.5$$

$\therefore$ By Poisson distribution,

$$P[X = x] = \frac{e^{-\lambda} \lambda^x}{x!}, x = 0, 1, 2, \dots$$

$$= \frac{e^{-1.5} \cdot (1.5)^x}{x!}; x = 0, 1, 2, \dots$$

Thus,

- i) The desired probability = $P[X = 3]$

$$= \frac{e^{-1.5} \cdot (1.5)^3}{3!}$$

$$= \frac{(0.2231)(3.375)}{6} = 0.1255$$

$$\left[\begin{array}{l} \because e^{-0.5} = 0.6065, e^{-1} = 0.3679, \text{ so} \\ e^{-1.5} = e^{-1} \times e^{-0.5} = (0.3679)(0.6065) = 0.2231 \\ \text{See the table given in the Appendix} \\ \text{at the end of this unit} \end{array} \right]$$

- ii) The desired probability = $P[X > 2]$

$$= 1 - P[X \leq 2]$$

$$= 1 - [P[X = 2] + P[X = 1] + P[X = 0]]$$

$$= 1 - \left[\frac{e^{-1.5} \cdot (1.5)^2}{2!} + \frac{e^{-1.5} \cdot (1.5)^1}{1!} + \frac{e^{-1.5} \cdot (1.5)^0}{0!} \right]$$

$$= 1 - e^{-1.5} \left[\frac{2.25}{2} + 1.5 + 1 \right] = 1 - (3.625)e^{-1.5}$$

$$= 1 - (3.625)(0.2231) = 1 - 0.8087 = 0.1913$$

Example 9: If the mean of a Poisson distribution is 1.44, find the values of variance and the central moments of order 3 and 4.

Solution: Here, mean = 1.44

$$\Rightarrow \lambda = 1.44$$

Hence, Variance = $\lambda = 1.44$

$$\mu_3 = \lambda = 1.44$$

$$\mu_4 = 3\lambda^2 + \lambda = 3(1.44)^2 + 1.44 = 7.66.$$

Example 10: If a Poisson variate X is such that $P[X = 1] = 2P[X = 2]$, find the mean and variance of the distribution.

Solution: Let λ be the mean of the distribution, hence by Poisson distribution,

$$P[X = x] = \frac{e^{-\lambda} \lambda^x}{x!}; x = 0, 1, 2, \dots$$

$$\text{Now, } P[X = 1] = 2P[X = 2]$$

$$\Rightarrow \frac{e^{-\lambda} \cdot \lambda^1}{1!} = 2 \frac{e^{-\lambda} \cdot \lambda^2}{2!}$$

$$\Rightarrow \lambda = \lambda^2 \Rightarrow \lambda^2 - \lambda = 0 \Rightarrow \lambda(\lambda - 1) = 0 \Rightarrow \lambda = 0, 1$$

But $\lambda = 0$ is rejected

[$\because$ if $\lambda = 0$ then either $n = 0$ or $p = 0$ which implies that Poisson distribution does not exist in this case.]

$$\therefore \lambda = 1$$

Hence mean = $\lambda = 1$, and

Variance = $\lambda = 1$.

Check Your Progress 2

- 1) Assume that the chance of an individual coal miner being killed in a mine accident during a year is $\frac{1}{1400}$. Use the Poisson distribution to calculate the probability that in a mine employing 350 miners, there will be at least one fatal accident in a year. (use $e^{-0.25} = 0.78$)

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- 2) The mean and standard deviation of a Poisson distribution are 6 and 2 respectively. Test the validity of this statement.

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- 3) For a Poisson distribution, it is given that $P[X = 1] = P[X = 2]$, find the value of the mean of the distribution. Hence find $P[X = 0]$ and $P[X = 4]$.

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6.5 DISCRETE UNIFORM DISTRIBUTION

Discrete uniform distribution can be conceived in practice if under the given experimental conditions, the different values of the random variable are equally likely. For example, the number on an unbiased die when thrown may be 1 or 2 or 3 or 4 or 5 or 6. These values of random variable, “the number on an unbiased die when thrown” are equally likely and for such an experiment, the discrete uniform distribution is appropriate.

Definition: A random variable X is said to have a discrete uniform (rectangular) distribution if it takes any positive integer value from 1 to n , and its probability mass function is given by

$$P[X = x] = \begin{cases} \frac{1}{n} & \text{for } x = 1, 2, \dots, n \\ 0, & \text{otherwise.} \end{cases}$$

where n is called the parameter of the distribution.

For example, the random variable X , “the number on the unbiased die when thrown”, takes on the positive integer values from 1 to 6 follows discrete uniform distribution having the probability mass function.

$$P[X = x] = \begin{cases} \frac{1}{6}, & \text{for } x = 1, 2, 3, 4, 5, 6. \\ 0, & \text{otherwise.} \end{cases}$$

Mean and Variance of the Distribution

$$\begin{aligned} \text{Mean} = E(X) &= \sum_{x=1}^n x p(x) = \sum_{x=1}^n x \cdot \left(\frac{1}{n}\right) = \frac{1}{n} \sum_{x=1}^n x \\ &= \frac{1}{n} [1 + 2 + 3 + \dots + n] \\ &= \frac{1}{n} \cdot \frac{n(n+1)}{2} \left[\because \text{sum of first } n \text{ natural numbers} = \frac{n(n+1)}{2} \right] \\ &\quad \left[\text{(see Unit 3 of Course MST – 001)} \right] \\ &= \frac{n+1}{2}. \end{aligned}$$

$$\text{Variance} = E(X^2) - [E(X)]^2 \quad \left[\because \mu_2 = \mu_2' - (\mu_1')^2 \right]$$

where

$$E(X) = \frac{n+1}{2} \quad [\text{Obtained above}]$$

$$E(X^2) = \sum_{x=1}^n x^2 \cdot p(x)$$

$$\text{and } E(X^2) = \sum_{x=1}^n x^2 \cdot \frac{1}{n}$$

$$= \frac{1}{n} [1^2 + 2^2 + 3^2 + \dots + n^2]$$

$$= \frac{1}{n} \left[\frac{n(n+1)(2n+1)}{6} \right] \left[\begin{array}{l} \therefore \text{sum of squares of first } n \\ \text{natural numbers} = \frac{n(n+1)(2n+1)}{6} \\ \text{(see Unit 3 of Course MST - 001)} \end{array} \right]$$

$$= \frac{(n+1)(2n+1)}{6}$$

$$\therefore \text{Variance} = \frac{(n+1)(2n+1)}{6} - \left(\frac{n+1}{2} \right)^2$$

$$= \frac{(n+1)}{12} [2(2n+1) - 3(n+1)]$$

$$= \frac{n+1}{12} [4n+2-3n-3] = \frac{(n+1)}{12} (n-1) = \frac{n^2-1}{12}$$

Example 11: Find the mean and variance of a number on an unbiased die when thrown.

Solution: Let X be the number on an unbiased die when thrown,

$\therefore X$ can take the values 1, 2, 3, 4, 5, 6 with

$$P[X = x] = \frac{1}{6}; x = 1, 2, 3, 4, 5, 6.$$

Hence, by uniform distribution, we have

$$\text{Mean} = \frac{n+1}{2} = \frac{6+1}{2} = \frac{7}{2}, \text{ and}$$

$$\text{Variance} = \frac{n^2-1}{12} = \frac{(6)^2-1}{12} = \frac{35}{12}.$$

Uniform Frequency Distribution

If an experiment, satisfying the requirements of discrete uniform distribution, is repeated N times, then expected frequency of a value of random variable is given by

$$f(x) = N \cdot P[X = x]; x = 1, 2, \dots, n$$

$$= N \cdot \frac{1}{n}; x = 1, 2, 3, \dots, n.$$

Example 12: If an unbiased die is thrown 120 times, find the expected frequency of appearing 1, 2, 3, 4, 5, 6 on the die.

Solution: Let X be the uniform discrete random variable, “the number on the unbiased die when thrown”.

$$\therefore P[X = x] = \frac{1}{6}; x = 1, 2, \dots, 6$$

Hence, the expected frequencies of the value of random variable are given as computed in the following table:

X	$P[X = x]$	Expected/Theoretical frequencies $f(x) = N \cdot P[X = x] = 120 \cdot P[X = x]$
1	$\frac{1}{6}$	$120 \times \frac{1}{6} = 20$
2	$\frac{1}{6}$	$120 \times \frac{1}{6} = 20$
3	$\frac{1}{6}$	$120 \times \frac{1}{6} = 20$
4	$\frac{1}{6}$	$120 \times \frac{1}{6} = 20$
5	$\frac{1}{6}$	$120 \times \frac{1}{6} = 20$
6	$\frac{1}{6}$	$120 \times \frac{1}{6} = 20$

6.6 HYPERGEOMETRIC DISTRIBUTION

In a discrete uniform probability distribution, the probability distribution is obtained for the possible outcomes in a single trial, however, if there are more than one but finite trials with only two possible outcomes in each trial, we apply some other distribution. One such distribution which is applicable in such a situation is binomial distribution. The binomial distribution deals with finite and independent trials, each of which has exactly two possible outcomes (Success or Failure) with constant probability of success in each trial. For example, if we again consider the example of drawing ticket randomly from an urn containing 10 tickets bearing numbers from 1 to 10. Then, the probability that the drawn ticket bears an odd number is $\frac{5}{10} = \frac{1}{2}$. If we replace the ticket back, then the

probability of drawing a ticket bearing an odd number is again $\frac{5}{10} = \frac{1}{2}$. So, if we draw ticket again and again with replacement, trials become independent and

probability of getting an odd number is same in each trial. Suppose, it is asked that what is the probability of getting 2 tickets bearing odd number in 3 draws then we apply binomial distribution as follows:

Let X be the number of times an odd number appears in 3 draws, then by binomial distribution,

$$P[X = 2] = {}^3C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{3-2} = (3) \left(\frac{1}{2}\right) \left(\frac{1}{2}\right) = \frac{3}{8}.$$

But, if in the example discussed above, we do not replace the ticket after any draw the probability of getting an odd number gets changed in each trial and the trials remain no more independent and hence in this case binomial distribution is not applicable. Suppose, in this case also, we are interested in finding the probability of getting ticket bearing odd number twice in 3 draws, then it is computed as follows:

Let A_i be the event that i^{th} ticket drawn bears odd number and $\bar{A}_i$ be the event that i^{th} ticket drawn does not bear odd number.

$\therefore$ Probability of getting ticket bearing odd number twice in 3 draws

$$\begin{aligned} &= P[A_1 \cap A_2 \cap \bar{A}_3] + P[A_1 \cap \bar{A}_2 \cap A_3] + P[\bar{A}_1 \cap A_2 \cap A_3] \\ &= P[A_1]P[A_2 | A_1]P[\bar{A}_3 | A_1 \cap A_2] + P[A_1]P[\bar{A}_2 | A_1]P[A_3 | A_1 \cap \bar{A}_2] \\ &\quad + P[\bar{A}_1]P[A_2 | \bar{A}_1]P[A_3 | \bar{A}_1 \cap A_2] \\ &\quad \text{[Multiplication theorem for dependent events]} \\ &= \frac{5}{10} \cdot \frac{4}{9} \cdot \frac{5}{8} + \frac{5}{10} \cdot \frac{5}{9} \cdot \frac{4}{8} + \frac{5}{10} \cdot \frac{5}{9} \cdot \frac{4}{8} \\ &= 3 \times \frac{5 \times 5 \times 4}{10 \times 9 \times 8} \end{aligned}$$

This result can be written in the following form also:

$$\begin{aligned} &= \frac{5 \times 4 \times 5 \times 3 \times 2}{2 \times 10 \times 9 \times 8} \quad \text{[Multiplying and Dividing by 2]} \\ &= \frac{5 \times 4}{2} \times 5 \times \frac{1}{10 \times 9 \times 8} = {}^5C_2 \times {}^5C_1 \times \frac{1}{{}^{10}C_3} = \frac{{}^5C_2 \times {}^5C_1}{{}^{10}C_3} \end{aligned}$$

In the above result, 5C_2 is representing the number of ways of selecting 2 out of 5 tickets bearing odd number, 5C_1 is representing the number of ways of selecting 1 out of 5 tickets bearing even number i.e. not bearing odd number, and ${}^{10}C_3$ is representing the number of ways of selecting 3 out of total 10 tickets.

Let us consider another similar example of a bag containing 20 balls out of which 5 are white and 15 are black. Suppose 10 balls are drawn at random one by one without replacement, then as discussed in the above example, the probability that in these 10 draws, there are 2 white and 8 black balls is

$$\frac{{}^5C_2 \times {}^{15}C_8}{{}^{20}C_{10}}.$$

Note: The result remains exactly same whether the items are drawn one by one without replacement or drawn at once.

Let us now generalise the above argument for N balls, of which M are white and $N - M$ are black. Of these, n balls are chosen at random without replacement. Let X be a random variable that denote the number of white balls drawn. Then, the probability of $X = x$ white balls among the n balls drawn is given by

$$P[X = x] = \frac{{}^MC_x \cdot {}^{N-M}C_{n-x}}{{}^NC_n}$$

$$[\text{For } x = 0, 1, 2, \dots, n (n \leq M) \text{ or } x = 0, 1, 2, \dots, M (n > M)]$$

The above probability function of discrete random variable X is called the Hypergeometric distribution.

Remark 6: We have a hypergeometric distribution under the following conditions:

- There are finite number of dependent trials
- A single trial results in one of the two possible outcomes-Success or Failure
- Probability of success and hence that of failure is not same in each trial i.e. sampling is done without replacement

Remark 7: If number (n) of balls drawn is greater than the number (M) of white balls in the bag, then if $n \leq M$, the number (x) of white balls drawn cannot be greater than n and if $n > M$, then number of white balls drawn cannot be greater than M . So, x can take the values upto n (if $n \leq M$) and M (if $n > M$) i.e. x can take the value upto n or M , whichever is less, i.e. $x = \min \{n, M\}$.

The discussion leads to the following definition

Definition: A random variable X is said to follow the hypergeometric distribution with parameters N , M and n if it assumes only non-negative integer values and its probability mass function is given by

$$P[X = x] = \begin{cases} \frac{{}^MC_x \cdot {}^{N-M}C_{n-x}}{{}^NC_n} & \text{for } x = 0, 1, 2, \dots, \min \{n, M\} \\ 0, & \text{otherwise} \end{cases}$$

where n , M , N are positive integers such that $n \leq N$, $M \leq N$.

Mean and Variance

$$\begin{aligned} \text{Mean} = E(X) &= \sum_{x=0}^n x \cdot p[X = x] \\ &= \sum_{x=1}^n x \cdot \frac{{}^MC_x \cdot {}^{N-M}C_{n-x}}{{}^NC_n} \\ &\dots \end{aligned}$$

$$= \frac{M}{N} \binom{N-1}{C_{n-1}}$$

[This result is obtained using properties of binomial coefficients and involves lot of calculations and hence its derivation may be skipped. It may be noticed that in this result the left upper suffix and also the right lower suffix is the sum of the corresponding suffices of the binomial coefficients involved in each product term.]

Also,

$$\begin{aligned} V(X) &= E(X^2) - [E(X)]^2 \\ &= \frac{NM(N-M)(N-n)}{N^2(N-1)} \quad \text{[Given without proof]} \end{aligned}$$

Example 13: A jury of 5 members is drawn at random from a voters' list of 100 persons, out of which 60 are non-graduates and 40 are graduates. What is the probability that the jury will consist of 3 graduates?

Solution: The computation of the actual probability is hypergeometric, which is shown as follows:

$$\begin{aligned} P[2 \text{ non-graduates and } 3 \text{ graduates}] &= \frac{{}^{60}C_2 \cdot {}^{40}C_3}{{}^{100}C_5} \\ &= \frac{60 \times 59 \times 40 \times 39 \times 38 \times 5 \times 4 \times 3 \times 2}{2 \times 6 \times 100 \times 99 \times 98 \times 97 \times 96} \\ &= 0.2323 \end{aligned}$$

Example 14: Let us suppose that in a lake there are N fish. A catch of 500 fish (all at the same time) is made, and these fish are returned alive into the lake after marking each with a red spot. After two days, assuming that during this time these 'marked' fish have been distributed themselves 'at random' in the lake and there is no change in the total number of fish, a fresh catch of 400 fish (again, all at once) is made. What is the probability that of these 400 fish, 100 will be having red spots?

Solution: The computation of the probability is hypergeometric and is shown as follows: As marked fish in the lake are 500 and other are $N - 500$,

$$\therefore P[100 \text{ marked fish and } 300 \text{ others}] = \frac{{}^{500}C_{100} \cdot {}^{N-500}C_{300}}{{}^N C_{400}}.$$

We cannot numerically evaluate this if N is not given. Though N can be estimated using method of Maximum likelihood estimation which you will read in later units. We are not going to estimate it.

You will agree that the exact computation of this probability is complicated. Such problem is normally there with the use of hypergeometric distribution, especially, if N and M are large. However, if n is small compared to N i.e. if n

is such that $\frac{n}{N} < 0.05$, say then there is not much difference between sampling

with and without replacement and hence in such cases, the probability obtained by binomial distribution comes out to be approximately equal to that obtained using hypergeometric distribution.

Check Your Progress 3

- 1) Obtain the mean, variance of the discrete uniform distribution for the random variable, “the number on a ticket drawn randomly from an urn containing 10 tickets numbered from 1 to 10”. Also obtain the expected frequencies if the experiment is repeated 150 times.

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- 2) A lot of 25 units contains 10 defective units. An engineer inspects 2 randomly selected units from the lot. He/She accepts the lot if both the units are found in good condition, otherwise all the remaining units are inspected. Find the probability that the lot is accepted without further inspection.

.....
.....

We now conclude this unit by giving a summary of what we have covered in it.

6.7 SUMMARY

The following main points have been covered in this unit:

- 1) A discrete random variable X is said to follow **Bernoulli distribution** with parameter p if its probability mass function is given by

$$P[X = x] = \begin{cases} p^x (1-p)^{1-x} & ; x = 0, 1 \\ 0 & ; \text{elsewhere} \end{cases}$$

Its **mean** and **variance** are p and $p(1-p)$, respectively.

- 2) A discrete random variable X is said to follow **binomial distribution** if it assumes only a finite number of non-negative integer values and its probability mass function is given by

$$P[X = x] = \begin{cases} {}^n C_x p^x q^{n-x} & ; x = 0, 1, 2, \dots, n \\ 0 & ; \text{elsewhere} \end{cases}$$

where, n is the number of independent trials, x is the number of successes in n trial, p is the probability of success in each trial, and $q = 1 - p$ is the probability of failure in each trial.

- 3) The **constants of Binomial distribution** are:

Mean = np , Variance = npq ,

- 4) For a binomial distribution, **Mean > Variance**.

- 5) A random variable X is said to follow **Poisson distribution** if it assumes indefinite number of non-negative integer values and its probability mass function is given by:

$$p(x) = P(X = x) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!} & ; x = 0, 1, 2, 3, \dots \text{ and } \lambda > 0. \\ 0 & ; \text{elsewhere} \end{cases}$$

- 6) For Poisson distribution, **Mean = Variance** = λ .
- 7) A random variable X is said to have a **discrete uniform (rectangular)** distribution if it takes any positive integer value from 1 to n , and its probability mass function is given by

$$P[X = x] = \begin{cases} \frac{1}{n} & \text{for } x = 1, 2, \dots, n \\ 0, & \text{otherwise.} \end{cases}$$

where n is called the parameter of the distribution.

- 8) For **discrete uniform** distribution, **mean** = $\frac{n+1}{2}$ and **variance** = $\frac{n^2-1}{12}$.
- 9) A random variable X is said to follow the **hypergeometric distribution** with parameters N , M and n if it assumes only non-negative integer values and its probability mass function is given by

$$P(X = x) = \begin{cases} \frac{{}^M C_x \cdot {}^{N-M} C_{n-x}}{{}^N C_n} & \text{for } x = 0, 1, 2, \dots, \min\{n, M\} \\ 0, & \text{otherwise} \end{cases}$$

where n , M , N are positive integers such that $n \leq N$, $M \leq N$.

- 10) For **hypergeometric** distribution, **mean** = $\frac{nM}{N}$ and
variance = $\frac{NM(N-M)(N-n)}{N^2(N-1)}$.

6.8 SOLUTIONS/ANSWERS

Check Your Progress 1

- 1) Let p be the probability of hitting the target (success) in a trial.

$$\therefore n = 5, p = \frac{1}{4}, q = 1 - \frac{1}{4} = \frac{3}{4},$$

and hence by binomial distribution, we have

$$P[X = x] = {}^n C_x p^x q^{n-x} = {}^5 C_x \left(\frac{1}{4}\right)^x \left(\frac{3}{4}\right)^{5-x}; x = 0, 1, 2, 3, 4, 5.$$

$$\therefore \text{Required probability} = P[X \geq 2]$$

$$= P[X = 2] + P[X = 3] + P[X = 4] + P[X = 5]$$

$$= 1 - (P[X = 0] + P[X = 1])$$

$$= 1 - \left[{}^5 C_0 \left(\frac{1}{4}\right)^0 \left(\frac{3}{4}\right)^{5-0} + {}^5 C_1 \left(\frac{1}{4}\right)^1 \left(\frac{3}{4}\right)^{5-1} \right]$$

$$= 1 - \left[\frac{243}{1024} + \frac{405}{1024} \right] = \frac{376}{1024} = \frac{47}{128}$$

- 2) Let p be the probability that the dacoit will be killed (success) by a bullet.

$\therefore n = 6, p = 0.6, q = 1 - p = 1 - 0.6 = 0.4$, and hence by binomial distribution, we have

$$P[X = x] = {}^nC_x p^x q^{n-x}; x = 0, 1, 2, \dots, n$$

$$= {}^6C_x (0.6)^x (0.4)^{6-x}; x = 0, 1, 2, \dots, 6.$$

$$\begin{aligned}\therefore \text{The required probability} &= P[\text{The dacoit is still alive}] \\ &= P[\text{No bullet kills the dacoit}] \\ &= P[\text{Number of successes is zero}] \\ &= P[X = 0] = {}^6C_0 (0.6)^0 (0.4)^6 \\ &= 0.0041\end{aligned}$$

$$3) \text{ Mean} = np = 3 \quad \dots (1)$$

$$\text{Variance} = npq = 4 \quad \dots (2)$$

$\therefore$ Dividing (2) by (1), we have

$$q = \frac{4}{3} > 1 \text{ and hence not possible}$$

[$\because$ q, being probability, cannot be greater than 1]

4) Let $X \sim B(n, p)$, then

$n = 5$ and

$$np + npq = 4.8 \quad [\because \text{given that Mean} + \text{Variance} = 4.8]$$

$$\Rightarrow 5p + 5pq = 4.8$$

$$\Rightarrow 5[p + p(1-p)] = 4.8$$

$$\Rightarrow 5[p + p - p^2] = 4.8$$

$$\Rightarrow 5p^2 - 10p + 4.8 = 0$$

$$\Rightarrow 25p^2 - 50p + 24 = 0 \quad [\text{Multiplying by 5}]$$

$$\Rightarrow 25p^2 - 30p - 20p + 24 = 0$$

$$\Rightarrow 5p(5p - 6) - 4(5p - 6) = 0$$

$$\Rightarrow (5p - 6)(5p - 4) = 0$$

$$\Rightarrow p = \frac{6}{5}, \frac{4}{5}$$

The first value $p = \frac{6}{5}$ is rejected [$\because$ probability can never exceed 1]

$$\therefore p = \frac{4}{5} \text{ and hence } q = 1 - p = \frac{1}{5}.$$

Thus, the binomial distribution is

$$P[X = x] = {}^nC_x p^x q^{n-x}$$

$$= {}^5C_x \left(\frac{4}{5}\right)^x \left(\frac{1}{5}\right)^{5-x}; x = 0, 1, 2, 3, 4, 5.$$

The binomial distribution in tabular form is given as

X	p(x)
0	${}^5C_0 \left(\frac{4}{5}\right)^0 \left(\frac{1}{5}\right)^5 = \frac{1}{3125}$
1	${}^5C_1 \left(\frac{4}{5}\right)^1 \left(\frac{1}{5}\right)^4 = \frac{20}{3125}$
2	${}^5C_2 \left(\frac{4}{5}\right)^2 \left(\frac{1}{5}\right)^3 = \frac{160}{3125}$
3	${}^5C_3 \left(\frac{4}{5}\right)^3 \left(\frac{1}{5}\right)^2 = \frac{640}{3125}$
4	${}^5C_4 \left(\frac{4}{5}\right)^4 \left(\frac{1}{5}\right)^1 = \frac{1280}{3125}$
5	${}^5C_5 \left(\frac{4}{5}\right)^5 \left(\frac{1}{5}\right)^0 = \frac{1024}{3125}$

- 5) Given that Mean = 30 and S.D. = 5

$$\text{Thus, } np = 30, \sqrt{npq} = 5$$

$$\Rightarrow np = 30, npq = 25$$

$$\frac{npq}{np} = \frac{25}{30} = \frac{5}{6} \Rightarrow q = \frac{5}{6}, p = 1 - q = 1 - \frac{5}{6} = \frac{1}{6}, n \left(\frac{1}{6}\right) = 30 \Rightarrow n = 180$$

Check Your Progress 2

- 1) Let X be the Poisson variable “Number of fatal accidents in a year”.

$$\text{Here } n = 350, p = \frac{1}{1400}$$

$$\Rightarrow \lambda = np = (350) \left(\frac{1}{1400}\right) = 0.25.$$

By Poisson distribution,

$$P[X = x] = \frac{e^{-\lambda} \cdot \lambda^x}{|x|}, x = 0, 1, 2, \dots$$

$$= \frac{e^{-0.25} (0.25)^x}{|x|}, x = 0, 1, 2, \dots$$

Therefore, P [at least one fatal accident]

$$= P[X \geq 1] = 1 - P[X < 1] = 1 - P[X = 0]$$

$$= 1 - \frac{e^{-0.25} (0.25)^0}{|0|} = 1 - e^{-0.25} = 1 - 0.78 = 0.22$$

- 2) As mean = 6, therefore, $\lambda = 6$.

As standard deviation is 2, therefore, variance = 4 $\Rightarrow \lambda = 4$.

We get two different values of λ , which is impossible. Hence, the statement is invalid.

- 3) Let λ be the mean of the distribution,
 $\therefore$ by Poisson distribution, we have

$$P[X = x] = \frac{e^{-\lambda} \lambda^x}{x!}; x = 0, 1, 2, 3, \dots$$

Given that $P[X = 1] = P[X = 2]$,

$$\therefore \frac{e^{-\lambda} \lambda^1}{1!} = \frac{e^{-\lambda} \lambda^2}{2!}$$

$$\Rightarrow \lambda = \frac{\lambda^2}{2} \Rightarrow \lambda^2 - 2\lambda = 0 \Rightarrow \lambda(\lambda - 2) = 0$$

$$\Rightarrow \lambda = 0, 2.$$

$\lambda = 0$ is rejected,

$$\therefore \lambda = 2$$

Hence, Mean = 2.

$$\text{Now, } P[X = 0] = \frac{e^{-\lambda} \lambda^0}{0!} = e^{-\lambda} = e^{-2} = 0.1353, \text{ [See the Appendix 1]}$$

$$\begin{aligned} \text{and } P[X = 4] &= \frac{e^{-\lambda} \lambda^4}{4!} = \frac{e^{-2} (2)^4}{24} = \frac{e^{-2} (16)}{24} = \frac{2}{3} (0.1353) \\ &= 2(0.0451) \\ &= 0.0902. \end{aligned}$$

Check Your Progress 3

- 1) Let X be the number on the ticket drawn randomly from an urn containing tickets numbered from 1 to 10.

$\therefore X$ is a discrete uniform random variable having the values

1, 2, 3, 4, ..., 10 with probability of each of these values equal to $\frac{1}{10}$.

Thus, the expected frequencies for the values of X are obtained as in the following table:

X	$P(X = x)$	Expected/Theoretical frequency $f(x) = N.P[X = x]$ $= 150.P[X = x]$
1	$\frac{1}{10}$	$150 \times \frac{1}{10} = 15$
2	$\frac{1}{10}$	$150 \times \frac{1}{10} = 15$
3	$\frac{1}{10}$	$150 \times \frac{1}{10} = 15$
...(for 4, 5, 6, 7, 8, 9 and 10)	$\frac{1}{10}$ in each case	$150 \times \frac{1}{10} = 15$

- 2) Here $N = 25$, $M = 10$ and $n = 2$.

$$\text{The desired probability} = P \left[\begin{array}{l} \text{none of the 2 randomly selected} \\ \text{units is found defective} \end{array} \right]$$

$$= \frac{{}^{10}C_0 \cdot {}^{25-10}C_2}{{}^{25}C_2} = \frac{(1) \cdot {}^{15}C_2}{{}^{25}C_2} = \frac{15 \times 14}{25 \times 24} = 0.35.$$

6.9 REFERENCES

IGNOU Course Material

1. Post-Graduate Diploma in Applied Statistics (PGDAST), MST 003: Probability Theory, Block 3: Discrete Probability Distributions, UNIT 9: Binomial Distribution (for Sections 6.2, 6.3 and related portions in Sections 6.0 , 6.1, 6.7, 6.8 and Check Your Progress questions)
2. PGDAST, MST 003: Probability Theory, Block 3: Discrete Probability Distributions, UNIT 10: Poisson Distribution (for Section 6.4 and related portions in Sections 6.0 , 6.1, 6.7, 6.8 and Check Your Progress questions)
3. PGDAST, MST 003: Probability Theory, Block 3: Discrete Probability Distributions, UNIT 11: Discrete Uniform and Hypergeometric Distributions (for Sections 6.5, 6.6 and related portions in Sections 6.0 , 6.1, 6.7, 6.8 and Check Your Progress questions)

Reference Book

4. “Introduction to Probability and Statistics for Engineers and Scientists”, 3rd Edition, Sheldon M. Ross, Elsevier Academic Press, 2004

Appendix 1

Value of $e^{-\lambda}$ (For Computing Poisson Probabilities) ($0 < \lambda < 1$)

λ	0	1	2	3	4	5	6	7	8	9
0.0	1.0000	0.9900	0.9802	0.9704	0.9608	0.9512	0.9418	0.9324	0.9231	0.9139
0.1	0.9048	0.8958	0.8860	0.8781	0.8694	0.8607	0.8521	0.8437	0.8353	0.8270
0.2	0.7187	0.8106	0.8025	0.7945	0.7866	0.7788	0.7711	0.7634	0.7558	0.7483
0.3	0.7408	0.7334	0.7261	0.7189	0.7118	0.7047	0.6970	0.6907	0.6839	0.6771
0.4	0.6703	0.6636	0.6570	0.6505	0.6440	0.6376	0.6313	0.6250	0.6188	0.6125
0.5	0.6065	0.6005	0.5945	0.5886	0.5827	0.5770	0.5712	0.5655	0.5599	0.5543
0.6	0.5448	0.5434	0.5379	0.5326	0.5278	0.5220	0.5160	0.5113	0.5066	0.5016
0.7	0.4966	0.4916	0.4868	0.4810	0.4771	0.4724	0.4670	0.4630	0.4584	0.4538
0.8	0.4493	0.4449	0.4404	0.4360	0.4317	0.4274	0.4232	0.4190	0.4148	0.4107
0.9	0.4066	0.4026	0.3985	0.3946	0.3906	0.3867	0.3829	0.3791	0.3753	0.3716
($\lambda=1, 2, 3, \dots, 10$)										
λ	1	2	3	4	5	6	7	8	9	10
$e^{-\lambda}$	0.3679	0.1353	0.0498	0.0183	0.0070	0.0028	0.0009	0.0004	0.0001	0.00004

Note: To obtain values of $e^{-\lambda}$ for other values of λ , use the laws of exponents i.e.

$$e^{-(a+b)} = e^{-a} \cdot e^{-b} \text{ e. g. } e^{-2.25} = e^{-2} \cdot e^{-0.25} = (0.1353)(0.7788) = 0.1054.$$

UNIT 7 CONTINUOUS PROBABILITY DISTRIBUTIONS AND EXACT SAMPLING DISTRIBUTIONS

Structure

- 7.0 Introduction
- 7.1 Objectives
- 7.2 Normal Distribution
- 7.3 Continuous Uniform Distribution
- 7.4 Introduction to χ^2 -Distribution
- 7.5 Introduction to t-Distribution
- 7.6 Introduction to F-Distribution
- 7.7 Summary
- 7.8 Solutions/Answers
- 7.9 References

7.0 INTRODUCTION

In the previous unit, we have studied standard discrete distributions. In this unit, we are going to discuss standard continuous univariate distributions. This unit introduces normal distribution, uniform distribution, chi-square distribution, t-distribution and F-distribution. Normal distribution has widespread applications. It is being used in almost all data-based research in the field of agriculture, trade, business, industry and society. For instance, normal distribution is a good approximation to the distribution of heights of randomly selected large number of students studying at the same level in a university. The continuous uniform distribution is a simple distribution that is useful in random phenomenon.

Sometimes statistics such as sample mean, sample proportion, sample variance, etc. may follow a particular sampling distribution. Based on this fact Prof. R.A. Fisher, Prof. G. Snedecor and some other statisticians worked in this area and found some exact sampling distributions which are often followed by some of the statistics. In this unit, we shall discuss some important sampling distributions such as χ^2 (read as chi-square), t and F. Generally, these sampling distributions are named on the name of the originator, that is, Fisher's F-distribution is named on Prof. R.A. Fisher.

7.1 OBJECTIVES

After studying this unit, you would be able to:

- introduce and explain the normal distribution;
- state various characteristics of the normal distribution;
- define continuous uniform and state its properties;
- describe the probability curve and properties of χ^2 -distribution;
- describe the probability curve and properties of t-distribution;
- describe the probability curve and properties of F-distribution;

- solve various practical problems based on the properties of various distributions.

7.2 NORMAL DISTRIBUTION

The concept of normal distribution was initially discovered by English mathematician Abraham De Moivre (1667-1754) in 1733. De Moivre obtained this continuous distribution as a limiting case of binomial distribution. His work was further refined by Pierre S. Laplace (1749-1827) in 1774. But the contribution of Laplace remained unnoticed for long till it was given concrete shape by Karl Gauss (1777-1855) who first made reference to it in 1809 as the distribution of errors in Astronomy. That is why the normal distribution is sometimes called Gaussian distribution. Though, normal distribution can be used as approximation to most of the other distributions, here we are going to discuss (without proof) its approximation to (i) binomial distribution and (ii) Poisson distribution.

Normal Distribution as a Limiting Case of Binomial Distribution

Normal distribution is a limiting case of binomial distribution under the following conditions:

- n , the number of trials, is indefinitely large i.e. $n \rightarrow \infty$;
- neither p (the probability of success) nor q (the probability of failure) is too close to zero.

Under these conditions, the binomial distribution can be closely associated by a normal distribution with standardised variable given by $Z = \frac{X - np}{\sqrt{npq}}$. The approximation becomes better with increasing n . In practice, the approximation is very good if both np and nq are greater than 5.

The normal distribution is symmetrical, and the curve of the distribution is mesokurtic, which is the main feature of the normal distribution.

Normal Distribution as a Limiting Case of Poisson Distribution

As we have discussed above that there is a relation between the binomial and normal distributions. It can, in fact, be shown that the Poisson distribution approaches a normal distribution with standardized variable given by $Z = \frac{X - \lambda}{\sqrt{\lambda}}$ as λ increases indefinitely.

Normal Distribution

Normal distribution is defined as follows:

Definition: A continuous random variable X is said to follow normal distribution with parameters μ ($-\infty < \mu < \infty$) and $\sigma^2 (>0)$ if it takes on any real value and its probability density function is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, -\infty < x < \infty;$$

which may also be written as

$$= \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right\}, -\infty < x < \infty.$$

Remark

- (i) The probability function represented by $f(x)$ may also be written as $f(x; \mu, \sigma^2)$.
- (ii) If a random variable X follows normal distribution with mean μ and variance σ^2 , then we may write, “ X is distributed to $N(\mu, \sigma^2)$ ” and is expressed as $X \sim N(\mu, \sigma^2)$.
- (iii) No continuous probability function, including the normal distribution, can be used to obtain the probability of occurrence of a particular value of the random variable. This is because such probability is very small, so instead of specifying the probability of taking a particular value by the random variable, we specify the probability of its lying within an interval.
- (iv) If $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X-\mu}{\sigma}$ is standard normal variate having mean ‘0’ and variance ‘1’. The values of mean and variance of standard normal variate are obtained as under, for which properties of expectation and variance are used.

$$\begin{aligned} \text{Mean of } Z \text{ i.e. } E(Z) &= E\left(\frac{X-\mu}{\sigma}\right) = \frac{1}{\sigma}[E(X-\mu)] \\ &= \frac{1}{\sigma}[E(X)-\mu] \\ &= \frac{1}{\sigma}[\mu-\mu] = 0 \quad [\because E(X) = \text{Mean of } X = \mu] \end{aligned}$$

$$\begin{aligned} \text{Variance of } Z \text{ i.e. } V(Z) &= V\left(\frac{X-\mu}{\sigma}\right) \\ &= \frac{1}{\sigma^2}[V(X-\mu)] = \frac{1}{\sigma^2}[V(X)] \\ &= \frac{1}{\sigma^2}(\sigma^2) \quad [\because \text{variance of } X \text{ is } \sigma^2] \\ &= 1. \end{aligned}$$

- (v) The probability density function of standard normal variate $Z = \frac{X-\mu}{\sigma}$ is given by $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$, $-\infty < z < \infty$.

This result can be obtained on replacing $f(x)$ by $\phi(z)$, x by z , μ by 0 and σ by 1 in the probability density function of normal variate X i.e. in

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, -\infty < x < \infty$$

- (vi) The graph of the normal probability function $f(x)$ with respect to x is famous 'bell-shaped' curve. The top of the bell is directly above the mean μ . For large value of σ , the curve tends to flatten out and for small values of σ , it has a sharp peak as shown in (Fig. 7.1):

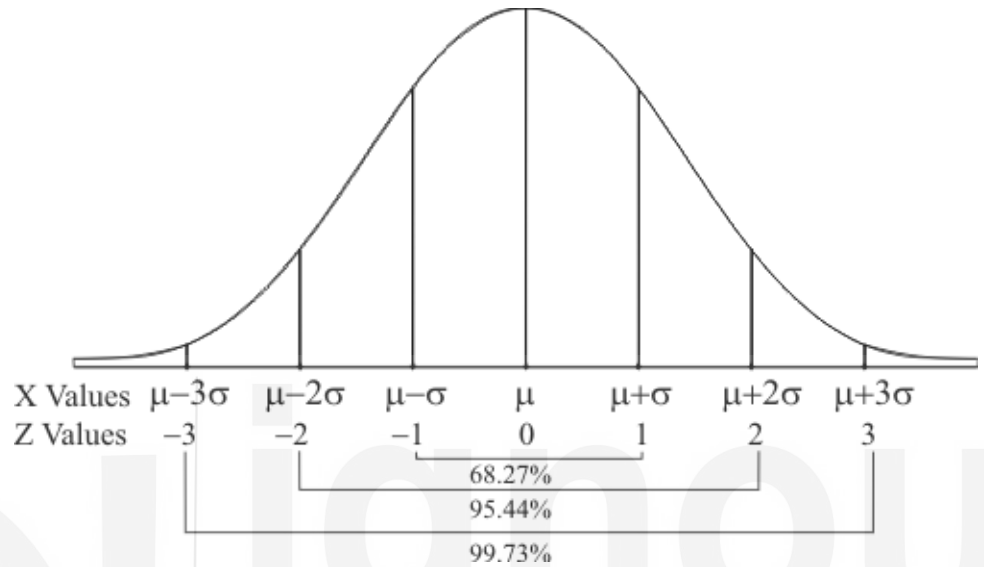


Fig. 7.1: The Normal Distribution

Normal distribution has various properties and large number of applications. It can be used as an approximation to most of the other distributions and hence is most important probability distribution in statistical analysis. Theory of estimation of population parameters and testing of hypotheses on the basis of sample statistics (to be discussed in the next unit) have also been developed using the concept of normal distribution as most of the sampling distributions tend to normality for large samples. Normal distribution has become widely and uncritically accepted on the basis of much practical work. As a result, it holds a central position in Statistics.

Let us now take some examples of writing the probability function of normal distribution when mean and variance are specified, and vice-versa.

Example 1: (i) If $X \sim N(40, 25)$ then write down the p.d.f. of X

(ii) If $X \sim N(-36, 20)$ then write down the p.d.f. of X

(iii) If $X \sim N(0, 2)$ then write down the p.d.f. of X

Solution: (i) Here we are given $X \sim N(40, 25)$

$\therefore$ in usual notations, we have

$$\mu = 40, \sigma^2 = 25 \Rightarrow \sigma = \pm\sqrt{25}$$

$$\Rightarrow \sigma = 5 \quad [\because \sigma > 0 \text{ always}]$$

Now, the p.d.f. of random variable X is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$= \frac{1}{5\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-40}{5}\right)^2}, \quad -\infty < x < \infty$$

(ii) Here we are given $X \sim N(-36, 20)$.

$\therefore$ in usual notations, we have

$$\mu = -36, \sigma^2 = 20 \Rightarrow \sigma = \sqrt{20}$$

Now, the p.d.f. of random variable X is given by

$$\begin{aligned} f(x) &= \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \\ &= \frac{1}{\sqrt{20}\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-(-36)}{\sqrt{20}}\right)^2} = \frac{1}{\sqrt{40\pi}} e^{-\frac{1}{40}(x+36)^2} \\ &= \frac{1}{2\sqrt{10\pi}} e^{-\frac{1}{40}(x+36)^2}, \quad -\infty < x < \infty \end{aligned}$$

(iii) Here we are given $X \sim N(0, 2)$.

$\therefore$ in usual notations, we have

$$\mu = 0, \sigma^2 = 2 \Rightarrow \sigma = \sqrt{2}$$

Now, the p.d.f. of random variable X is given by

$$\begin{aligned} f(x) &= \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} = \frac{1}{\sqrt{2}\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-0}{\sqrt{2}}\right)^2} \\ &= \frac{1}{2\sqrt{\pi}} e^{-\frac{1}{4}x^2}, \quad -\infty < x < \infty \end{aligned}$$

Example 2: Below, in each case, there is a p.d.f. of a normally distributed random variable. Obtain the parameters (mean and variance) of the variable.

$$(i) f(x) = \frac{1}{6\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-46}{6}\right)^2}, \quad -\infty < x < \infty$$

$$(ii) f(x) = \frac{1}{4\sqrt{2\pi}} e^{-\frac{1}{32}(x-60)^2}, \quad -\infty < x < \infty$$

Solution: (i) $f(x) = \frac{1}{6\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-46}{6}\right)^2}, \quad -\infty < x < \infty$

Comparing it with,

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

we have

$$\mu = 46, \quad \sigma = 6$$

$$\therefore \text{Mean} = \mu = 46, \text{ variance} = \sigma^2 = 36$$

$$\begin{aligned} \text{(ii) } f(x) &= \frac{1}{4\sqrt{2\pi}} e^{-\frac{1}{32}(x-60)^2}, \quad -\infty < x < \infty \\ &= \frac{1}{4\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-60}{4}\right)^2} \end{aligned}$$

Comparing it with,

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

we get

$$\mu = 60, \quad \sigma = 4$$

$$\therefore \text{Mean} = \mu = 60, \text{ variance} = \sigma^2 = 16$$

Now, we are going to state some important properties of Normal distribution in the next section.

Main Characteristics of Normal Distribution

The normal probability distribution with mean μ and variance σ^2 has the following properties:

- i) The curve of the normal distribution is bell-shaped as shown in Fig. 7.1.
- ii) The curve of the distribution is completely symmetrical about $x = \mu$, i.e. if we fold the curve at $x = \mu$, both the parts of the curve are the mirror images of each other.
- iii) For normal distribution, **Mean = Median = Mode**
- iv) $f(x)$, being the probability, can never be negative and hence no portion of the curve lies below x-axis.
- v) Though x-axis becomes closer and closer to the normal curve as the magnitude of the value of x goes towards ∞ or $-\infty$, yet it never touches it.
- vi) Normal curve has only one mode.
- vii) Central moments of Normal distribution are

$$\mu_1 = 0, \mu_2 = \sigma^2, \mu_3 = 0, \mu_4 = 3\sigma^4$$

The distribution is symmetrical and curve is always mesokurtic.

Note: Not only μ_1 and μ_3 are 0 but all the odd order central moments are zero for a normal distribution.

- viii) For normal curve,

$$Q_3 - \text{Median} = \text{Median} - Q_1$$

i.e. First and third quartiles of normal distribution are equidistant from median.

ix) Quartile Deviation (Q.D.) = $\frac{Q_3 - Q_1}{2}$ is approximately equal to $\frac{2}{3}$ of the standard deviation.

x) Mean deviation is approximately equal to $\frac{4}{5}$ of the standard deviation.

xi) If $X_1, X_2, \dots, X_n$ are independent normal variables with means $\mu_1, \mu_2, \dots, \mu_n$ and variances $\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2$ respectively, then the linear combination $a_1X_1 + a_2X_2 + \dots + a_nX_n$ of $X_1, X_2, \dots, X_n$ is also a normal variable with mean

$$a_1\mu_1 + a_2\mu_2 + \dots + a_n\mu_n \text{ and variance } a_1^2\sigma_1^2 + a_2^2\sigma_2^2 + \dots + a_n^2\sigma_n^2.$$

xii) Particularly, sum or difference of two independent normal variates is also a normal variate. If X and Y are two independent normal variates with means μ_1, μ_2 and variances σ_1^2, σ_2^2 , then

$$X + Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2) \text{ and } X - Y \sim N(\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2).$$

Also, if $X_1, X_2, \dots, X_n$ are independent variates each distributed as:

$$N(\mu, \sigma^2), \text{ then their mean } \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

xiii) Area property:

$$P[\mu - \sigma < X < \mu + \sigma] = \int_{\mu - \sigma}^{\mu + \sigma} f(x) dx = 0.6827,$$

$$\text{Or } P[-1 < Z < 1] = \int_{-1}^1 \phi(z) dz = 0.6827,$$

$$P[\mu - 2\sigma < X < \mu + 2\sigma] = \int_{\mu - 2\sigma}^{\mu + 2\sigma} f(x) dx = 0.9544,$$

$$\text{Or } P[-2 < Z < 2] = \int_{-2}^2 \phi(z) dz = 0.9544, \text{ and}$$

$$P[\mu - 3\sigma < X < \mu + 3\sigma] = \int_{\mu - 3\sigma}^{\mu + 3\sigma} f(x) dx = 0.9973.$$

$$\text{Or } P[-3 < Z < 3] = \int_{-3}^3 \phi(z) dz = 0.9973.$$

Let us now take up some problems based on properties of Normal Distribution in the next section.

Example 3: If X_1 and X_2 are two independent variates each distributed as $N(0, 1)$, then write the distribution of (i) $X_1 + X_2$. (ii) $X_1 - X_2$.

Solution: We know that, if X_1 and X_2 are two independent normal variates s.t.

$$X_1 \sim N(\mu_1, \sigma_1^2) \text{ and } X_2 \sim N(\mu_2, \sigma_2^2)$$

then

$$X_1 + X_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2), \text{ and}$$

$$X_1 - X_2 \sim N(\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2) \quad [\text{See Property xii (Section 7.3)}]$$

Here, $X_1 \sim N(0,1)$, $X_2 \sim N(0,1)$

$$\therefore \text{ i) } X_1 + X_2 \sim N(0+0, 1+1)$$

$$\text{ i.e. } X_1 + X_2 \sim N(0, 2), \text{ and}$$

$$\text{ ii) } X_1 - X_2 \sim N(0-0, 1+1)$$

$$\text{ i.e. } X_1 - X_2 \sim N(0, 2)$$

Example 4: If $X \sim N(30, 25)$, find the mean deviation about mean.

Solution: Here $\mu = 30$, $\sigma^2 = 25 \Rightarrow \sigma = 5$.

$$\therefore \text{ Mean deviation about mean} = \sqrt{\frac{2}{\pi}} \sigma = \sqrt{\frac{2}{\pi}} \cdot 5 = 5\sqrt{\frac{2}{\pi}}$$

You can now try some exercises based on the properties of normal distribution which you have studied in the present unit.

Check Your Progress 1

1) Write down the p.d.f. of r. v. X in each of the following cases:

$$\text{ (i) } X \sim N\left(\frac{1}{2}, \frac{4}{9}\right)$$

$$\text{ (ii) } X \sim N(-40, 16)$$

.....
.....

2) Below, in each case, is given the p.d.f. of a normally distributed random variable. Obtain the parameters (mean and variance) of the variable.

$$\text{ (i) } f(x) = \frac{1}{2\sqrt{2\pi}} e^{-\frac{x^2}{8}}, \quad -\infty < x < \infty$$

$$\text{ (ii) } f(x) = \frac{1}{2\sqrt{\pi}} e^{-\frac{1}{4}(x-2)^2}, \quad -\infty < x < \infty$$

.....
.....

3) If X_1 and X_2 are two independent normal variates with means 30, 40 and variances 25, 35 respectively. Find the mean and variance of

$$\text{ i) } X_1 + X_2$$

$$\text{ ii) } X_1 - X_2$$

.....
.....

7.3 CONTINUOUS UNIFORM DISTRIBUTION

The uniform (or rectangular) distribution is a very simple distribution. It provides a useful model for a few random phenomena like having random number from the interval $[0, 1]$, then one is thinking of the value of a uniformly distributed random variable over the interval $[0, 1]$.

Definition: A random variable X is said to follow a continuous uniform (rectangular) distribution over an interval (a, b) if its probability density function is given by

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a < x < b \\ 0, & \text{otherwise} \end{cases}$$

The distribution is called uniform distribution since it assumes a constant (uniform) value for all x in (a, b) . If we draw the graph of $y = f(x)$ over x -axis and between the ordinates $x = a$ and $x = b$ (say), it describes a rectangle as shown in Fig. 7.2

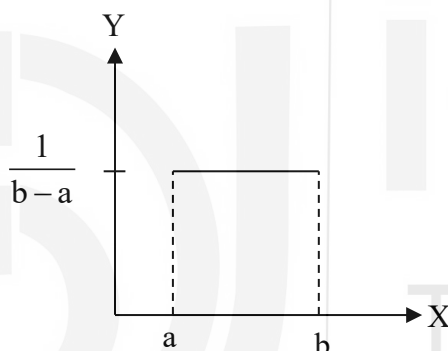


Fig. 7.2: Graph of a uniform function

A uniform variate X on the interval (a, b) is written as $X \sim U[a, b]$

Cumulative Distribution Function

The cumulative distribution function of the uniform random variate over the interval (a, b) is given by:

$$F(x) = \begin{cases} 0 & \text{for } x \leq a \\ \frac{x-a}{b-a} & \text{for } a < x < b \\ 1 & \text{for } x \geq b \end{cases}$$

On plotting its graph, we have

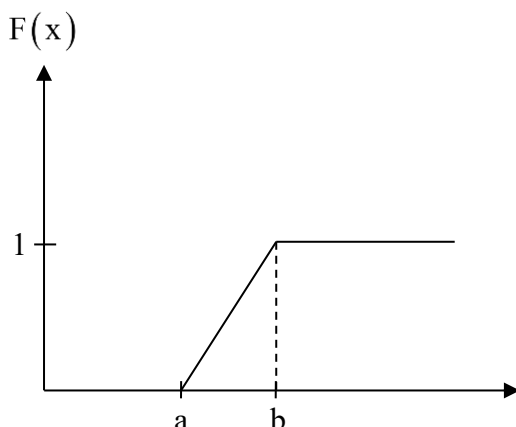


Fig. 7.3: Graph of distribution function

Mean and Variance of Uniform Distribution

$$\text{Mean} = \frac{a+b}{2} \text{ and Variance} = \frac{(b-a)^2}{12}.$$

Let us now take up some examples of continuous uniform distribution.

Example 5: If X is uniformly distributed with mean 2 and variance 12, find $P[X < 3]$.

Solution: Let $X \sim U[a, b]$

$\therefore$ probability density function of X is

$$f(x) = \frac{1}{b-a}, \quad a < x < b.$$

Now as Mean = 2

$$\Rightarrow \frac{a+b}{2} = 2$$

$$\Rightarrow a+b = 4 \quad \dots (1)$$

Variance = 12

$$\Rightarrow \frac{(b-a)^2}{12} = 12$$

$$\Rightarrow (b-a)^2 = 144$$

$$\Rightarrow b-a = \pm 12$$

$$\Rightarrow b-a = 12 \quad \dots (2)$$

$\left[\begin{array}{l} \because b-a = -12, \text{ being negative is} \\ \text{rejected as } b \text{ should be greater than } a \\ \Rightarrow b-a \text{ should be positive} \end{array} \right]$

Adding (1) and (2), we have

$$2b = 16$$

$$\Rightarrow b = 8 \text{ and hence } a = 4 - 8 = -4$$

$$\therefore f(x) = \frac{1}{b-a} = \frac{1}{8-(-4)} = \frac{1}{12} \text{ for } -4 < x < 8.$$

$$\begin{aligned} \text{Thus, the desired probability} &= P[X < 3] = \int_{-4}^3 \frac{1}{12} dx = \frac{1}{12} \int_{-4}^3 1 dx = \frac{1}{12} [x]_{-4}^3 \\ &= \frac{1}{12} [3 - (-4)] = \frac{7}{12}. \end{aligned}$$

Example 6: Metro trains are scheduled every 5 minutes at a certain station. A person comes to the station at a random time. Let the random variable X count the number of minutes he/she has to wait for the next train. Assume X has a uniform distribution over the interval $(0, 5)$. Find the probability that he/she has to wait at least 3 minutes for the train.

Solution: As X follows uniform distribution over the interval $(0, 5)$,

∴ probability density function of X is

$$f(x) = \frac{1}{b-a} = \frac{1}{5-0} = \frac{1}{5}, 0 < x < 5$$

Thus, the desired probability

$$\begin{aligned} P[X \geq 3] &= \int_3^5 f(x) dx = \int_3^5 \frac{1}{5} dx = \frac{1}{5} \int_3^5 (1) dx \\ &= \frac{1}{5} [x]_3^5 = \frac{1}{5} (5-3) = \frac{2}{5} = 0.4 \end{aligned}$$

Check Your Progress 2

1) Suppose that X is uniformly distributed over $(-a, a)$. Determine 'a' so that

i) $P[X > 4] = \frac{1}{3}$

ii) $P[X < 1] = \frac{3}{4}$

2) A random variable X has a uniform distribution over $(-2, 2)$. Find k for which $P[X > k] = \frac{1}{2}$.

7.4 INTRODUCTION TO χ^2 -DISTRIBUTION

Before describing the chi-square (χ^2) distribution, first, we will discuss a very useful concept “degrees of freedom”. The exact sampling distributions are described with the help of degrees of freedom.

Degrees of Freedom (df)

The term degree of freedom (df) is related to the independence of sample observations. In general, the number of degree of freedom is the total number of observations minus the number of independent constraints or restrictions imposed on the observations.

For a sample of n observations, if there are k restrictions among observations ($k < n$), then the degrees of freedom will be $(n-k)$.

For example, suppose there are two observations X_1, X_2 and a restriction (condition) is imposed that their total should be equal to 100. Then we can arbitrarily assign value to any one of these two as 30, 98, 52, etc. but the value

of rest is automatically determined by a simple adjustment like $X_2 = (100 - X_1)$. Therefore, we can say that only one observation is independent or there is one degree of freedom.

Consider another example, suppose there are 50 observations as $X_1, X_2, \dots, X_{50}$ and the restriction (condition) are imposed on these, like:

$$(i) \sum_{i=1}^{50} X_i = 2000 \quad (ii) \sum_{i=1}^{50} X_i^2 = 10000 \quad (iii) X_5 = 2X_4$$

The restriction (i) reveals that the sum of observations equal to 2000 (ii) reveals that the sum of squares of observations equal to 10000 and restriction (iii) implies that the fifth observation is double of the fourth observation.

With these three restrictions (i), (ii) and (iii), there shall be only 47 observations independently chosen and rest three could be obtained by solving equations or by adjustments. The degrees of freedom of this set of observations of size 50 is now $(50 - 3) = 47$.

Chi-square Distribution

The chi-square distribution is first discovered by Helmer in 1876 and later independently explained by Karl- Pearson in 1900. The chi-square distribution was discovered mainly as a measure of goodness of fit in case of frequency.

If a random sample $X_1, X_2, \dots, X_n$ of size n is drawn from a normal population having mean μ and variance σ^2 then the sample variance can be defined as

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

or

$$\sum_{i=1}^n (X_i - \bar{X})^2 = (n-1)S^2 = vS^2$$

where, $v = n - 1$ and the symbol v read as 'nu'.

Generally, v symbol is used to represent the general degrees of freedom. Its value may be $n, n - 1$, etc.

Thus, the variate $\chi^2 = \frac{vS^2}{\sigma^2}$, which is the ratio of sample variance multiplied by its degrees of freedom and the population variance follows the χ^2 -distribution with v degrees of freedom.

Properties of χ^2 -distribution

In the previous section, we have discussed the χ^2 -distribution and its probability density function. Now, in this section, we shall discuss some important properties of χ^2 -distribution as given below:

1. The probability curve of the chi-square distribution lies in the first quadrant because the range of χ^2 -variate is from 0 to ∞ .
2. Chi-square distribution has only one parameter n , that is, the degrees of freedom.
3. Chi-square probability curve is highly positively skewed.
4. Chi-square distribution is a uni-modal distribution, that is, it has a single mode.

5. The mean and variance of the chi-square distribution with n df are n and $2n$ respectively.

After deliberating the properties of χ^2 -distribution above, some of these are discussed in detail in the subsequent sub-sections.

Probability Curve of χ^2 -distribution

The probability distribution of all possible values of a statistic is known as the sampling distribution of that statistic. The shape of the probability distribution of a statistic can be shown by the probability curve. In this sub-section, we will discuss the probability curve of chi-square distribution.

A rough sketch of the probability curve of chi-square distribution is shown in Fig. 7.4 at different values of degrees of freedom as $n = 1, 4, 10$ and 22 .

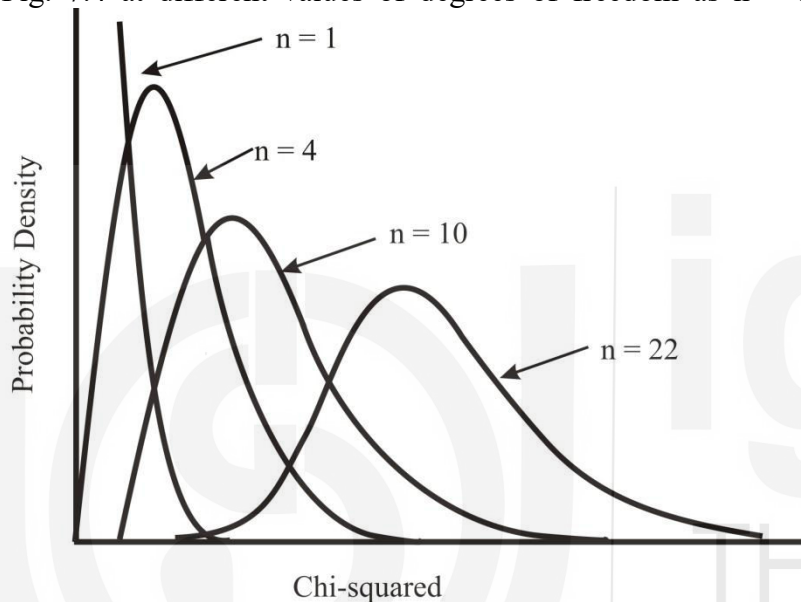


Fig. 7.4: Chi-square probability curves for $n = 1, 4, 10$ and 22

After looking at the probability curves of χ^2 -distribution for $n = 1, 4, 10$ and 22 , one can understand that probability curve takes shape of inverse J for $n = 1$. The probability curve of chi-square distribution gets skewed more and more to the right as n becomes smaller and smaller. It becomes more and more symmetrical as n increases because as n tends to ∞ , the χ^2 -distribution becomes a normal distribution.

Mean and Variance of χ^2 -distribution

The mean and variance of χ^2 -distribution are derived with the help of the moment about origin. The mean and variance of the chi-square distribution with n df are given by

$$\text{Mean} = n \text{ and Variance} = 2n$$

Hence, we have observed that the variance is twice the mean of the chi-square distribution.

Example 7: What are the mean and variance of the chi-square distribution with 5 degrees of freedom?

Solution: We know that the mean and variance of the chi-square distribution with n degrees of freedom are

$$\text{Mean} = n \text{ and Variance} = 2n$$

In our case, $n = 5$, therefore, Mean = 5 and Variance = 10.

Applications of χ^2 -Distribution

The applications of chi-square distribution are very wide in Statistics. Some of them are listed below. The chi-square distribution is used:

1. To test the hypothetical value of population variance.
2. To test the goodness of fit, that is, to judge whether there is a discrepancy between theoretical and experimental observations.
3. To test the independence of two attributes, that is, to judge whether the two attributes are independent.

Try the following check your progress to make sure that you have understood the chi-square distribution.

Check Your Progress 3

- 1) What are the mean and variance of the chi-square distribution with 10 degrees of freedom?

.....
.....

- 2) List the applications of the chi-square distribution.

.....
.....

7.5 INTRODUCTION TO t-DISTRIBUTION

The t-distribution was discovered by W.S. Gosset in 1908. He was better known by the pseudonym ‘Student’ and hence t-distribution is called ‘Student’s t-distribution’.

If a random sample $X_1, X_2, \dots, X_n$ of size n is drawn from a normal population having mean μ and variance σ^2 then we know that the sample mean $\bar{X}$ is distributed normally with mean μ and variance σ^2 / n , that is, if $X_i \sim N(\mu, \sigma^2)$

then $\bar{X} \sim N(\mu, \sigma^2 / n)$, and also the variate

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

is distributed normally with mean 0 and variance 1, i.e. $Z \sim N(0, 1)$.

In general, the standard deviation σ is not known and in such a situation the only alternative left is to estimate the unknown σ^2 . The value of sample variance (S^2) is used to estimate the unknown σ^2 where,

$$S^2 = \frac{1}{(n-1)} \sum_{i=1}^n (X_i - \bar{X})^2$$

Thus, in this case, the variate $\frac{\bar{X} - \mu}{S / \sqrt{n}}$ is not normally distributed, whereas it

follows t-distribution with $(n-1)$ df i.e.

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{(n-1)} \quad \dots (3)$$

$$\text{where } S^2 = \frac{1}{(n-1)} \sum_{i=1}^n (X_i - \bar{X})^2$$

The t-variate is a widely used variable and its distribution is called student's t-distribution on the pseudonym name 'Student' of W.S. Gosset.

After describing the t-distribution, we try to calculate the value of t-variate as in the given example.

Example 8: The life of light bulbs manufactured by company A is known to be normally distributed. The CEO of the company claims that an average lifetime of the light bulbs is 300 days. A researcher randomly selects 25 bulbs for testing the lifetime and he observed the average lifetime of the sampled bulbs is 290 days with standard deviation of 50 days. Calculate the value of t-variate.

Solution: Here, we are given that

$$\mu = 300, n = 25, \bar{X} = 290 \text{ and } S = 50$$

The value of t-variate can be calculated by the formula given below

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

Therefore, we have

$$t = \frac{290 - 300}{50/\sqrt{25}} = \frac{-10}{10} = -1$$

Properties of t-Distribution

In the previous section, we have discussed the t-distribution briefly. Now, we shall discuss some of the important properties of the t-distribution. The t-distribution has the following properties:

1. The t-distribution is a uni-modal distribution, that is, t-distribution has a single mode.
2. The mean and variance of the t-distribution with n df are zero and $\frac{n}{n-2}$ respectively for $n > 2$.
3. The probability curve of t-distribution is similar in shape to the standard normal distribution and is symmetric about $t = 0$ line but flatter than the normal curve.
4. The probability curve is bell-shaped and asymptotic to the horizontal axis.

The properties deliberated above are some of the important properties of the t-distribution. We shall discuss some of these properties such as probability curve, mean and variance of the t-distribution in detail in the subsequent sub-sections.

Probability Curve of t-distribution

The probability curve of t-distribution is bell-shaped and symmetric about $t = 0$ line. The probability curves of t-distribution are shown in Fig. 7.5 at two different values of degrees of freedom, $n = 4$ and $n = 12$.

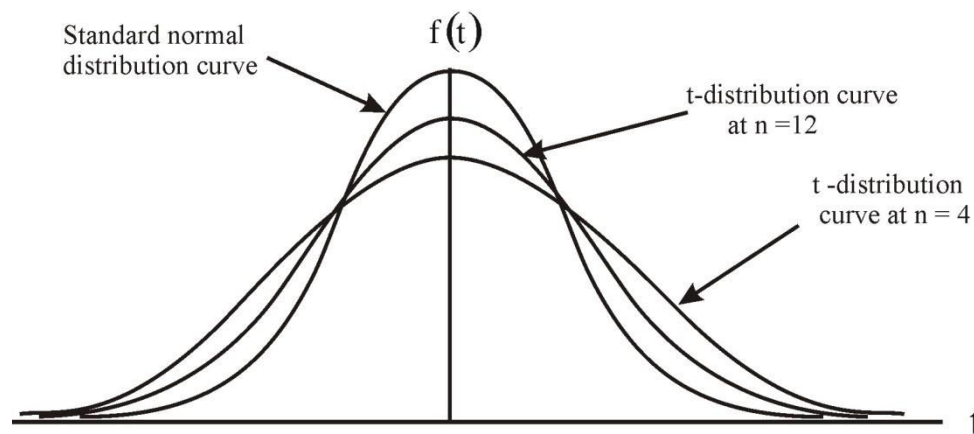


Fig. 7.5: Probability curves for t-distribution at $n = 4, 12$ along with standard normal curve

In the figure given above, we have drawn the probability curves of t-distribution at two different values of degrees of freedom with the probability curve of standard normal distribution. By looking at the figure, one can easily understand that the probability curve of t-distribution is similar in shape to that of normal distribution and asymptotic to the horizontal-axis whereas it is flatter than the standard normal curve. The probability curve of the t-distribution tends to the normal curve as the value of n increases. Therefore, for a sufficiently large value of sample size $n(> 30)$, the t-distribution tends to the normal distribution.

Mean and Variance of t-distribution

The probability curve shows that t-distribution is symmetrical about $t = 0$ line. Thus, mean of the t-distribution is zero.

The mean and variance of the t-distribution with n df are given by

$$\text{Mean} = 0 \text{ and Variance} = \frac{n}{n-2} \text{ for } n > 2$$

Applications of t-Distribution

After learning the important properties of t-distribution and its mean and variance. Now, you may be interested to know the applications of the t-distribution. The t-distribution has a wide number of applications in Statistics. The t-distribution is used:

1. To test the hypothesis about the population mean.
2. To test the hypothesis about the difference of two population means of two normal populations.
3. To test the hypothesis that the population correlation coefficient is zero.

Check Your Progress 4

- 1) If the scores on an IQ test of the students of a class are assumed to be normally distributed with mean of 60. From the class, 15 students are randomly selected and an IQ test of a similar level is conducted. The average test score and standard deviation of test scores in the sample group are found to be 65 and 12 respectively. Calculate the value of t-variate.

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.....

.....

.....

- 2) What are the mean and variance of t-distribution with 20 degrees of freedom?

.....

- 3) Write any three applications of t-distribution.

.....

7.6 INTRODUCTION TO F-DISTRIBUTION

F-distribution was introduced by Prof. R. A. Fisher and defined as the ratio of two independent chi-square variates when divided by their respective degrees of freedom. If we draw a random sample $X_1, X_2, \dots, X_{n_1}$ of size n_1 from a normal population with mean μ_1 and variance σ_1^2 and another independent random sample $Y_1, Y_2, \dots, Y_{n_2}$ of size n_2 from another normal population with mean μ_2 and variance σ_2^2 respectively. Also, $v_1 S_1^2 / \sigma_1^2$ is distributed as chi-square variate with v_1 df i.e.

$$\chi_1^2 = \frac{v_1 S_1^2}{\sigma_1^2} \sim \chi_{(v_1)}^2 \quad \dots (4)$$

$$\text{where, } v_1 = n_1 - 1, \bar{X} = \frac{1}{n_1} \sum_{i=1}^{n_1} X_i \text{ and } S_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (X_i - \bar{X})^2$$

Similarly, $v_2 S_2^2 / \sigma_2^2$ is distributed as chi-square variate with v_2 df i.e.

$$\chi_2^2 = \frac{v_2 S_2^2}{\sigma_2^2} \sim \chi_{(v_2)}^2 \quad \dots (5)$$

$$\text{where, } v_2 = n_2 - 1, \bar{Y} = \frac{1}{n_2} \sum_{i=1}^{n_2} Y_i \text{ and } S_2^2 = \frac{1}{n_2 - 1} \sum_{i=1}^{n_2} (Y_i - \bar{Y})^2$$

Now, if we take the ratio of the above chi-square variates given in equations (4) and (5), then we get

$$\begin{aligned} \frac{\chi_1^2}{\chi_2^2} &= \frac{v_1 S_1^2 / \sigma_1^2}{v_2 S_2^2 / \sigma_2^2} \\ \Rightarrow \frac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2} &= \frac{\chi_1^2 / v_1}{\chi_2^2 / v_2} \sim F_{(v_1, v_2)} \quad \dots (6) \end{aligned}$$

By observing the above form given in equation (6) we reveal that the ratio of two independent chi-square variates when divided by their respective degrees of freedom follows F-distribution where, v_1 and v_2 are called the degrees of freedom of F-distribution.

Now, if variances of both the populations are equal i.e. $\sigma_1^2 = \sigma_2^2$ then F-variate is written in the form of ratio of two sample variances i.e.

$$F = \frac{S_1^2}{S_2^2} \sim F_{(v_1, v_2)} \quad \dots (4)$$

After describing the F-distribution, we try to calculate the value of F-variate as in the given example.

Example 9: A statistician selects 7 women randomly from the population of women, and 12 men from a population of men. The table given below shows the standard deviation of each sample and population:

Population	Population Standard Deviation	Sample Standard Deviation
Women	40	45
Men	80	75

Compute the value of F-variate.

Solution: Here, we are given that

$$n_1 = 7, n_2 = 12, \sigma_1 = 40, \sigma_2 = 80, S_1 = 45, S_2 = 75$$

The value of F-variate can be calculated by the formula given below

$$F = \frac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2}$$

where, S_1^2 & S_2^2 are the sample variances.

Therefore, we have

$$\begin{aligned} F &= \frac{(45)^2 / (40)^2}{(75)^2 / (80)^2} \\ &= \frac{1.27}{0.88} = 1.93 \end{aligned}$$

For the above calculation, the degrees of freedom v_1 for women's data are $7 - 1 = 6$ and the degrees of freedom v_2 for men's data are $12 - 1 = 11$.

After introducing the F-distribution, one may be interested in knowing the properties of this distribution. Now, we shall discuss some of the important properties of F-distribution.

Properties of F-Distribution

The F-distribution has wide properties in Statistics. Some of them are as follow:

1. The probability curve of F-distribution is positively skewed. The curve becomes highly positive skewed when v_2 is smaller than v_1 .
2. F-distribution curve extends on abscissa from 0 to ∞ .
3. F-distribution is a uni-modal distribution, that is, it has a single mode.
4. The square of t-variate with v df follows F-distribution with 1 and v degrees of freedom.

5. The mean of F-distribution with (v_1, v_2) df is $\frac{v_2}{v_2 - 2}$ for $v_2 > 2$.
6. The variance of F-distribution with (v_1, v_2) df is

$$\frac{2v_2^2(v_1 + v_2 - 2)}{v_1(v_2 - 2)^2(v_2 - 4)} \text{ for } v_2 > 4.$$

7. If we interchange the degrees of freedom v_1 and v_2 then there exists a very useful relation as

$$F_{(v_1, v_2), (1-\alpha)} = \frac{1}{F_{(v_2, v_1), \alpha}}$$

Probability Curve of F-distribution

F-variate is the ratio of two sample variances when population variances are equal and follows F-distribution with v_1 and v_2 df i.e.

$$F = \frac{S_1^2}{S_2^2} \sim F_{(v_1, v_2)} \quad \dots (6)$$

where, S_1^2 and S_2^2 are the sample variances of two normal populations. As you have studied that the F-variate varies from 0 to ∞ , therefore, it is always positive so probability curve of F-distribution wholly lies in the first quadrant. These v_1 and v_2 are the degrees of freedom and are called the parameters of F-distribution. Hence, the shape of the probability curve F-distribution depends on v_1 and v_2 . The probability curves of F-distribution for various pairs of degrees of freedom (v_1, v_2) i.e. (5, 5), (5, 20) and (20, 5) are shown in Fig. 7.6 given below:

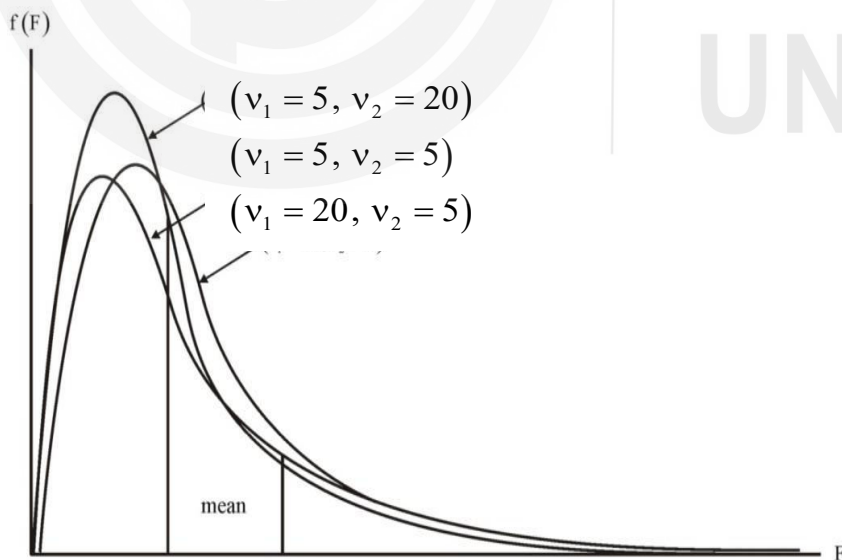


Fig. 7.6: Probability curves of F-distribution for (5, 5), (5, 20) and (20, 5) degrees of freedom.

After looking at the probability curves of F-distribution drawn above, we can observe that the probability curve of the F-distribution is a uni-model curve. And by increasing in the first degrees of freedom from $v_1 = 5$ to $v_1 = 20$ the mean of the distribution (shown by the vertical line) does not change but probability

curve shifted from the tail to the centre of the distribution whereas increasing in the second degrees of freedom from $v_2 = 5$ to $v_2 = 20$ the mean of the distribution (shown by the vertical line) decrease and the probability curve shifted from the tail to the centre of the distribution. One can also get an idea about the skewness of the F-distribution, we have observed from the probability curve of the F-distribution that it is positively skewed curve and it becomes very highly positive skewed if v_2 becomes small.

Mean and Variance of F-distribution

The mean and variance of F-distribution are given below:

Mean

$$= \frac{v_2}{v_2 - 2} \quad \text{for } v_2 > 2$$

Variance

$$= \frac{2v_2^2(v_1 + v_2 - 2)}{v_1(v_2 - 2)^2(v_2 - 4)} \quad \text{for } v_2 > 4$$

From the above derivation of mean and variance, we can conclude that mean of F-distribution is independent of v_1 .

Applications of F-Distribution

Though F-distribution has many applications some of them are listed here.

The F-distribution has the following applications:

- F-distribution is used to test the hypothesis about equality of the variances of two normal populations.
- F-distribution is used to test the hypothesis about multiple correlation coefficients.
- F-distribution is used to test the hypothesis about correlation ratio.
- F-distribution is used to test the equality of means of k-populations, when one characteristic of the population is considered i.e. F-distribution is used in one-way analysis of variance.
- F-distribution is used to test the equality of k-population means for two characteristics at a time i.e. F-distribution is used in two-way analysis of variance.

Relation Between t , χ^2 and F-Distributions

After describing the exact sampling distributions χ^2 , t & F , we shall explain the relationship between t & F -distributions and χ^2 & F -distributions. These relations can be described in terms of the following two statements:

Statement 1: If a variate t follows Student's t -distribution with v df, then square of t follows F-distribution with $(1, v)$ df i.e. if $t \sim t_{(v)}$ then $t^2 \sim F(1, v)$.

Statement 2: If F -variate follows F-distribution with (v_1, v_2) df and if

$v_2 \rightarrow \infty$ then $\chi^2 = v_1 F$ follows a χ^2 -distribution with v_1 df.

The proofs of these statements are beyond the scope of this course.

Check Your Progress 5

- 1) For a survey, 15 students are selected randomly from class A and 10 students are selected randomly from class B. At the stage of the analysis of the sample data, the following information is available:

Class	Population Standard Deviation	Sample Standard Deviation
Class A	65	60
Class B	45	50

Calculate the value of F-variate.

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- 2) Write any five properties of F-distribution.

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- 3) What are the mean and variance of F-distribution with $v_1 = 5$ and $v_2 = 10$ degrees of freedom?

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- 4) Write four applications of F-distribution.

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Tabulated Values Of Z, t, χ^2 and F

The area or probability of normal, t, χ^2 and F-distributions can be calculated with the help of tables (See Appendix). You can also find the area or probability of a variate using tables. You can refer to the references for the same.

We now end this unit by giving a summary of what we have covered in it.

7.7 SUMMARY

The following main points have been covered in this unit:

- 1) A continuous random variable X is said to follow normal distribution with parameters μ ($-\infty < \mu < \infty$) and $\sigma^2(>0)$ if it takes on any real value and its probability density function is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty$$

- 2) If $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X-\mu}{\sigma}$ is standard normal variate.
- 3) The curve of the normal distribution is bell-shaped and is completely symmetrical about $x = \mu$.
- 4) For normal distribution, Mean = Median = Mode.
- 5) Sum of independent normal variables is also a normal variable.
- 6) A random variable X is said to follow a continuous uniform (rectangular) distribution over an interval (a, b) if its probability density function is given by

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a < x < b \\ 0, & \text{otherwise} \end{cases}$$

- 7) For **continuous uniform distribution**, **Mean is $(a+b)/2$** and **variance is $(b-a)^2/12$**
- 8) The unit introduced χ^2 -distribution, its probability curve, its properties and applications.
- 9) The unit introduced t-distribution, its probability curve, its properties and applications.
- 10) The unit introduced F-distribution, its probability curve, its properties and applications.

7.8 SOLUTIONS/ANSWERS

Check Your Progress 1

- 1) (i) Here we are given $X \sim N\left(\frac{1}{2}, \frac{4}{9}\right)$

$\therefore$ in usual notations, we have

$$\mu = \frac{1}{2}, \quad \sigma^2 = \frac{4}{9} \quad \Rightarrow \quad \sigma = \frac{2}{3}$$

Now, p.d.f. of r.v. X is given by

$$\begin{aligned} f(x) &= \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty \\ &= \frac{1}{\frac{2}{3}\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-1/2}{2/3}\right)^2} \\ &= \frac{3}{2\sqrt{2\pi}} e^{-\frac{9}{2}\left(\frac{2x-1}{4}\right)^2}, \quad -\infty < x < \infty \end{aligned}$$

- (ii) Here we are given $X \sim N(-40, 16)$

∴ in usual notations, we have

$$\mu = -40, \quad \sigma^2 = 16 \Rightarrow \sigma = 4$$

Now, p.d.f. of r.v. X is given by

$$\begin{aligned} f(x) &= \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty \\ &= \frac{1}{4\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-(-40)}{4}\right)^2} \\ &= \frac{1}{4\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x+40}{4}\right)^2}, \quad -\infty < x < \infty \end{aligned}$$

$$2) (i) f(x) = \frac{1}{2\sqrt{2\pi}} e^{-\frac{x^2}{8}}, \quad -\infty < x < \infty$$

$$\begin{aligned} &= \frac{1}{2\sqrt{2\pi}} e^{-\frac{1}{2}\frac{x^2}{4}} \\ &= \frac{1}{2\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-0}{2}\right)^2} \quad \dots(1), -\infty < x < \infty \end{aligned}$$

Comparing (1) with,

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty$$

we get

$$\mu = 0, \quad \sigma = 2$$

$$\therefore \text{Mean} = \mu = 0 \text{ and variance} = \sigma^2 = (2)^2 = 4$$

$$(ii) f(x) = \frac{1}{2\sqrt{\pi}} e^{-\frac{1}{4}(x-2)^2}, \quad -\infty < x < \infty$$

$$\begin{aligned} &= \frac{1}{\sqrt{2} \times \sqrt{2} \sqrt{\pi}} e^{-\frac{1}{2 \times 2}(x-2)^2} \\ &= \frac{1}{\sqrt{2} \sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-2}{\sqrt{2}}\right)^2} \quad \dots(1), -\infty < x < \infty \end{aligned}$$

Comparing (1) with,

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty$$

we get

$$\mu = 2, \quad \sigma = \sqrt{2}$$

$$\therefore \text{Mean} = \mu = 2 \text{ and variance} = \sigma^2 = (\sqrt{2})^2 = 2$$

$$3) \text{ i) } X_1 + X_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

$$\Rightarrow X_1 + X_2 \sim N(30 + 40, 25 + 35)$$

$$\Rightarrow X_1 + X_2 \sim N(70, 60)$$

$$\text{ii) } X_1 - X_2 \sim N(30 - 40, 25 + 35)$$

$$\Rightarrow X_1 - X_2 \sim N(-10, 60)$$

Check Your Progress 2

$$1) \text{ As } X \sim U[-a, a],$$

$\therefore$ probability density function of X is

$$f(x) = \frac{1}{a - (-a)} = \frac{1}{a + a} = \frac{1}{2a}, \quad -a < x < a.$$

$$\text{i) Given that } P[X > 4] = \frac{1}{3}$$

$$\Rightarrow \int_4^a \frac{1}{2a} dx = \frac{1}{3}$$

$$\Rightarrow \frac{1}{2a} [x]_4^a = \frac{1}{3}$$

$$\Rightarrow \frac{a - 4}{2a} = \frac{1}{3}$$

$$\Rightarrow 3a - 12 = 2a$$

$$\Rightarrow a = 12.$$

$$\text{ii) } P[X < 1] = \frac{3}{4}$$

$$\Rightarrow \int_{-a}^1 \frac{1}{2a} dx = \frac{3}{4}$$

$$\Rightarrow \frac{1}{2a} [x]_{-a}^1 = \frac{3}{4}$$

$$\Rightarrow \frac{1}{2a} [1 + a] = \frac{3}{4}$$

$$\Rightarrow 1 + a = \frac{3}{2} a$$

$$\Rightarrow 2 + 2a = 3a$$

$$\Rightarrow a = 2$$

$$2) \text{ As } X \sim U[-2, 2],$$

$$\therefore f(x) = \frac{1}{4}, \quad -2 < x < 2.$$

Now $P[X > k] = \frac{1}{2}$

$$\Rightarrow \int_k^2 \frac{1}{4} dx = \frac{1}{2}$$

$$\Rightarrow \frac{2-k}{4} = \frac{1}{2}$$

$$\Rightarrow 2 - k = 2$$

$$\Rightarrow k = 0.$$

Check Your Progress 3

- 1) We know that the mean and variance of chi-square distribution with n degrees of freedom are

$$\text{Mean} = n \text{ and Variance} = 2n$$

In our case, $n = 10$, therefore, Mean = 10 and Variance = 20.

- 2) Refer to Section 7.4.

Check Your Progress 4

- 1) Here, we are given that

$$\mu = 60, n = 15, \bar{X} = 65 \text{ and } S = 12$$

We know that the t-variate is

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

Therefore, we have

$$t = \frac{65 - 60}{12/\sqrt{15}} = \frac{5}{4.39} = 1.14$$

- 2) We know that the mean and variance of t-distribution with n degrees of freedom are

$$\text{Mean} = 0 \text{ and Variance} = \frac{n}{n-2} \text{ for } n > 2$$

$$\text{In our case, } n=20, \text{ therefore, Mean} = 0 \text{ \& Variance} = \frac{20}{20-2} = \frac{10}{9}$$

- 3) Refer to Section 7.5

Check Your Progress 5

- 1) Here, we are given that

$$n_1 = 15, n_2 = 10, \sigma_1 = 65, \sigma_2 = 45, S_1 = 60, S_2 = 50$$

The value of F-variate can be calculated as follows:

$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}$$

where, S_1^2 & S_2^2 are the values of sample variances.

Therefore, we have

$$F = \frac{(60)^2 / (65)^2}{(50)^2 / (45)^2}$$
$$= \frac{0.85}{1.23} = 0.69$$

- 2) Refer to Section 7.6.
- 3) We know that the mean and variance of F-distribution with v_1 and v_2 degrees of freedom are

$$\text{Mean} = \frac{v_2}{v_2 - 2} \quad \text{for } v_2 > 2 \quad \text{and}$$

$$\text{Variance} = \frac{2v_2^2(v_1 + v_2 - 2)}{v_1(v_2 - 2)^2(v_2 - 4)} \quad \text{for } v_2 > 4.$$

In our case, $v_1 = 5$ and $v_2 = 10$, therefore,

$$\text{Mean} = \frac{v_2}{v_2 - 2} = \frac{10}{10 - 2} = \frac{5}{4} \quad \text{and}$$

$$\text{Variance} = \frac{2(10)^2(5 + 10 - 2)}{5(10 - 2)^2(10 - 4)} = \frac{65}{48}$$

- 4) Refer to Section 7.6

7.9 REFERENCES

IGNOU Course Material

1. Post-Graduate Diploma in Applied Statistics (PGDAST), MST 003: Probability Theory, Block 4: Continuous Probability Distributions, UNIT 12: Normal Distribution (for Section 7.2 and related portions in Sections 7.0, 7.1, 7.7, 7.8 and Check Your Progress questions)
2. PGDAST, MST 003: Probability Theory, Block 4: Continuous Probability Distributions, UNIT 15: Continuous Uniform and Exponential Distributions (for Section 7.3 and related portions in Sections 7.0, 7.1, 7.7, 7.8 and Check Your Progress questions)
3. PGDAST, MST 004: Statistical Inference, Block 1: Sampling Distributions, UNIT 3: Standard Sampling Distributions - I (for Sections 7.4, 7.5 and related portions in Sections 7.0, 7.1, 7.7, 7.8 and Check Your Progress questions)
4. PGDAST, MST 004: Statistical Inference, Block 1: Sampling Distributions, UNIT 4: Standard Sampling Distributions - II (for Section 7.6 and related portions in Sections 7.0, 7.1, 7.7, 7.8 and Check Your Progress questions)

Reference Book

5. "Introduction to Probability and Statistics for Engineers and Scientists", 3rd Edition, Sheldon M. Ross, Elsevier Academic Press, 2004